Call: HORIZON-CL4-2021-DIGITAL-EMERGING-01

(Digital and emerging technologies for competitiveness and fit for the green deal)

Topic: HORIZON-CL4-2021-DIGITAL-EMERGING-01-09

Type of Action: HORIZON-IA

Proposal number: 101070524

Proposal acronym: RECLAIM

Type of Model Grant Agreement: HORIZON Action Grant Budget-Based

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Proposal ID **101070524**

Acronym **RECLAIM**

1 - General information

Fields marked * are mandatory to fill.

Topic	HORIZON-CL4-2021-DIGITAL-EMERGING-01-0	Type of Action	HORIZON-IA
Call	HORIZON-CL4-2021-DIGITAL-EMERGING-01	Type of Model Grant Agreement	HORIZON-AG

Acronym RECLAIM

Proposal title AI-powered Robotic Material Recovery in a Box

Note that for technical reasons, the following characters are not accepted in the Proposal Title and will be removed: < > " &

Duration in months 36

Fixed keyword 1 Robotics and automatic control

Fixed keyword 2 Artificial intelligence, intelligent systems, multi agent systems

Free keywords *Portable Material Recovery, Intelligent Autonomous Material Sorting, Solid Waste Management*

Abstract *

Recyclable materials recovery is a key element of the circular economy and the EU Green Deal. It is typically performed manually at large scale Material Recovery Facilities (MRFs) installed close to dense urban areas. Recent advances in AI and robotics have enabled the automation of several MRF activities. However, they target large waste volumes and are not cost-effective for smaller, less accessible areas.
To accommodate the latter, portable material recovery units can be deployed nearby. Despite the increasing demand for portable units, offerings lack intelligent, automated components that could significantly increase their productivity.
To fill this gap, RECLAIM will develop a portable, robotic MRF (prMRF) tailored to small-scale material recovery. The proposal exploits well-tested technology in robotics, AI and data analytics which is improved to facilitate distributed material recovery.
RECLAIM adopts a modular multi-robot/multi-gripper approach for material recovery, based on low cost Robotic Recycling Workers (RoReWos). An AI module combines imaging in the visual and infrared domain to identify, localize and categorize recyclables. The output of this module is used by a multi-RoReWo team that implements efficient and accurate material sorting. Further, a citizen science approach will increase social sensitivity to the Green Deal. This is accomplished via a novel Recycling Data-Game that enables and encourages citizens to participate in project RTD activities by providing annotations to be used in deep learning for the re-training of the AI module.
RECLAIM developments will be implemented and repeatedly assessed in demanding, real material recovery tasks. Three different scenarios will attest its effectiveness and applicability in a broad range of locations that face material recovery challenges. This will pave the way for the prMRF market uptake and provide a major boost in making Europe zero polluting, climate-neutral, sustainable and globally competitive.

Remaining characters 1

Has this proposal (or a very similar one) been submitted in the past 2 years in response to a call for proposals under any EU programme, including the current call? ○ Yes ◉ No

Please give the proposal reference or contract number.

Administrative forms

Proposal ID **101070524**

Acronym **RECLAIM**

Declarations

Field(s) marked * are mandatory to fill.

1) We declare to have the explicit consent of all applicants on their participation and on the content of this proposal. * ☒

2) We confirm that the information contained in this proposal is correct and complete and that none of the project activities have started before the proposal was submitted (unless explicitly authorised in the call conditions). ☒

3) We declare:
- to be fully compliant with the eligibility criteria set out in the call
- not to be subject to any exclusion grounds under the EU Financial Regulation 2018/1046
- to have the financial and operational capacity to carry out the proposed project.

☒

4) We acknowledge that all communication will be made through the Funding & Tenders Portal electronic exchange system and that access and use of this system is subject to the Funding & Tenders Portal Terms and Conditions. ☒

5) We have read, understood and accepted the Funding & Tenders Portal Terms & Conditions and Privacy Statement that set out the conditions of use of the Portal and the scope, purposes, retention periods, etc. for the processing of personal data of all data subjects whose data we communicate for the purpose of the application, evaluation, award and subsequent management of our grant, prizes and contracts (including financial transactions and audits). ☒

6) We declare that the proposal complies with ethical principles (including the highest standards of research integrity as set out in the ALLEA European Code of Conduct for Research Integrity, as well as applicable international and national law, including the Charter of Fundamental Rights of the European Union and the European Convention on Human Rights and its Supplementary Protocols. Appropriate procedures, policies and structures are in place to foster responsible research practices, to prevent questionable research practices and research misconduct, and to handle allegations of breaches of the principles and standards in the Code of Conduct. ☒

7) We declare that the proposal has an exclusive focus on civil applications (activities intended to be used in military application or aiming to serve military purposes cannot be funded). If the project involves dual-use items in the sense of Regulation 428/2009, or other items for which authorisation is required, we confirm that we will comply with the applicable regulatory framework (e.g. obtain export/import licences before these items are used). ☒

8) We confirm that the activities proposed do not
- aim at human cloning for reproductive purposes;
- intend to modify the genetic heritage of human beings which could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed), or
- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer.
- lead to the destruction of human embryos (for example, for obtaining stem cells)

These activities are excluded from funding. ☒

9) We confirm that for activities carried out outside the Union, the same activities would have been allowed in at least one EU Member State. ☒

The coordinator is only responsible for the information relating to their own organisation. Each applicant remains responsible for the information declared for their organisation. If the proposal is retained for EU funding, they will all be required to sign a declaration of honour.

False statements or incorrect information may lead to administrative sanctions under the EU Financial Regulation.

2 - Participants

List of participating organisations

#	Participating Organisation Legal Name	Country	Action
1	IDRYMA TECHNOLOGIAS KAI EREVNAS	EL	
2	UNIVERSITA TA MALTA	MT	
3	KATHOLIEKE UNIVERSITEIT LEUVEN	BE	
4	HELLENIC RECOVERY RECYCLING CORPORATION	EL	
5	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	
6	RobEnSo P.C.	Greece	
7	AIMPLAS - ASOCIACION DE INVESTIGACION DE MATERIALES PLASTICOS Y CONEXAS	Spain	
8	AXIA INNOVATION UG	DE	
9	INTERNATIONAL SOLID WASTE ASSOCIATION	NL	
10	PERIFERIAKOS FOREAS DIAXIRISIS STEREON APOVLITON IONION NISON ANONIMI ETERIA TON OTA	EL	

Organisation data

PIC	Legal name
999995893	*IDRYMA TECHNOLOGIAS KAI EREVNAS*

Short name: FORTH

Address

Street	N PLASTIRA STR 100
Town	IRAKLEIO
Postcode	70013
Country	Greece
Webpage	www.forth.gr

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	11/05/2016	- no
SME self-assessment	unknown	
SME validation	25/09/2008	- no

Departments carrying out the proposed work

Department 1

Department name	Institute of Computer Science	☐ not applicable
	☒ Same as proposing organisation's address	
Street	N PLASTIRA STR 100	
Town	IRAKLEIO	
Postcode	70013	
Country	Greece	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Michail**

Last name* **MANIADAKIS**

E-Mail* **mmaniada@ics.forth.gr**

Position in org. Researcher

Department Institute of Computer Science, Computational Vision and Robotics Laboratory

☐ Same as organisation name

☒ Same as proposing organisation's address

Street N PLASTIRA STR 100

Town IRAKLEIO

Post code 70013

Country Greece

Website http://users.ics.forth.gr/~mmaniada/index.htm

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Maria	Pateraki	pateraki@ics.forth.gr	+xxx xxxxxxxxx
Emmanouil	Hourdakis	ehourdak@ics.forth.gr	+xxx xxxxxxxxx
Emmanouil	Lourakis	lourakis@ics.forth.gr	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Michail	Maniadakis	Man	Greece	mmaniada@ics.forth.gr	Category B Senior le	Leading	https://orcid.org/0000-0002-5903-7778	Orcid ID
Prof	Maria	Pateraki	Woman	Greece	pateraki@ics.forth.gr	Category A Top grad	Team member	https://orcid.org/0000-0002-8943-4598	Orcid ID
Dr	Manolis	Lourakis	Man	Greece	lourakis@ics.forth.gr	Category A Top grad	Team member	https://orcid.org/0000-0003-4596-5773	Orcid ID
Dr	Manolis	Hourdakis	Man	Greece	ehourdak@ics.forth.gr	Category C Recognis	Team member		
Mr	Giorgos	Papadopoulos	Man	Greece	gpaps@ics.forth.gr	Category D First stag	Team member		
Mr	Manos	Papadakis	Man	Greece	manospapad@ics.forth.gr	Category D First stag	Team member		
Mr	Vasilis	Tzounakos	Man	Greece	tzounvas@ics.forth.gr	Category D First stag	Team member		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*M. Koskinopoulou, F. Raptopoulos, G. Papadopoulos, N. Mavrakis, M. Maniadakis, Robotic Waste Sorting Technology: Toward a Vision-Based Categorization System for the Industrial Robotic Separation of Recyclable Waste. IEEE Robotics Autom. Mag 2021; 28(2): 50-60.*
Good	*A Delta Robot Pair implemented and deployed in the Material Recovery Facilities of Chania, Crete, and another Delta Robot Pair implemented and deployed in the Material Recovery Facilities of Heraklion, Crete*
Good	*Two RoReWo Team based robotic systems are under development to be deployed in GreenDot Ltd, Cyprus and the Material Recovery Facilities of Heraklion, Crete.*
Publication	*E. Papadakis, F. Raptopoulos, M. Koskinopoulou and M. Maniadakis, On the Use of Vacuum Technology for Applied Robotic Systems, 2020 6th International Conference on Mechatronics and Robotics Engineering (ICMRE), Barcelona, Spain, 2020, pp. 73-77. DOI: 10.1109/ ICMRE49073.2020.9065189*
Publication	*M. Lourakis, G. Terzakis, A Globally Optimal Method for the PnP Problem with MRP Rotation Parameterization, IEEE International Conference on Pattern Recognition (ICPR'20), pp. xx-xx, Milan, Italy, January 10-15, 2021*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
InterRecycle (InterReg)	*InterRecycle is a recently founded project that considers the use of RoReWos for material recovery in industrial setups. InterRecycle develops two RoReWo Teams to be deployed in two MRFs in Cyprus and in Greece. The first versions of these systems will be ready and operational in the first half of 2022. The experience gained by InterRecycle provides important guidance for the successful implementation of the RECLAIM project.*
ANASA (GSRT)	*The ANASA project aims at developing, integrating and commercializing autonomous robotic systems accomplishing the categorization and separation of recyclable materials. The implemented system combines images in the visual and infrared domain to identify and categorize materials which are physically separated by a Delta Robot. The implemented system has been deployed in two industrial sites in the island of Crete, Greece.*
FELICE(H2020)	*FELICE addresses one of the greatest challenges in robotics, i.e. that of coordinated interaction and combination of human and robot skills. The project targets the application priority area of agile production and aspires to design the next generation assembly processes required to effectively address current and pressing needs in manufacturing. To this end, it envisages adaptive workspaces and a cognitive robot collaborating with workers in assembly lines.*
sustAGE(H2020)	*sustAGE is developing an IoT-based solution aiming to support prevention of workplace risks and ill-health, improvement of physical and mental health and communication with peers, whilst allowing effective human resource management via informed recommendations for job role or task changes. sustAGE is applied in two industry domains automotive manufacturing and transportation/logistics.*
VipGPU (GSRT)	*VipGPU develops hardware and software technologies to facilitate the implementation of algorithms for (i) Simultaneous Localization and Mapping and (ii) 3D rendering, using newly designed GPUs (Systems On Chip). It aims to design and implement a multidisciplinary platform consisting of multiple GPUS, that will execute the aforementioned algorithms with very low power consumption. The optimization methods that will be developed will utilize the acceleration properties provided by the GPUs*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Compact robotic material recovery facility	*FORTH has developed a compact, stationary material recovery setup at the municipal waste processing plant of Heraklion, Crete, Greece. This setup provides FORTH with significant experience and know-how for implementing the prMRF targeted in RECLAIM.*
Computational Vision and Robotics Lab	*Fully equipped Robotics lab with significant expertise in material recovery automation already applied in the real world.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999887059	*UNIVERSITA TA MALTA*

Short name: UoM

Address

Street	TAL OROQQ
Town	MSIDA
Postcode	2080
Country	Malta
Webpage	www.um.edu.mt

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	30/09/2015	- no
SME self-assessment	30/09/2015	- no
SME validation	unknown	

Departments carrying out the proposed work

Department 1

Department name	Institute of Digital Games (IDG)	☐ not applicable
	☐ Same as proposing organisation's address	
Street	20 Triq L-Esperanto	
Town	Msida	
Postcode	2080	
Country	Malta	

Links with other participants

Type of link	Participant
Same Group	

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Antonios**

Last name* **Liapis**

E-Mail* **antonios.liapis@um.edu.mt**

Position in org. Senior Lecturer

Department Institute of Digital Games

☐ Same as organisation name

☐ Same as proposing organisation's address

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Town Msida

Post code 2080

Country Malta

Website http://antoniosliapis.com/

Phone +356 2340 3510

Phone 2 *+xxx xxxxxxxxx*

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Antonios	Liapis	Man	Greece	antonios.liapis@um.edu.mt	Category B Senior le	Leading	0000-0001-5554-1961	Orcid ID
Prof	Georgios N.	Yannakakis	Man	Greece	georgios.yannakakis@um.edu.mt	Category A Top grad	Team member	0000-0001-7793-1450	Orcid ID
Dr	Iro	Voulgari	Woman	Greece	iro.voulgari@um.edu.mt	Category C Recognis	Team member	0000-0002-9593-5364	Orcid ID

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Gabriella A. B. Barros, Michael Cerny Green, Antonios Liapis and Julian Togelius: "Who Killed Albert Einstein? From Open Data to Murder Mystery Games," in Transactions on Games, vol. 11, no 1, pp. 79-89, 2019. The first instance of complete game generation based on real-world data, which can be an inspiration for data games produced in RECLAIM (cited 14 times)*
Publication	*G. N. Yannakakis, A. Liapis and C. Alexopoulos: "Mixed-Initiative Co-Creativity," in Proceedings of the 9th Conference on the Foundations of Digital Games, 2014. A vision paper on how AI can augment the creative process of users and how (cited 176 times)*
Publication	*J. Zhu, A. Liapis, S. Risi, R. Bidarra and G. Michael Youngblood: "Explainable AI for Designers: A Human-Centered Perspective on Mixed-Initiative Co-Creation," in Proceedings of the IEEE Conference on Computational Intelligence and Games, 2018. A vision paper for AI-assisted design tools to help users uncover how the AI took creative decisions (cited 109 times)*
Publication	*Georgios N. Yannakakis and Julian Togelius: "Artificial Intelligence and Games," Springer, 2018. The first handbook on AI as applied to games (cited 332 times)*
Publication	*K Makantasis, A Liapis, GN Yannakakis: "The Pixels and Sounds of Emotion: General-Purpose Representations of Arousal in Games". IEEE Transactions on Affective Computing (accepted). A first attempt at fusing gameplay video and soundscape information in a supervised learning task for modelling emotion (cited 7 times)*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
AI4Media	*AI4Media (H2020-ICT funding; no: 951911) aspires to become a centre of excellence and a wide network of researchers across Europe and beyond, with a focus on delivering the next generation of core AI advances to serve the key sector of Media, to make sure that the European values of ethical and trustworthy AI are embedded in future AI deployments, and to reimagine AI as a crucial beneficial enabling technology in the service of Society and Media. https://ai4media.eu/*
LearnML - Learn to Machine Learn	*The LearnML project (Erasmus+ funding; no: 2019-1-MT01-KA201-051220) brings state-of-the-art innovation into schools, transfers the notion of AI literacy to primary and secondary education and introduces students to the core principles of AI and machine learning through a uniquely designed game-based educational toolbox. http://learnml.eu/*
CoM_n_Play-Science	*The COM_n_PLAY Project (H2020-EU.5.d.; no: 787476) investigates the nature and impact of the informal science learning gained through coding, making and play activities. It identifies diverse practices and looks deeper into a sample of them, whereby participants of real-life activities are surveyed, observed, and gamefully engaged in intensive research. https://comnplayscience.eu/*
PrismArch	*PrismArch (H2020 ICT funding no: 952002) aims to design and develop a VR-aided design environment that hosts both architects and engineers towards the effective realization of an architectural project. As a platform it will allow for dynamic collaboration and joint decision-making, and as an interface it will allow users to preview, evaluate, simulate their work-in-progress with the assistance of suggestions and feedback provided by artificial intelligence. https://prismarch-h2020.eu/*
CrossCult	*During the CrossCult Project (H2020-EU.3.6.3 funding; no: 693150), 4 real-world flagship pilots involving a total of 8 culture heritage sites across Europe were implemented. The pilots used cutting-edge technology to connect existing digital cultural assets and to combine them with interactive experiences that all together are intended to increase retention, stimulate reflection and help European citizens appreciate their past and present in a holistic manner. https://www.crosscult.eu/*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
High-end computing	*UoM has latest state-of-the-art infrastructure and facilities available for performing the proposed research. UM will make available its supercomputing cluster, which contains 64 nodes, each with 12 cores and 32 GB of main memory (thus 768 cores and 2TB of main memory, in total).*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999991334	*KATHOLIEKE UNIVERSITEIT LEUVEN*

Short name: KUL

Address

Street	OUDE MARKT 13
Town	LEUVEN
Postcode	3000
Country	Belgium
Webpage	www.kuleuven.be

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	12/08/1911 - no
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name	Department of Mechanical Engineering	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Celestijnenlaan 300	
Town	Leuven	
Postcode	3001	
Country	Belgium	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Prof.

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Jef**

Last name* **Peeters**

E-Mail* **jef.peeters@kuleuven.be**

Position in org. Tenure Track Professor - Re & Demanufacturing

Department Department of Mechanical Engineering

☐ Same as organisation name

☐ Same as proposing organisation's address

Street Celestijnenlaan 300

Town Leuven

Post code 3001

Country Belgium

Website https://www.mech.kuleuven.be/en

Phone *+xxx xxxxxxxxx*

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Other contact persons

First Name	Last Name	E-mail	Phone
Karel	Kellens	karel.kellens@kuleuven.be	0032495695176
Isiah	Zaplana	isiah.zaplana@kuleuven.be	+xxx xxxxxxxxx
Sanne	D'haen	sanne.dhaen@kuleuven.be	+xxx xxxxxxxxx
Lukas	Lanneau	lukas.lanneau@kuleuven.be	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Isiah	Zaplana	Man	Spain	isiah.zaplana@kuleuven.be	Category C Recognis	Team member	0000-0002-0862-3240	Orcid ID
Prof	Karel	Kellens	Man	Belgium	Karel.Kellens@kuleuven.be	Category B Senior le	Leading	0000-0002-0340-6669	Orcid ID
Prof	Jef	Peeters	Man	Belgium	jef.peeters@kuleuven.be	Category B Senior le	Leading	0000-0003-1356-6508	Orcid ID
Mr	Bart	Engelen	Man	Belgium	Bart.Engelen@kuleuven.be	Category D First stag	Team member	0000-0002-6812-8719	Orcid ID
Mr	Dillam	Diaz	Man	Colombia	dillam.diaz@kuleuven.be	Category D First stag			
Dr	Ellen	Bracquene	Woman	Belgium	Ellen.Bracquene@kuleuven.be	Category C Recognis			
Mr	Jeroen	Cramer	Man	Belgium	Jeroen.Cramer@kuleuven.be	Category D First stag	Team member	0000-0002-3679-0556	Orcid ID
Mr	Gert	Schouterden	Man	Belgium	Gert.Schouterden@kuleuven.be	Category D First stag	Team member	0000-0002-8391-0514	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Diaz Romero, D.J., Sterkens, W., Van den Eynde, S., Goedemé, T., Dewulf, W., Peeters, J. with Diaz Romero, D.J. (corresp. author) (2021). Deep learning computer vision for the separation of Cast- and Wrought-Aluminum scrap. Resources Conservation And Recycling, 172, Art.No. 105685, 1-9. doi: 10.1016/j.resconrec.2021.105685*
Publication	*Cramer, M., Kellens, K., Demeester, E. with Cramer, M. (corresp. author) (2021). Probabilistic decision model for adaptive task planning in human-robot collaborative assembly based on designer and operator intents. IEEE Robotics and Automation Letters, 6 (4), 7325-7332. doi: 10.1109/LRA.2021.3095513*
Publication	*Duflou, J., Sutherland, J.W., Dornfeld, D., Herrmann, C., Jeswiet, J., Kara, S., Hauschild, M., Kellens, K. (2012). Towards Energy and Resource Efficient Manufacturing: A Processes and Systems Approach. CIRP Annals. Manufacturing Technology, 61 (2), 587-609. doi: 10.1016/ j.cirp.2012.05.002 Open Access*
Publication	*Cramer J, Cramer M, Demeester E, Kellens K. Exploring the potential of magnetorheology in robotic grippers. Procedia cirp. 2018 Jan 1;76:127-32.*
Publication	*Wagner, F., Peeters, J.R., De Keyzer, J., Janssens, K., Duflou, J.R., Dewulf, W. (2019). Towards a more circular economy for WEEE plastics - Part A: Development of innovative recycling strategies. WASTE MANAGEMENT, 100, 269-277. doi: 10.1016/j.wasman.2019.09.026 Open Access*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
AUSOM Project	*The AUSOM project aims at bringing a robust cost effective robotic sorting technique based on deep learning, computer vision and laser induced breakdown spectroscopy (LIBS) for sorting of shredded scrap metal, to the market.*
ACROGRIP	*The ACROGRIP project supported Flemish companies in their search for the most suitable robotic grippers by (1) conducting an extensive academic and industrial technology review, (2) developing a selection method and accompanying open web tool, and (3) organizing use cases and an industrial test lab for evaluating the applicability of flexible and innovative gripping techniques.*
SUDE Innovation mandate, 4.0 and Pilot	*3 consecutive projects bringing a deep learning computer vision controlled robotic system for the fully automatic grasping and dismantling of electronics to a TRL8, meaning a pilot system currently being finalized in a 20 feet container to be moved for testing at the location of a recycling plant before the end of 2021*
Random bin picking of unknown objects	*Efficient use of materials, reducing waste, and using end-of-life (EOL) product streams as resources are important pillars in order to evolve towards a circular economy, providing both better resource efficiency and improved security of materials supply. Considering these challenges, this PhD-research project focuses on product recognition and manipulation strategies required to deal with non-homogeneous bin or conveyor picking of unknown products.*
Robust robotic grasping of hetrogenious objects	*This PhD research targets to combine the advantages of both (hard and soft) gripper types in hybrid stiffness-adaptable grippers, by finding the most appropriate region within the continuous spectrum between soft and hard grippers. For this purpose, magnetorheological (MR) fluids, whose aggregation state can be continuously controlled from a free-flowing liquid to a viscoelastic solid by a magnetic field, are used in the development of hybrid MR-based robotic grippers.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Closed converyor setup with scara robot	*Software and hardware has been installed for the robotic picking and sorting of scrap metal from 0.8*6m recirculating conveyor setup with variable speed of 0-4m/s, a scara robot mounted on top and a pneumatic ejection system at the end of the conveyor*
2k depth and 4k 4 channel RGBN camera and software	*API have been developed and state of the art camera systems have been installed and neural networks have been trained for the real time segmentation and classification, as well as weight prediction of shredded metals and plastics*
Conveyor with multiple cobot and robot setups	*Multiple robot setups (e.g. UR5, KUKA LBR iiwa 14, Fanuc Deltarobot, Stäubli TX90...) and related peripheral equipment (e.g. conveyors, bins, sensors, camera's, grippers...) are available in the ACRO research lab.*
50+ commercial and self-developed grippers	*Within the ACROGRIP test lab, a wide variety (50+) of both commercial and in house developed robotic grippers is available for test in this project. Furthermore the involved ACRO research group has a broad industrial network including most of the major technology providers of robotic grippers.*
Material characterization lab	*A state of the art Laser Induced Breakdown Spectrometer (LIBS), X-ray fluorecent and Fourier Transformation (FTIR) system are available for material characterisation and a dedicated system has been developed to automate these analysis in the PolyCe H2020 projects*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
889187458	*HELLENIC RECOVERY RECYCLING CORPORATION*

Short name: HRRC

Address

Street	5, Himaras str., Marousi
Town	Athens
Postcode	151 25
Country	Greece
Webpage	www.herrco.gr

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	22/09/2021 - no
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **Panagiotis**

Last name* **Mavroeidis**

E-Mail* **pmavroeidis@herrco.gr**

Position in org. Technical Manager

Department HELLENIC RECOVERY RECYCLING CORPORATION

☒ Same as organisation name

☒ Same as proposing organisation's address

Street 5, Himaras str., Marousi

Town Athens

Post code 151 25

Country Greece

Website www.herrco.gr

Phone +30 210 8010962-3

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Ioannis	Pallas	ioannis.pallas@gr.ey.com	+30 697 209 5161
Panagiotis	Mavroeidis	management@herrco.gr	+30 210 8010962-3

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	George	Karamanlis	Man	Greece	gkaramanlis@herrco.gr	Category D First stage	Team member		
Mr	Achileas	Gougos	Man	Greece	agougos@herrco.gr	Category D First stage	Team member		
Mr	Lambros	Petropoulos	Man	Greece	lpetropoulos@herrco.gr	Category D First stage	Team member		
Mr	Panagiotis	Mavroeidis	Man	Greece	pmavroeidis@herrco.gr	Category D First stage	Team member		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☒
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Other achievement	*In 2020, HERRCO contributed to the achievement of the national target for packaging recycling by 96%, through the collection of 490,000 tons of packaging waste (out of 511,000 tons which is the national target for packaging recycling)*
	To date, HERRCO has provided 259 Greek municipalities with 174.800 "Blue Bins" for the collection of packaging waste (paper, plastic, glass, aluminum, tinplate) and 10,169 "Blue Bells" for the collection of glass packaging.
	In 2020, 532,000 tons of packaging materials and printed paper were recycled, including Industrial and Commercial Packaging Waste ICPW quantities

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
LIFE07 ENV/CY/000081 | Acronym: R.E.P.T.)	*The project aimed to address problems related to waste management in small islands and support the MS' authorities for the better design and implementation of recycling policies to meet the requirements of the EU Directives on Packaging Waste and WEEE. A Decision Support Tool was developed to calculate the environmental benefit and financial cost of alternative waste management systems on islands and take decisions to enhance environmental performance in the waste recovery and recycling sectors.*
The Blue Bin project (self-financed by HERRCO)	*The project constitutes the main process applied by the company to meet the national target for packaging recycling of municipal waste and is implemented in close collaboration with Greek municipalities, in accordance with the institutional framework. Citizens are called to place their packaging waste in the Blue Bins and then the recyclable materials are collected and transferred to the Materials Recovery Facilities, where they are sorted and prepared for the recycling process.*
The Blue Bell project (self-financed by HERRCO)	*The project serves exclusively the collection of glass packaging originated from municipal waste. It mainly addresses business activities in the food service industry sector related to the consumption of goods packaged in glass. The aim of this project is to develop a separate glass packaging waste stream*
Special Recycling Projects(self-financed by HERRCO	*HERRCO implements complementary projects to achieve national objectives and targets for recycling in the fields of recycling certification of Industrial and Commercial Packaging Waste (ICPW), recycling of wood packaging, monitoring of wood packaging reuse, collection and recycling of PET plastic bottles, etc.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Materials Recovery Facilities	*HERRCO currently owns 10 Materials Recovery Facilities and also is responsible for the monitoring and coordination of the activities of other 22 owned by third parties.*
Waste Sorting equipment	*The main equipment used in the waste sorting process that takes place in HERRCO's Materials Recovery Facilities comprises: bag openers, conveyor belts, sorting lines, optical sorting systems, densification systems, etc.*
Waste trucks	*HERRCO has equipped the Greek municipalities with 529 waste trucks for the collection of municipal recycling materials (packaging) collected in the municipal recycling bins (Blue Bins, Blue Bells)*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
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- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
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    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998993786	*IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA*

Short name: IRIS

Address

Street	CALLE VELAZQUEZ, NO 94 PRIMERA PLANTA
Town	MADRID
Postcode	28006
Country	Spain
Webpage	www.iris.cat

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	unknown
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2019	- yes
SME self-assessment	31/12/2019	- yes
SME validation	24/02/2009	- yes

Departments carrying out the proposed work

Department 1

Department name	IRIS Innovation Lab	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Carretera d'Esplugues 39-41	
Town	Cornellà de Llobregat	
Postcode	08940	
Country	Spain	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Ioannis**

Last name* **Kakogiannos**

E-Mail* **ikakogiannos@iris-eng.com**

Position in org. Project and Innovation Manager

Department Project Management Office

☐ Same as organisation name

☐ Same as proposing organisation's address

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Post code 08940

Country Spain

Website https://www.iris-eng.com/

Phone +34 93 554 25 00

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Albert	Torres	albert.torres@iris-eng.com	+34 93 554 25 00
Alexandra	Poch	apoch@iris-eng.com	+34 93 554 25 00
Colm	Digby	cdigby@iris.cat	+34 93 554 25 00
Joan	Puig	jpuig@iris-eng.com	+34 93 554 25 00
Ana Maria	Lopez	alopez@iris-eng.com	+34 93 554 25 00

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	David	Nettleton	Man	United Kingdom	david.nettleton@iris-eng.com	Category B Senior le	Leading	https://orcid.org/0000-0002-5852-7716	Orcid ID
Dr	Sara	Sanchez	Woman	Spain	ssanchez@iris-eng.com	Category B Senior le	Leading	https://orcid.org/0000-0002-1154-165X	Orcid ID
Dr	Laurent	Philippet	Man	France	lphilippet@iris-eng.com	Category B Senior le	Leading		
Ms	Diana	Galilea	Woman	Spain	diana.galilea@iris-eng.com	Category D First stage	Team member		
Mr	Jose Luis	Angulo	Man	Mexico	jlangulo@iris-eng.com	Category D First stage	Team member		
Mr	Martin	Folger	Man	Germany	jlangulo@iris-eng.com	Category D First stage	Team member		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Araujo-Andrade, C.; Bugnicourt, E.; Philippet, L.; Rodriguez-Turienzo, L.; Nettleton, D.; Hoffmann, L.; Schlummer, M. Review on the photonic techniques suitable for automatic monitoring of the composition of multi-materials wastes in view of their posterior recycling. Waste Management & Research. 2021;39(5):631-651. doi:10.1177/0734242X21997908. Int. Solid Wastes Public Clean. Assoc. ISWA 2021* *Year: 2021*
Publication	*An overview of the recent trends on the waste valorization techniques for food wastes, Jan 2019, Journal of Environmental Management, Arunima Nayak, Brij Bhushan, https://doi.org/10.1016/j.jenvman.2018.12.041* *Year: 2019*
Software	*Scadalytics is a Process Analytical Technology data management software. Machine Learning & Chemometrics tools for complex process control. Ability to intelligently interpret & connect metadata. Communication drivers for main PAT sensors on market, simplifying integration in plant & consolidation of real-time monitoring information for optimal running of processes & production plant. Connectivity with ERP, MES & SCADA; elevate chain of communication to plant managers & directors.*
Good	*VISUM: Non-destructive NIR Analysers for in-line inspection of critical process parameters. Real-time monitoring of multiple quality parameters, Product standardisation, Fraud control, Higher quality end product, Lead time reduction, Automated quality control, Higher performance, Reduction of production costs, Brand reputation protection through lower incidences of claims, Application Programming Interface, among others.*
Good	*VISUM InLine: Industrial NIR analyzer for real-time and continuous chemical composition monitoring adapted to several types of geometries and unit operations (conveyor belts, pipes, tubes, etc.) and unit process machinery (e.g. blenders). It is designed for: (1) Continuous monitoring of Quality Attributes (CQAs); (2) Real-time and simultaneous measurement of Critical Control Parameters (CCPs); (3) Anomaly Detection.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
ROSSINI, H2020 FOF, GA 818087	*RObot enhanced SenSing, INtelligence and actuation to Improve job quality in manufacturing.*
MULTICYCLE, H2020 SPIRE, GA 820695	*IRIS is the coordinator. With Hyperspectral Imaging we are classifying multilayer films of: polyamide (PA), polypropylene (PP), polyethylene (PE) and polyethylene terephthalate (PET). We can identify the majority layer and sometimes the two majority layers.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Spectroscopic Laboratory	*Equipped with several instruments, namely, 1 FTIR-ATR, 2 UV-VIS, 1 Extended-SWIR, 1 NIR spectrophotometer as well as a SWIR Hyperspectral imager . The staff are properly trained and skilled in chemometrics modelling by using a commercial suite as well as ad hoc non-commercial chemometric tools.*
Electronics, mechanical, optical workshop and lab	*Equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others.*

Optical photonics technologies and PAT systems	*HSI, NIR based cameras suitable for the surface and composition analysis of different materials and for the optimization and monitoring of numerous kinds of processes. The described tools can be applied for the digitalisation of waste valorisation processes, adding value to the circular economy.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
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- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
892386130	*RobEnSo P.C.*

Short name: RBNS

Address

Street	Eirinis kai Filias 24
Town	Heraklion, Crete
Postcode	715 00
Country	Greece
Webpage	

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	19/01/2021	- yes
SME self-assessment	19/01/2021	- yes
SME validation	unknown	

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **Freiderikos**

Last name* **Raptopoulos**

E-Mail* **robenso.contact@gmail.com**

Position in org. CTO

Department RobEnSo P.C.

☒ Same as organisation name

☒ Same as proposing organisation's address

Street Eirinis kai Filias 24

Town Heraklion, Crete

Post code 715 00

Country Greece

Website *Please enter website*

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Nikitas	Mavrakis	nikitasmavrakis@gmail.com	+xxx xxxxxxxxx
Maria	Koskinopoulou	koskinop@gmail.com	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	Freiderikos	Raptopoulos	Man	Greece	fredyraptis@hotmail.com	Category D First stage	Leading	https://orcid.org/0000-0001-7020-9932	Orcid ID
Dr	Maria	Koskinopoulou	Woman	Greece	koskinop@gmail.com	Category C Recognised	Team member	https://orcid.org/0000-0002-8040-2995	Orcid ID
Mr	Nikitas	Mavrakis	Man	Greece	nikitasmavrakis@gmail.com	Category D First stage	Team member		
Mr	Giorgos	Ploumis	Man	Greece	gploumis@hotmail.com	Category D First stage	Team member		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Good	*ROBENSO has implemented a new low-cost / high performance robot for material recovery, namely, the Robotic Recycling Worker (RoReWo). The RoReWo is now at TRL5 (validation in real industrial environment) and on the way to the market.*
Good	*RoReWo Teaming. ROBENSO has implemented control hubs and standardized function blocks to effectively control multiple RoReWo units that co-operate for the accomplishment of tasks. This facilitates flexible, modular design and quick installation for the composite system.*
Good	*ROBENSO has put to the recycling industry market a new Delta Robot solution for both material recovery and quality control of manually recovered materials.*
Publication	*F. Raptopoulos, M. Koskinopoulou and M. Maniadakis, Robotic Pick-and-Toss Facilitates UrbanWaste Sorting, 2020 IEEE 16th International Conference on Automation Science and Engineering (CASE), HongKong, 2020, pp. 1149-1154.*
Publication	*M. Koskinopoulou, F. Raptopoulos, G. Papadopoulos, N. Mavrakis, M. Maniadakis, Robotic Waste Sorting Technology: Toward a Vision-Based Categorization System for the Industrial Robotic Separation of Recyclable Waste. IEEE Robotics Autom. Mag 2021; 28(2): 50-60.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
ANASA (GSRT)	*The ANASA project has targeted the implementation of autonomous Delta Robots for material recovery in industrial setups. The implemented system combines visual and infrared sensing to recover and separate materials with high accuracy accomplishing more than 60 sortings per minute.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
R&D LAb	*A fully equipped industrial-like setup for experimentation on robotic waste treatment and material recycling in particular. It consists of a conveyor belt, various robotic platforms (Delta robot + RoReWo) and various grippers.*
RVM	*Prototype reverse vending machine for the identification and sorting of packages and containers received in bulk.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
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    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999513415	*AIMPLAS - ASOCIACION DE INVESTIGACION DE MATERIALES PLASTICOS Y CONEXAS*

Short name: AIMPLAS

Address

Street	VALENCIA PARC TECHNOLOGIC CALLE GUSTAVE
Town	PATERNA VALENCIA
Postcode	46980
Country	Spain
Webpage	www.aimplas.net

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2019	- yes
SME self-assessment	31/12/2019	- yes
SME validation	08/01/2009	- yes

Departments carrying out the proposed work

Department 1

Department name	Mechanical Recycling	☐ not applicable
	☒ Same as proposing organisation's address	
Street	VALENCIA PARC TECHNOLOGIC CALLE GUSTAVE	
Town	PATERNA VALENCIA	
Postcode	46980	
Country	Spain	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **Jesus**

Last name* **LATORRE ZACARES**

E-Mail* **ecdgpro@aimplas.es**

Position in org. Project's Director

Department Projects Department

☐ Same as organisation name

☒ Same as proposing organisation's address

Street VALENCIA PARC TECHNOLOGIC CALLE GUSTAVE EIFFEL 4

Town PATERNA VALENCIA

Post code 46980

Country Spain

Website www.aimplas.net

Phone +34961366040

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Ferran	Marti	fmarti@aimplas.es	+34961366040
Sonia	Albein-Urios	salbein@aimplas.es	+34961366040

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mrs	Sonia	Albein-Urios	Woman	Spain	salbein@aimplas.es	Category B Senior re	Leading	0000-0002-9083-5292	Orcid ID
Mrs	Vanessa	Gutierrez	Woman	Spain	vgutierrez@aimplas.es	Category B Senior re	Team member		
Ms	Adrián	Morales	Man	Spain	amorales@aimplas.es	Category D First stage	Team member		
Dr	Nuria	Garcia-Bastida	Man	Spain	ngarcia@aimplas.es	Category B Senior re	Team member		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Service	*EuCertPlast audits for plastic recycling plants and Recyclass audits for recyclability of plastic products. Ecolabelling of recycled products (e.g. Blue Angel, Distintiu de Garantia Qualitat Ambiental, RecyClass). Test laboratory accredited by ENAC in accordance with UNE-EN ISO/IEC 17025 standard to test recycled plastic materials in accordance with European standards.*
Publication	*Beiras, Ricardo; Verdejo, Eva; Campoy-Lopez, Pedro, Vidal-Linan, Leticia: Aquatic toxicity of chemically defined microplastics can be explained by functional additives. J. Hazardous Materials, 406, 2021, 124338 https://www.sciencedirect.com/science/article/abs/pii/S0304389420323281*
Other achievement	*Huge expertise in development and courses delivery (university master's degree, and for companies), e.g.: University Master's Degree in Polymeric Materials and Composites, Recycling of Plastic Materials, Chemical Recycling of Plastic Materials, Characterization and Quality Control of Recycled Material, Recycled Plastic in Food Contact. Participation in standardization committees, e.g.: AEN / CTN 53 /SC SC08- Recycled Plastics, CTN53 Plastics and Rubber or CEN/TC 249/WG 11- Plastics recycling.*
Publication	*VIDAL Rosario, MOLINER Enrique, MARTIN Pedro P., FITA Sergio, WONNEBERGER Maik, VERDEJO EVA, VANFLETEREN Francois, LAPENA Nieves, GONZALEZ ANA: Life Cycle Assessment of Novel Aircraft Interior Panels Made From Renewable or Recyclable Polymers With Natural Fiber Reinforcements and Non-Halogenated Flame Retardants. Journal of Industrial Ecology 2018, 22:132-144*
Publication	*CASTALDO Rachele, DE FALCO Francesca, AVOLIO Roberto, BOSSANNE Emilie, FERNANDES Felipe Cicaroni, COCCA Mariacristina, DI PACE Emilia, ERRICO Maria Emanuela, GENTILE Gennaro, JASINSKI Dominik, SPINELLI Daniele, URIOS Sonia Albein, VILKKI Markku, AVELLA Maurizio: Critical Factors for the Recycling of Different End-of-Life Materials: Wood Wastes, Automotive Shredded Residues, and Dismantled Wind Turbine Blades. Polymers 2019, 11*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
DECOAT	*DECOAT: Recycling of coated and painted textile and plastic materials --H2020-- Project – Decoat GA ID: 814505*
URBANREC	*URBANREC: New approaches for the valorisation of URBAN bulky waste into high added value RECycled products--WASTE H2020-- www.urbanrec-project.eu/ GA ID: 690103*
Multicycle	*Multicycle: Advanced and sustainable recycling processes and value chains for plastic-based multi-materials --H2020 SPIRE-- http://multicycle-project.eu/ GA ID: 820695*
CIMPA	*CIMPA: A CIrcular Multilayer Plastic Approach for value retention of end-of-life multilayer films –H2020-SC5 -- GA ID: 101003864*
SOL-REC2	*SOL-REC2: Innovative digital watermarks and green solvents for the recovery and recycling of multi-layer materials --H2020-SC5-- GA ID: 101003532*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Equipment	*Electrostatic/triboelectric separation. Elutriation. Equipment for separation of materials (plastic and non-plastic) based on electrostatic and triboelectric properties. Zig-zag air separation system using combined differences in particle density, size and shape, with a capacity from 20kg/h to 400kg*

Equipment	*Optical and NIR plastic material identifier and separator. Equipment for sorting of a mixture of plastic waste. Identifies and separates waste using NIR and VIS technologies with a hyperspectral camera*
Equipment	*Chromatography and other characterization techniques. Identification and quantification of residual monomers, solvents, VOCs, Analysis of contaminants and impurities in plastic materials and other media; Gas composition in packaging (O2, CO2, N2). Non-invasive analysis of oxygen content. Recycled ma*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
917429493	*AXIA INNOVATION UG*

Short name: AXIA

Address

Street	TURKENSTRASSE 29
Town	MUNCHEN
Postcode	80799
Country	Germany
Webpage	www.axia-innovation.com

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2019	- yes
SME self-assessment	31/12/2019	- yes
SME validation	unknown	

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Dr

Gender: (•) Woman ( ) Man ( ) Non Binary

First name*: **Ioanna**

Last name*: **Deligkizi**

E-Mail*: **ide@axia-innovation.com**

Position in org.: CEO

Department: AXIA INNOVATION UG

☒ Same as organisation name

☒ Same as proposing organisation's address

Street: TURKENSTRASSE 29

Town: MUNCHEN

Post code: 80799

Country: Germany

Website: https://www.axia-innovation.com

Phone: +49 89 24588469

Phone 2: *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Alexandros	Karathanasis	azk@axia-innovation.com	+xxx xxxxxxxxx
Giannis	Maris	info@axia-innovation.com	+xxx xxxxxxxxx
Vassilis	Maris	vma@axia-innovation.com	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Ioanna	Deligkiozi	Woman	Greece	ide@axia-innovation.com	Category A Top grade	Leading	0000-0002-6612-4140	Orcid ID
Ms	Raquel	Moreno Martin	Woman	Spain	rmo@axia-innovation.com	Category C Recognis	Team member	0000-0003-0631-3860	Orcid ID
Ms	Cristina	Onorato	Woman	Italy	com@axia-innovation.com	Category B Senior re	Team member	0000-0002-9192-377X	Orcid ID

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Best practices handbook for personalized interventions for treating Osteoarthritis, developed as part of our former H2020 project OActive. (https://www.oactive.eu/downloads/publicdeliverables/)*
Service	*AXIA is organizing and carrying out a number of internal and external training workshops. For this purpose, a set of materials, including training manuals and guidelines are developed and prepared.*
Service	*AXIA is developing marketing campaigns, producing dissemination and communication material including flyers brochures, and leaflets, as well as videos (https://www.youtube.com/channel/UCZs2b9y4zxUq6XyY0_XZqHA/videos) to effectively reach its target audiences.*
Service	*AXIA is specialized in offering its Innovation Management services in collaborative EU-funded research projects. Those services include exploitation and business planning, as well as intellectual property rights management for the consortium members and the exploitable project results.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
MASTRO, H2020	*The MASTRO project targets to develop intelligent bulk materials for smart applications in the transport sector incorporating several self-responsiveness properties, increasing consumer safety, component life-span, and performance, while reducing maintenance and manufacturing costs and through-life greenhouse gas emissions.AXIA was in charge of dissemination/communication, business plan and exploitation.*
MultiCycle, H2020	*MultiCycle will deliver an industrial recycling pilot plant for fossil and bio-based thermoplastic multilayer packaging and fibre reinforced composites using a novel solvent based selective extraction process which allows recovering pure plastics and additives in mixed wastes for their later reprocessing into value-added applications. AXIA is in charge of developing and decision support tool, business and exploitation.*
LightCoce, H2020	*LightCoce project aims in offering an open access Ecosystem for upscaling and testing multifunctional lightweight concrete and ceramic materials through 5 upgraded Pilot Lines (PLs) (targeting to Concrete, Conventional and Advanced Ceramics), as well as characterisation facilities, process modelling, quality assurance and monitoring, and standardisation, safety and innovation management services that will be accessible to the interested customers at fair conditions and cost.*
PureNano, H2020	*PureNano proposes an innovative, low cost, sustainable method for purification of plating baths which will lead to a significant extension of their lifetime (up to 10 times) that will have as a result a decrease of operation expenditures of plating shops, decrease of environmental risks from transportation of spent baths and increase of process efficiency and products quality. AXIA is responsible to define recycling and reuse paths for circular economy.*
BIOMAC, H2020	*BIOMAC project will create an Open Innovation Test Bed (OITB) ecosystem for nanomaterials openly accessible by SMEs and the industry though a single-entry point. OITB services include regulation, safety, sustainability, circularity and market potential with modelling, process control, standardisation and characterisation. AXIA working on circularity business models.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Premises	*We have all the necessary equipment to accelerate our consortia and clients? success. AXIA owns licenses for various software tools that will deliver high quality innovation management services, including market research, IPR and business management.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
899789752	*INTERNATIONAL SOLID WASTE ASSOCIATION*

Short name: ISWA

Address

Street	STATIONSPLEIN 45, A4.004
Town	ROTTERDAM
Postcode	3013 AK
Country	Netherlands
Webpage	www.iswa.org

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is no (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name	International Solid Waste Association	☐ not applicable
	☒ Same as proposing organisation's address	
Street	STATIONSPLEIN 45, A4.004	
Town	ROTTERDAM	
Postcode	3013 AK	
Country	Netherlands	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **Alan**

Last name* **Encinas**

E-Mail* **aencinas@iswa.org**

Position in org. Technical Coordinator

Department INTERNATIONAL SOLID WASTE ASSOCIATION

☒ Same as organisation name

☒ Same as proposing organisation's address

Street STATIONSPLEIN 45, A4.004

Town ROTTERDAM

Post code 3013 AK

Country Netherlands

Website www.iswa.org

Phone +31 6 18 39 44 79

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Aditi	Ramola	aramola@iswa.org	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Ms	Aditi	Ramola	Woman	Malaysia	aramola@iswa.org	Category D First stage	Team member		
Mr	Alan	Encinas	Man	Mexico	aencinas@iswa.org	Category D First stage	Leading		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Multicycle	*Advanced and sustainable recycling processes and value chains for plastic-based multi-materials --H2020 SPIRE-- http://multicycle-project.eu/ GA ID: 820695*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
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- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
889393292	*PERIFERIAKOS FOREAS DIAXIRISIS STEREON APOVLITON IONION NISON ANONIMI ETERIA TON OTA*

Short name: ION

Address

Street	P.VALIANOU-DIOIKITIRIO KEFALONIAS 0
Town	ARGOSTOLI GREECE
Postcode	281 00
Country	Greece
Webpage	

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	09/09/2021 - no
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name	PERIFERIAKOS FOREAS DIAXIRISIS STEREON APOVLITON IONION NISON ANO	☐ not applicable
	☒ Same as proposing organisation's address	
Street	P.VALIANOU-DIOIKITIRIO KEFALONIAS 0	
Town	ARGOSTOLI GREECE	
Postcode	281 00	
Country	Greece	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Vassileios**

Last name* **Analytis**

E-Mail* **banalytis@gmail.com**

Position in org. General Director

Department PERIFERIAKOS FOREAS DIAXIRISIS STEREON APOVLITON IONION NISON ANONIMI E

☒ Same as organisation name

☒ Same as proposing organisation's address

Street P.VALIANOU-DIOIKITIRIO KEFALONIAS 0

Town ARGOSTOLI GREECE

Post code 281 00

Country Greece

Website *Please enter website*

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Vassileios	Analytis	Man	Greece	banalytis@gmail.com	Category C Recognised	Leading		
Mr	Ioannis	Tavoularis	Man	Greece	gtavoularis@hotmail.com	Category D First stage	Team member		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☒
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*V. Analytis, A. Andreopoulos, K. Aravosis, (2016) "Effect of Public Policies on the Solid Waste Management Industry and Entrepreneurship in Greece", SDP Journal*
Publication	*E. Stamatakis, E. Zoulias, G. Tzamalis, Z. Massina, V. Analytis, C. Christodoulou, A. Stubos, "Metal hydride hydrogen compressors: Current developments & early markets", Renewable Energy 127, 850-862 (2018)*
Publication	*V. Analytis, A. Andreopoulos, K. Aravosis, (2009) "Treatment of Municipal Solid Waste & Recycling", International Conference of the Hellenic Solid Waste Management Association (HSWMA).*
Publication	*I. Kapageridis, I. Tavoularis, (2002), Mine Access Road Design and Modeling with IROAD, in Mine Planning and Equipment Selection 2002 (MPES)*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
	Full scale environmental management plans for 5 Greek airports (Athens, Thessaloniki, Rhodes, Corfu, and Heraklion) in the period 2008-2012

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

Proposal ID **101070524**

Acronym **RECLAIM**

3 - Budget

?

No.	Name of beneficiary	Country	Role	Personnel costs/€	Subcontracting costs/€	Purchase costs - Travel and substistence /€	Purchase costs - Equipment/€	Purchase costs - Other goods, works and services/€	Financial support to third parties/€ (actual cost)	Internally invoiced goods and services/€ (Unit costs-usual accounting practices)	Indirect costs/€	Total eligible costs	Funding rate	Maximum EU contribution to eligible costs	Requested EU contribution to eligible costs/€	Max grant amount	Income generated by the action	Financial contributions	Own resources	Total estimated income
1	Idryma Technologias Kai Erevnas	EL	Coordinator	299,200		45,000	40,000	15,000			99800.00	499000.00	100	499000.00	494,000	494000.00				494000.00
2	Universita Ta Malta	MT	Partner	240,000		27,000		5,000			68000.00	340000.00	100	340000.00	340,000	340000.00				340000.00
3	Katholieke Universiteit Leuven	BE	Partner	250,000		16,000		16,000			70500.00	352500.00	100	352500.00	352,500	352500.00				352500.00
4	Hellenic Recovery Recycling Corporation	EL	Partner	128,000		12,000	130,000				67500.00	337500.00	100	337500.00	337,500	337500.00				337500.00
5	Iris Technology Solutions, Sociedad Limitada	ES	Partner	222,600		18,000		88,000			82150.00	410750.00	70	287525.00	287,525	287525.00				287525.00
6	Robenso P.c.	EL	Partner	228,000		30,000	40,000	20,000			79500.00	397500.00	70	278250.00	278,250	278250.00				278250.00
7	Aimplas - Asociacion De Investigacion De Materiales Plasticos Y Conexas	ES	Partner	140,160		10,000		10,000			40040.00	200200.00	100	200200.00	200,200	200200.00				200200.00
8	Axia Innovation Ug	DE	Partner	241,200		16,000					64300.00	321500.00	70	225050.00	225,050	225050.00				225050.00

Proposal ID **101070524**

Acronym **RECLAIM**

9	International Solid Waste Association	NL	Partner	105,820		20,000		12,000			34455.00	172275.00	100	172275.00	172,275	172275.00				172275.00
10	Periferiakos Foreas Diaxirisis Stereon Apovliton Ionion Nison Anonimi Eteria Ton Ota	EL	Partner	128,000		20,000	15,000	35,000			49500.00	247500.00	100	247500.00	247,500	247500.00				247500.00
			TOTAL	1,982,980	0	214,000	225,000	201,000	0	0	655745.00	3278725.00		2939800.00	2,934,800	2934800.00	0	0	0	2934800.00

Proposal ID **101070524**

Acronym **RECLAIM**

4 - Ethics & security

Ethics Issues Table

1. Human Embryonic Stem Cells and Human Embryos			Page
Does this activity involve Human Embryonic Stem Cells (hESCs)?	○ Yes	◉ No	
Does this activity involve the use of human embryos?	○ Yes	◉ No	
2. Humans			**Page**
Does this activity involve human participants?	◉ Yes	○ No	9, 38
Are they volunteers for non medical studies (e.g. social or human sciences research)?	◉ Yes	○ No	9, 38
Are they healthy volunteers for medical studies?	○ Yes	◉ No	
Are they patients for medical studies?	○ Yes	◉ No	
Are they potentially vulnerable individuals or groups?	○ Yes	◉ No	
Are they children/minors?	○ Yes	◉ No	
Are they other persons unable to give informed consent?	○ Yes	◉ No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	○ Yes	◉ No	
Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014)? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	○ Yes	◉ No	
3. Human Cells / Tissues (not covered by section 1)			**Page**
Does this activity involve the use of human cells or tissues?	○ Yes	◉ No	
4. Personal Data			**Page**
Does this activity involve processing of personal data?	○ Yes	◉ No	
Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)?	○ Yes	◉ No	
Is it planned to export personal data from the EU to non-EU countries? Specify the type of personal data and countries involved	○ Yes	◉ No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country? Specify the type of personal data and countries involved	○ Yes	◉ No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	○ Yes	◉ No	
5. Animals			**Page**
Does this activity involve animals?	○ Yes	◉ No	
6. Non-EU Countries			**Page**
Will some of the activities be carried out in non-EU countries?	○ Yes	◉ No	
In case non-UE countries are involved, do the activities undertaken in these countries raise potential ethics issues?	○ Yes	◉ No	

It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	○ Yes	◉ No	
Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4.	○ Yes	◉ No	
Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4.	○ Yes	◉ No	
Does this activity involve low and/or lower middle income countries, (if yes, detail the benefit-sharing actions planned in the self-assessment)	○ Yes	◉ No	
Could the situation in the country put the individuals taking part in the activity at risk?	○ Yes	◉ No	
7. Environment, Health and Safety			**Page**
Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	○ Yes	◉ No	
Does this activity deal with endangered fauna and/or flora / protected areas?	○ Yes	◉ No	
Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	○ Yes	◉ No	
8. Artificial Intelligence			**Page**
Does this activity involve the development, deployment and/or use of Artificial Intelligence? (if yes, detail in the self-assessment whether that could raise ethical concerns related to human rights and values and detail how this will be addressed).	◉ Yes	○ No	6, 35
9. Other Ethics Issues			**Page**
Are there any other ethics issues that should be taken into consideration?	○ Yes	◉ No	

I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines How to Complete your Ethics Self-Assessment ☒

Ethics Self-Assessment

Ethical dimension of the objectives, methodology and likely impact

Human Participants
The innovations to be developed in the project regard the recovery of recyclable materials from waste streams. As such it will not process sensitive or any other kind of human-related data (ie. only waste-related data are of interest to the project).
To proceed with the development of AI methods, human participants (volunteers) will be asked to annotate waste images via a newly developed Recycling Data Game (RDG).
Individuals will be recruited from the public with the support of consortium partners that will advertise the citizen-science approach adopted in RECLAIM. No restrictions are foreseen for the participation of citizens.
The data that the users will provide regard only the annotation of waste images.

Artificial Intelligence
The project considers the development of AI methods to facilitate the identification, localization and categorization of recyclable materials. Human participants to be involved in the annotation of waste images will be informed that the data they provide are going to be used for training and improving AI-driven robotic systems for material recovery.
All AI technologies considered In the project focus on efficient material recovery and not in the processing of any kind of personal or personally identifiable data.

Remaining characters 3716

Compliance with ethical principles and relevant legislations

To participate in the project, volunteers will provide the minimum possible information. The Recycling Data Game will assume only a simple registration of the users, without collecting or storing any kind of personal information. The participants will be anonymized and not known to each other.

While the data collection procedures and regulations are well understood by the consortium, and we believe there is no conflict with the EU ethical principles and relevant legislations, the partners will nonetheless appoint an Ethics Advisor during the operation of the project to provide oversight and guidance specifically related to aspects associated to data collection for AI purposes.

Remaining characters 4312

Proposal ID **101070524**

Acronym **RECLAIM**

Security issues table

1. EU Classified Information (EUCI)[2]			Page
Does this activity involve information and/or materials requiring protection against unauthorised disclosure (EUCI)?	◯ Yes	◉ No	
Does this activity involve non-EU countries?	◯ Yes	◉ No	
2. Misuse			**Page**
Does this activity have the potential for misuse of results?	◯ Yes	◉ No	
3. Other Security Issues			**Page**
Does this activity involve information and/or materials subject to national security restrictions? If yes, please specify: (Maximum number of characters allowed: 1000)	◯ Yes	◉ No	
Are there any other security issues that should be taken into consideration? If yes, please specify: (Maximum number of characters allowed: 1000)	◯ Yes	◉ No	

[2]According to the Commission Decision (EU, Euratom) 2015/444 of 13 March 2015 on the security rules for protecting EU classified information, "European Union classified information (EUCI) means any information or material designated by an EU security classification, the unauthorised disclosure of which could cause varying degrees of prejudice to the interests of the European Union or of one or more of the Member States".

[3]Classified background information is information that is already classified by a country and/or international organisation and/or the EU and is going to be used by the project. In this case, the project must have in advance the authorisation from the originator of the classified information, which is the entity (EU institution, EU Member State, third state or international organisation) under whose authority the classified information has been generated.

[4]EU classified foreground information is information (documents/deliverables/materials) planned to be generated by the project and that needs to be protected from unauthorised disclosure. The originator of the EUCI generated by the project is the European Commission.

5 - Other questions

Two-stage calls

The full stage-2 proposal must be consistent with the short outline proposal submitted the to stage 1 - in particular with respect to the proposal characteristics addressing the concepts of excellence and impact

Are there substantial differences compared to the stage-1 proposal? ◯ Yes ◯ No

Essential information to be provided for proposals including clinical Trials / studies / investigations

Clinical study means, for the purpose of this document, any systematic prospective or retrospective collection and analysis of health data obtained from individual patients or healthy persons in order to address scientific questions related to the understanding, prevention, diagnosis, monitoring or treatment of a disease, mental illness, or physical condition. It includes but it is not limited to clinical studies as defined by Regulation 536/2014 (on medicinal products), clinical investigation and clinical evaluation as defined by Regulation 2017/745 (on medical devices), performance study and performance evaluation as defined by Regulation 2017/746 (on in vitro diagnostic medical devices).

Are clinical studies / trials / investigations included in the work plan of this project? ◯ Yes ◯ No

Horizon Europe Programme

HORIZON-CL4-2021-DIGITAL-EMERGING-01-09:
AI, data and Robotics for the Green Deal (IA)

Technical description (Part B)

Title: AI-powered Robotic Material Recovery in a Box

Acronym: RECLAIM

List of participants

Part. No.	Short Name of Participant	Participant Organization name	Country
1	FORTH	Foundation for Research and Technology Hellas	EL
2	UoM	University of Malta	MT
3	KUL	Katholieke Universiteit Leuven	BE
4	HERCCO	Hellenic Recovery Recycling Corporation	EL
5	IRIS	Iris Technology Solutions, Sociedad Limitada	SP
6	RBNS	ROBENSO PC	EL
7	AIMPLAS	AIMPLAS - Technological Institute of Plastics	SP
8	AXIA	Axia Innovation UG	DE
9	ISWA	International Solid Waste Association	NL
10	ION	Periferiakos Foreas Diaxirisis Stereon Apovliton Ionion Nison Anonimi Eteria Ton Ota	EL

1. EXCELLENCE

RECLAIM There is no time to waste

1.1 Objectives & Ambition

Improved recycling policies and the increasing interest in the concept of circular economy for sustainable use of raw materials are key elements of modern and efficient waste management across the EU. Regardless of how waste collection and recycling is implemented in different EU countries, collected waste needs to be processed either to remove unwanted or hazardous objects from the main material (negative separation of objects) or, to sort the reusable materials from the partially mixed waste collected (positive separation). Sorting these waste streams is today still often performed mainly manually, which is a mind-numbing task. Further, workers are typically exposed to hazardous conditions with adverse health effects, e.g. pulmonary, gastrointestinal and musculoskeletal problems[1].

Recent technological developments[2] have facilitated the modernization of large scale industrial Material Recovery Facilities (MRFs), which can be crucially supported by smart and automated waste treatment equipment. This approach is now adopted by several MRFs that are typically installed near urban areas to carry out municipal waste treatment. Despite the high productivity of these industrial solutions, transportation of waste to these large plants is costly and makes material recovery more difficult due to the compression of waste during transport. Moreover, there are still several cases where waste treatment needs are not efficiently catered for by the existing model. These include natural disasters, occasional high volume waste, rural or distant areas[3] [4] and transport hubs[5]. To meet the recycling needs of such cases, a new market trend has recently emerged, focused on implementing light, flexible and portable waste management units that can be quickly deployed in the areas in need, to undertake waste management and particularly the recovery of valuable recyclable waste[6] [7] [8]. Despite the strong demand for these systems, to date they lack the integration of smart, high-tech solutions that could significantly enhance their productivity. As a result, waste sorting in these niche sectors is still today powered by manual sorting.

Therefore, the **RECLAIM** proposal aims to exploit mature and well-tested AI-driven robotic waste management technology that will be improved and embedded in a state of the art "portable, robotic MRF" (prMRF) that will be capable of significantly enhancing local-scale material recovery activities providing them with industrial-level efficiency.

Noticeably, the EU is progressively increasing its targets for the recycling of municipal waste (summarized in Box 1), in order to feed economically valuable waste materials back into the European economy and to protect not only the environment, but also human health. To achieve these ambitious but achievable goals, the use of advanced robotic sorting systems that accelerate material recovery should be realized beyond the current, urban-centered focus points, to target today's untreated, or ineffectively treated waste. It is, therefore, necessary to develop a powerful decentralized approach to the management of waste, which will enable the on-site, close-to-source recovery of recyclables. Interestingly, in an attempt to overcome environmental challenges, the European Green Deal aims to implement actions ensuring (i) no net emissions of greenhouse gases by 2050, (ii) economic growth decoupled from resource use, (iii) *no person and no place left behind* [9]. The prMRF, besides supporting the first two points, provides a new, viable solution that will act as a catalyst to ensure the third, that no person and no location will be left behind.

RECLAIM envisions the development of low cost, portable, easy to install and increased productivity prMRF, which (soon after the project end) will enable the construction of high-tech distributed waste treatment prMRF-fleets that will allow to achieve full material recovery anywhere, even in the most remote areas. Accordingly, prMRF is

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[1] Poulsen OM et al. Sorting and recycling of domestic waste. Review of occupational health problems and their possible causes. *Sci Total Environ*. May 1995 168(1):33-56.
[2] Wilts H, Garcia BR, Garlito RG, Gómez LS, Prieto EG. Artificial Intelligence in the Sorting of Municipal Waste as an Enabler of the Circular Economy. *Resources*. 2021; 10(4):28.
[3] Oliveira V, Sousa V, Vaz JM, Dias-Ferreira C. (2018) Model for the separate collection of packaging waste in Portuguese low-performing recycling regions. J Environmental Management 13-24.
[4] Demonstrating resource efficiency through innovative, integrated waste recycling schemes for remote areas, LIFE14 ENV/GR/000722, http://www.pavethewayste.eu/en/
[5] Argüello, G. (2020) Environmentally sound Management of Ship Wastes: challenges and opportunities for European ports. J. Shipping & Trade. **5,** 12.
[6] https://blog.kiverco.com/kiverco-introduces-unique-modular-recycling-plant
[7] https://wrtltd.co.uk/mobile-picking-station/
[8] https://ecoballa.com/?p=127&lang=en
[9] https://ec.europa.eu/info/strategy/priorities-2019-2024/european-green-deal_en

expected to have a key role in developing a global, leakage-free circular economy model benefiting businesses, the society, and the environment.

Box 1. The Waste Framework[10], Landfill[11] and Packaging Waste[12] Directives were amended in 2018 to include, inter alia, new targets and measures, which will move the EU to the circular economy model:

- targets to increase preparing for reuse and recycling of municipal waste to at least 55% of municipal waste by 2025, 60% by 2030 and 65% by 2035;
- targets to increase recycling of packaging waste to at least 65% by 31 December 2025 and 70% by 31 December 2030;
- a target to reduce landfill to a maximum of 10% of generated municipal waste by 2035;
- a ban on landfilling of waste suitable for recycling effective from 2030;
- mandatory separate collection of bio-waste by 31 December 2023 and of textiles and hazardous waste from households by 1st of January 2025;
- minimum requirements for all extended producer responsibility schemes;
- simplified and improved definitions and harmonized calculation methods for recycling rates throughout the EU.

The success of these targets and measures will be key to the medium- to long-term prospects for achieving an innovative circular economy in which nothing is wasted, by 2050.

Project Innovation Pillars

To achieve its goal, **RECLAIM** undertakes concrete steps to create new solutions and support innovation in four technology pillars that crucially facilitate the recovery of valuable materials and is expected to shape next generation waste management. The innovation pillars elaborated in **RECLAIM** are depicted in Figure 1.1, and include:

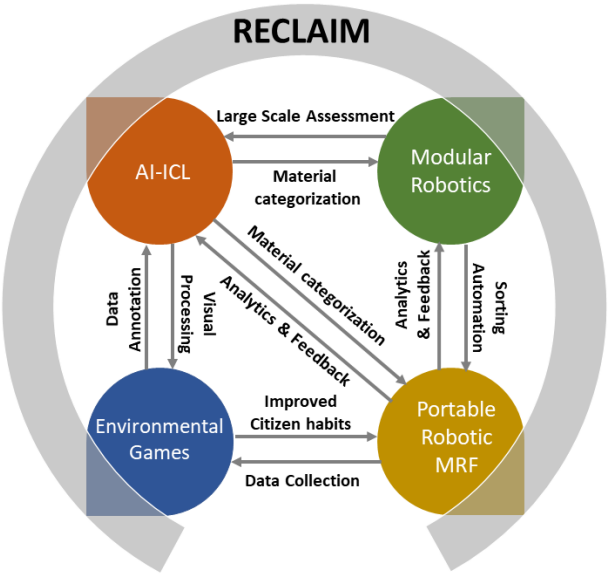

Figure 1.1. The four pillars supporting the **RECLAIM** project and their interactions.

PIL-1: Advanced AI for material identification, localization and categorization (AI-ILC) - exploit and improve existing AI technologies to develop effective solutions combining visual and hyperspectral information to accomplish high-performance recyclable waste identification, localization and categorization, applicable in real and harsh environments to support material recovery.

PIL-2: Modular, low-cost, high performance, multi-robot/multi-gripper recyclable recovery - RECLAIM promotes a new modular architecture for robotic recyclable recovery based on high productivity Robotic Recycling Workers (RoReWos) equipped with different grippers specializing in different material types. The use of RoReWo Teams installed in the prMRF is expected to double the "picks per invested euro" rate for the composite system.

PIL-3: Portable robotic Material Recovery Facility (prMRF) - the combined placement of established recycling processing mechanical equipment and the AI-powered robotic sorters in a container box that can be easily transported to points of interest, is ready for operation within few hours, is capable of efficient recovery of recyclable materials with a minimal number of recycling workers, is functional and sustainable for several years after the project end.

PIL-4: Environmental gaming for social awareness and data collection - increase social awareness to recycling via a novel Recycling Data-Game (RDG) that highlights the related challenges and encourages citizens to participate in project activities through a citizen science approach for providing data to AI-ILC training. Orthogonal to the above, RDG will provide the means to communicate the general principles of AI and Data Science to the public.

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[10] Directive (EU) 2018/851 of the European Parliament and of the Council of 30 May 2018 amending Directive 2008/98/EC on waste (OJ L 150, 14.06.2018, p. 109-140).

[11] Directive (EU) 2018/850 of the European Parliament and of the Council of 30 May 2018 amending Directive 1999/31/EC on the landfill of waste (OJ L 150, 14.06.2018, p. 100-108).

[12] Directive (EU) 2018/852 of the European Parliament and of the Council of 30 May 2018 amending Directive 94/62/EC on packaging and packaging waste (OJ L 150, 14.06.2018, p. 141-154).

Focused Deployments for Multifaceted System Assessment

The proposed work envisions deploying a prMRF at multiple pilot sites to assess its performance under real and particularly demanding conditions, recovering materials from post-consumer waste. In particular, the prMRF will be deployed in the group of Ionian islands, Greece, that provide the opportunity to assess its performance in multiple and complementary use-case scenarios. The Ionian group consist of seven main islands being very popular touristic destinations. All islands face major difficulties in waste management especially during the summer period. Still, the relatively limited size of such islands and the high seasonal characteristics of waste production[13] do not favor the investment on a full-size MRF. To date, the waste generated by residents and visitors is typically compacted and packaged to be transported to the mainland for material recovery. The residue of this waste treatment has to either return to the islands for landfilling or stay in the mainland for landfilling with particularly high costs[14]. In addition to the increased logistics expenses, especially in the case of roundtrip transfer (island-mainland-island), the quality of recyclable materials transferred to the mainland for processing decreases due to the temporary storage and compression. This causes increased material contamination/entanglement, resulting in significantly lower recycling rates and purities. As a result, a large percentage of valuable recyclables ends up in landfills instead of being recycled.

Similar to this particular case in Greece, the use of prMRF for the direct treatment of fresh, uncompressed, high-quality waste and the recovery/sorting of recyclables in a decentralized manner is of outmost importance for all rural areas. **RECLAIM** provides the only strategy to drastically increase the recovery rates for valuable materials by sorting at the source and to divide the logistics costs in half by avoiding back and forth transport.

The landscape of the Ionian islands group provides the unique opportunity to asses both the underlying technologies and business case for the prMRF. The **RECLAIM** partner ION has the responsibility of waste management in all the Ionian islands and will provide valuable support to explore how prMRF can be used to tackle the individual needs of different islands. In particular, ION has specified three scenarios of particularly high interest for the Ionian islands group but also for a broad range of areas in Europe facing similar challenges. The material recovery use case scenarios that will be explored in **RECLAIM** are summarized in Box 2.

Box 2. The material recovery scenarios to be explored in **RECLAIM**.

Scenario-1: Material recovery from mixed recyclables streams. The main recycling approach followed in Greece regards the use of bins where citizens collect mixed post-consumer municipal recyclables which are then separated in material recovery facilities. As mentioned above, especially for Ionian Islands this process has low efficiency, and entails high logistics expenses and high recoverable materials loss. **RECLAIM** will assess prMRF operation in the treatment of mixed recyclable waste streams, performing **positive separation of multiple material types**. This is the most challenging scenario due to the high material complexity to be investigated in RECLAIM and will be addressed during the whole duration of the project.

Scenario-2: Cleaning of citizen separated, material-specific streams. In a major part of Corfu (part of the Ionian islands group) the citizen-powered separation of recyclables is promoted, which assumes the use of multiple, material-dedicated bins for recyclable collection (this is the recycling approach also adopted by many North European countries). The collected materials need to be further processed to increase their purity and thus their value in the secondary market. **RECLAIM** will assess prMRF operation in the treatment of material-specific streams, performing **negative separation to remove unwanted objects**. This scenario will be explored during the months M13-M36, in parallel with Scenario-1, providing the opportunity to make prMRF appropriate for both positive and negative material selection. By comparing scenario 1 and 2 the advantages and disadvantages of separation by citizens will be evaluated, as separation by citizens also results in higher local collection costs for the separate waste streams.

Scenario-3: Immediate close to source material recovery. The inherent portable nature of the prMRF enables its installation close to areas with high productivity of waste, such as transportation hubs (e.g. ports, airports), festivals, sport events, exhibitions, etc. Unfortunately, so far, the fresh, high-quality waste produced in such areas is collected with garbage trucks to be integrated with ordinary municipality waste. **RECLAIM** will **exploit the portability of the prMRF** to explore the option of local, close to source robotic material recovery/sorting, and contrast its efficiency in relation to the current practice. This scenario will be explored in the last 6 months of the project, by installing the prMRF at a central location that will also serve as an open and live demonstrator to EU citizens (inhabitants and tourists), local authorities, governments, circular economy stakeholders and potential investors.

[13] N. Rogge, S. De Jaeger, C. Lavigne, (2017) Waste Performance of NUTS 2-regions in the EU: A Conditional Directional Distance Benefit-of-the-Doubt Model, Ecological Economics, 139, 2017.

[14] Fischer-Kowalski, M. Löw, M. Noll, D. Petridis, P. Skoulikidis, N., (2020) Samothraki in Transition: A Report on a Real-World Lab to Promote the Sustainability of a Greek Island. Sustainability 2020.

Ethical Issues

Ethics principles will be taken into consideration and applied in order to guarantee that the activity of **RECLAIM** in the field of AI and Gaming will not negatively affect the social pact or incorporate any techniques that could be used sinisterly in the future. Therefore, a list of ethical principles will be established by partners IRIS, RBNS and UoM. Ethics will be applied to ensure compliance with EU regulations and GDPR. Furthermore, a risk assessment and a data protection strategy will be defined and implemented throughout the entire duration of the project. FORTH, the coordinator, will be actively evaluating the process in order to ensure protection of rights and safety according to the ethical principles laid down in the Montreal Declaration for a Responsible Development of Artificial Intelligence.

1.1.1 OBJECTIVES

To realize **RECLAIM**'s aspirations and implement its outcomes, the proposed work is organized around six objectives described below. For each objective, we provide a short description of challenges, prior enablers and expected outputs. Outputs will be evaluated with a specific set of KPI (Key Performance Indicator) metrics.

Objective 1. Design and implement an eco-friendly prMRF (PIL-3)
Challenge. The implementation of prMRF assumes that mechanical Recycling Equipment[15] (R-Eq) such as trommels, flipl-flop separators, ballistic separators etc., is installed for solid waste pre-treatment and is operated in a way that guarantees eco-friendly operation and efficient cooperation throughout the waste treatment chain.
Enablers. This objective will be addressed in WP2: Application Requirements and system specification and WP5: prMRF implementation, demonstration & assessment, by combining: E-1-1: Environmentally friendly energy sources (e.g. solar panels) E-1-2: Commercial small-scale equipment typically used for the treatment of solid waste E-1-3: Available compact industrial research setup providing know-how on prMRF implementation
Output. Fully equipped prMRF installed in a container incorporating the R-Eq machinery, the AI-ILC and the robotic module, to accomplish fast and efficient recovery/sorting of recyclable materials. KPI-1-O1: Eco-friendly sources of energy provide >30% of the prMRF's power needs. KPI-1-O2: prMRF is continuously operating for >8 hours daily. KPI-1-O3: The R-Eq pre-processes waste ensuring that > 90% of it is accessible to the AI-ILC module. KPI-1-O4: The R-Eq provides enough feed for the robotic module to operate with <5% idle time.

Objective 2. Improve robotic recyclable recovery systems (PIL-2)
Challenge. For the robotic system to work within the prMRF, a change in the robotic technology commonly used to recover materials is required. The robotic solution should be low volume, modular, multi-functional and have a high pick per invested euro index.
Enablers. This objective will be addressed in WP4: RoReWo-Team for material recovery and WP5: prMRF implementation, demonstration & assessment, by combining: E-2-1: Robotic Recycling Workers (RoReWos), already successfully tested. E-2-2: Multiple available and under development grippers to be used in each RoReWo. E-2-3: Optimized RoReWo-Teams with a multi-robot/multi-gripper configuration
Output. RECLAIM develops an effective, modular robotic system based on multiple low-cost Robotic Recycling Workers (RoReWos), each exhibiting different material recovery skills. A team of cooperating RoReWos integrated in the prMRF to recover materials of different types. KPI-2-O1: The RoReWos Team combines ≥3 different grippers to recover multiple material types. KPI-2-O2: The RoReWos Team can treat >90% of the waste fed to the prMRF. KPI-2-O3: Installation volume <80% in comparison to the 2021 robotic waste sorter state of the art. KPI-2-O4: Pick per invested euro index >30% in comparison to the 2021 robotic waste sorter state of the art.

Objective 3. Advance and exploit state of the art AI technologies to effectively tackle the identification, categorization and localization of recyclable waste. (PIL-1)
Challenge. State of the art hyperspectral imaging together with RGB-based computer vision and Convolutional Neural Networks (CNN) technology coupled to implement an efficient AI module for recyclable Identification, Categorization and Localization (AI-ILC).

[15] The Evolution of Mixed Waste Processing Facilities- Technology and Equipment Guide (2016) The American Chemistry Council, prepared by Gershman, Brickner & Bratton, Inc.

Enablers. This objective will be addressed in WP3: Recyclable waste detection and categorization and WP5: prMRF implementation, demonstration & assessment, by combining:
E-3-1: Onsite human-driven fast data collection.
E-3-2: Misclassified object visual storage (to be used for re-training).
E-3-3: Hyperspectral extension and training of state of the art instance segmentation CNNs.

Output. A high-performance recyclable recognition module that can accurately guide the RoReWo Team in sorting recyclables. The automation of the data collection procedure, the powerful model training, and the periodic improvement (re-training) of the model will critically support the long-term sustainability of the prMRF.
KPI-3-O1: A recyclable waste-Data Base (rwDB) with annotated hyperspectral and RGB images.
KPI-3-O2: Each of the hyperspectral imaging and the RGB-based waste recognition modules can individually recognize >85% of the treatable recyclables from the mixed waste stream.
KPI-3-O3: In combination, the hyperspectral and the RGB-based waste recognition modules recognize >95% of the treatable recyclables from the mixed waste stream.
KPI-3-O4: The low confidence/misclassified waste stored in the rwDB is correctly annotated by >50%, before each cycle of AI-ILC retraining.
KPI-3-O5: The AI-ILC module processes waste streams at >5 fps (provide sufficient data to the RoReWo Team).

Objective 4. Deploy prMRF to realize the recovery and sorting of recyclables in targeted areas (PILs-1,2,3)

Challenge. Showcase the ability of prMRF to easily deploy and operate longitudinal recyclable material recovery processes in demanding real environments. Showcase and validate the ability of prMRF to adapt to waste streams with changing characteristics.

Enablers. This objective will be addressed in WP5: prMRF implementation, demonstration & assessment, by combining:
E1: Direct development of the prMRF based on the technology already available to project partners, to facilitate prompt real-world assessment of project developments (early prMRF deployment ready on M9).
E2: Ready-to-use prMRF to facilitate fast deployment.
E3: Analytics on the performance of the individual prMRF modules and informative evaluation metrics to adjust prMRF configuration on the fly.

Output. Demonstration and validation of the prMRF in multiple use-case scenarios with diverse operational requirements. Demonstration of efficient cooperation of R-Eq units with the intelligent waste treatment modules, accurate recognition of recyclable waste, effective waste recovery/sorting by the RoReWos Team, parameterized performance of the composite system.
KPI-4-O1: Pre-deployment prMRF configuration after considering waste distribution per area, use case and season and online configuration adjustment.
KPI-4-O2: Fast deployment of the prMRF, which can be fully functional in ≤2 days.
KPI-4-O3: The prMRF configuration can be modified in <30 minutes for each possible adjustment.
KPI-4-O4: The prMRF is capable of recovering >80% of the serviceable recyclable waste.
KPI-4-O5: Incrementally advanced prMRF performance based on the periodic AI-ILC re-training, to successfully treat >50% of the previously miscategorized recyclables.

Objective 5. Increase social sensitivity and engage citizens in material recovery research (PIL-4)

Challenge. Combine gamification and social computing technology with data collection procedures and the rwDB to develop a Recycling Data-Game (RDG) that promotes awareness, increases social sensitivity and facilitates the participation of citizens in data collection for prMRF implementation.

Enablers. This objective will be addressed in WP6: Recycling Data-Games, by combining:
E1: Gamification of recyclable sorting activities in MRFs
E2: Citizen-science approach for participatory data collection through the RDG
E3: Available techniques for procedural content generation, user modelling recommender systems

Output. Demonstrate the use of newly developed games that exploit rwDB data to implement easy-to-use recycling data-games, which operate as data collection hubs and enjoyable information diffusion cores about recycling.
KPI-5-O1: Use >2000 images and >20 recyclable sorting videos from the rwDB to develop the RDG
KPI-5-O3: The games will penetrate to >500 unique users
KPI-5-O4: The willingness to recycle will increase for >90% of the game users.

Objective 6. Maximize project Innovation impact (PIL-5)

Challenge. Shape the future of global recycling activities and ensure sustainable value for **RECLAIM** investments. Attract the interest of stakeholders and promote the project to academic peers, industry, policymakers

and the society as a whole. Exploit acquired knowledge beyond the project and pave the way for the commercialization of **RECLAIM** innovations.
Enablers. This objective will be addressed in WP7: Project impact maximization, as follows: E1- The SMEs participating in the **RECLAIM** will link their business-related networks with the project. E2 - Academic participants keen on market-oriented research with a focused interest in creating innovation. E3 - Capitalize on established global communication channels that will guarantee a major transformative impact on global recycling strategies. E4 - Multiple, concrete, well defined innovative assets with high market potential targeted by **RECLAIM**. E5 - Active collaboration with EU/national projects, existing and upcoming Horizon Europe Clusters and synergies with the European Partnership on AI, Data and Robotics
Output. Increased project impact, tangible dissemination activities, exploited project results, enhanced commercialization potential of project assets. KPI-O6-1: prMRF sustainability plan to assure operation for ≥ 5 years after the end of the project. KPI-O6-2: Targeted business models for (i) the prMRF, (ii) the RoReWos, (iii) the AI-ILC to justify their commercialization potential. KPI-O6-3: Seek and establish collaboration with ≥3 projects from the CSA HORIZON-CL4-2021-HUMAN-01-02 and Horizon Europe Clusters 4 and 6. *- Tables 2.4 & 2.5 provide additional lists of dissemination/communication criteria and KPIs*

1.1.2 Research and innovation maturity, advancements to the state of the art

The main result of the project will be a **fully functional prMRF** demonstrating material recovery in three different use case scenarios (see Box 2), to be validated at **TRL 7**. The prMRF will be demonstrated in demanding operational environments where it will function daily for many hours, achieving the recovery of multiple materials, from varying waste streams. Additionally, the project will create novel **recycling data games (RDG)** to engage the wider public and interested stakeholders in project research activities, which will be validated at **TRL 7**. Following the above, the project introduces novel scientific insights and technological solutions that are described below, organized into the project's four innovation pillars. For each pillar, we provide concrete statements on the advancement to the state of the art and a separate table showing the Technology Readiness Levels (TRL) of relevant, already available enabling components, as well as their envisioned TRL advancement at the end of **RECLAIM**.

PIL-1: Advanced AI for material identification, localization and categorization

Machine vision and image processing have experienced tremendous growth within the Industry 4.0 concept. On one hand, hardware advancements provide new opportunities for the identification of materials and localization of objects. On the other hand, algorithms powered by deep learning can effectively categorize materials and objects by taking into consideration and blending several complementary characteristics.

In terms of waste processing and material recovery, spectroscopy techniques are widely implemented in order to determine the consistency of objects. Images of the object in combination with chemometric models can be used for real-time identification and quantification of its composition[16], with high classification accuracy even for complex plastics. Complementary to hyperspectral cameras, in real world applications such as recyclable material sorting in the wild, the visual appearance of objects should be considered to separate adjacent or overlapping objects with same consistency. Interestingly, state of the art Deep Learning (DL) approaches have been recently used to successfully classify objects into predetermined material categories[17]. RECLAIM will implement a mixed approach that combines waste imaging in both the visual and infrared domain, to achieve high accuracy material identification, localization and categorization in real, particularly challenging conditions. Complementary to the mixed monitoring approach, RECLAIM will consider the performance of the individual monitoring systems, focusing particularly on the use of ordinary RGB cameras of much lower cost (in comparison to hyperspectral imaging). It is expected that RGB monitoring can be sufficient for the operation of prMRF in the second use case scenario (see Box 2), which assumes the detection of foreign bodies in a single-material stream, and their negative separation, i.e. the removal of foreign bodies. Moreover, to enhance the sustainability of project developments, RECLAIM will study methods that exploit human-provided visual and hyperspectral input to accomplish stable, incremental advancement of the AI-powered material identification, localization and categorization module.

[16] Araujo-Andrade, C.; Bugnicourt, E.; Philippet, L.; Rodriguez-Turienzo, L.; Nettleton, D.; Hoffmann, L.; Schlummer, M. Review on the photonic techniques suitable for automatic monitoring of the composition of multi-materials wastes in view of their posterior recycling. Waste Management & Research. 2021;39(5):631-651.

[17] M. Koskinopoulou, F. Raptopoulos, G. Papadopoulos, N. Mavrakis, M. Maniadakis, Robotic Waste Sorting Technology: Toward a Vision-Based Categorization System for the Industrial Robotic Separation of Recyclable Waste. IEEE Robotics Autom. Mag 2021; 28(2): 50-60.

Advancements to the state of the art

- Explore the limits of RGB- and hyperspectral-imaging in real-world material recovery tasks.
- Develop new DL methods combining RGB- and hyperspectral- imaging to accomplish superior performance on material identification, localization and categorization.
- Develop automated methods that exploit RDG data (see PIL-4, below) for the repeating advancement of DL-powered computer vision modules.

The targeted advancements will be supported by partner-contributed assets that will be elaborated and improved further in terms of the Technology Readiness Level (TRL), as summarized in the following table.

Table 1.1.2a Asset description and advancement table with respect to PIL 1.

Partner - Asset description and advancement	TRL: From → To
FORTH - Deep Learning powered material recognition. Fully functional Mask-RCNN and Yolact++ models are available for real-time material identification, localization and categorization on waste flows. The models will be trained with new waste images to support real-world prMRF operation.	TRL7 → TRL7 (WP3, WP5)
IRIS - Hyperspectral imaging for material identification/categorization. A Hyperspectral imaging sensor has been developed, capable of identifying and categorizing materials depending on their chemical consistency and shape. Models will be developed to address difficult real-world conditions and used as a reference for the further processing of data.	TRL6 → TRL7 (WP3, WP5)
RBNS - Quick annotation tool. It allows users to select the entire pixel area of the objects appearing in complex industrial images, with a single click. The tool will be extended to enable the annotation of hyperspectral images.	TRL6 → TRL7 (WP2, WP6)
AIMPLAS - Separation of Plastics. There methods available targeting the sorting of complex plastics (WEEE, automotive & marine littering). The method will be expanded to address food-packaging plastics that can be sorted and sold separately in the secondary market.	TRL6 →TRL7 (WP5)

PIL-2, New, modular, low-cost, high performance, multi-robot/multi-gripper recyclable recovery

In recent years, many robots have been used in commercial and research applications aimed at recovering and sorting recyclable materials. The implemented systems are based on general purpose robotic technology -most frequently Delta robots, but also robot arms and multi-axes gantry robots. These approaches offer features not necessarily important for post-consumer waste treatment (e.g. handling of heavy objects), which results in a significant and unjustified increase in the cost of the composite solution. On the contrary, it is more efficient and sustainable to focus on developing low-cost robots specialized in material recovery applications, which will be very productive, but also very robust and durable to operate in wet, dusty/dirty and highly congested waste management environments. Moreover, instead of using a single high-cost robotic unit with a single end-effector which restricts the application of the system in handling a limited number of material types, a distributed solution can be implemented, with multiple low-cost robots featuring different skills, each undertaking a different role in waste recovery and sorting tasks. This is the approach adopted in **RECLAIM** which aims to develop a new modular, low-cost, high performance robotic solution for recyclable material recovery and sorting.

Advancements to the state of the art

- Implement modular robotic systems for recyclable recovery based on low cost Robotic Recyclable Workers (RoReWos), equipped with grippers specialized to different material types.
- Implement RoReWo Teams accomplishing positive and negative material separation.
- Significantly increase the “picks per invested euro” rate for the composite robotic system.

The targeted advancements will be supported by available HW/SW assets that will be further evolved in terms of their Technology Readiness Level, as summarized in the following table.

Table 1.1b Asset description and advancement table with respect to PIL 2.

Partner - Asset description and advancement	TRL: From → To
FORTH - Material recovery optimization. An approach is available to select the next material to pick, based on proximity, ecological and economic criteria. This approach will scale up to select multiple materials for all RoReWos in a Team.	TRL6 → TRL7 (WP4, WP5)
RBNS - RoReWo units. Fully functional RoReWo units equipped with vacuum grippers have been implemented and assessed in an industrially relevant material recovery environment. RoReWos will be expanded with new grippers and will be evaluated in real operational environments.	TRL5 → TRL7 (WP4, WP5)

RBNS - Parametrized pick and place programming. A framework that facilitates pick and place programming by non-experts. The framework will be advanced to store reproducible parameterized behaviors for RoReWo-gripper pairs.	TRL5 → TRL7 (WP4, WP5)
KUL - Grasping mechanisms. Collaborative algorithms for joint grasping and load balancing have been implemented. These algorithms will be customized to support the design and implementation of new grippers for robotic sorting of objects/waste with unknown content.	TRL4 → TRL7 (WP4, WP5)

PIL-3, Portable robotic Material Recovery Facility (prMRF)

Portable material recovery and sorting facilities are used by waste management companies, demolition companies and local authorities/local authority contractors that have an active role in the circular economy value chain. They cover waste processing needs at places where installation of stationary waste sorting stations is impossible/impractical. Portable material recovery facilities are compact, easy to transfer and install recovery stations, which still rely on manual sorting. The increased workforce required for the operation of the facilities, which assumes humans to be transported to remote workplaces, trained in the recovery and sorting of materials, and finally work in difficult and unhealthy conditions with their efficiency constantly decreasing during a shift, reduces the productivity of the stations and limits their widespread use despite the increased market needs. The proposed work draws on the extensive experience of **RECLAIM** partners in intelligent automated waste treatment based on AI, computational vision, hyperspectral imaging, and robotic sorting, to build the first high-tech portable robotic material recovery facility that accomplishes increased productivity with a minimal need of people working in unhealthy conditions. A concept image of the proposed prMRF is shown in Figure 1.2.

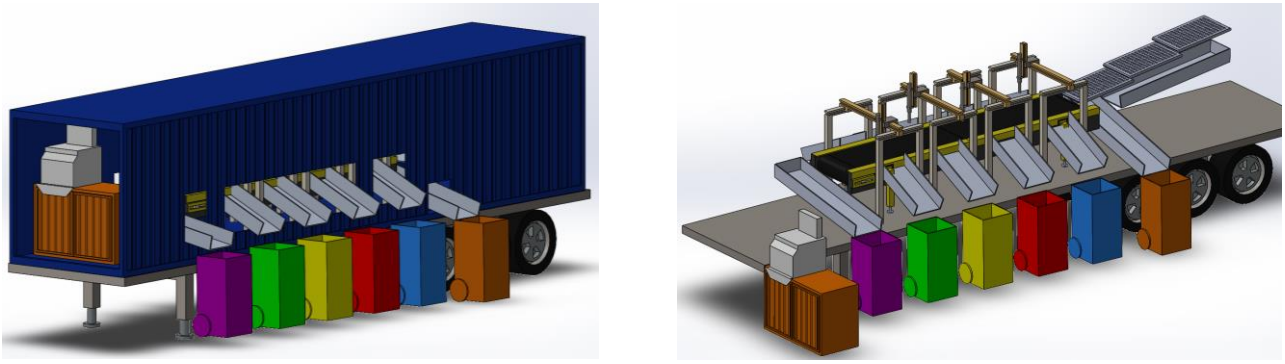

Figure 1.2. Concept image of the prMRF showing an outside and inside view. In the inside view, from right to left, there is a flip-flop screen to remove small and wet materials, a belt to transfer waste, 4 low-cost RoReWos installed above the conveyor belt to pick multiple materials which are then temporally stored in bins with different colours and at the end, a press for compacting the recovered materials.

Advancements to the state of the art

- Implement the prMRF, the first portable fully equipped robotic material recovery facility to be assessed in real and challenging operational environments.
- Exploit performance analytics to enable informed, on the fly prMRF configuration adaptations that address changing characteristics of the processed waste streams.

The accomplishment of the advancements summarized above will be supported by the experience gained during the development of a compact, prMRF-like industrial research setup in Heraklion, Crete, Greece, which hosts two cooperating ABB IRB-360 Delta robots that undertake material recovery. This compact setup has significantly facilitated data collection, experimentation and the assessment of robots and grippers in material recovery in highly realistic and demanding industrial environments. The expertise gained from the implementation of this research setup will significantly boost the proposed work and the implementation of the prMRF in particular.

Figure 1.3. The compact industrial research setup developed by FORTH, operating at a waste processing plant in Heraklion, Crete, Greece.

Table 1.1c Asset description and advancement table with respect to PIL 3.

Partner - Asset description and advancement	TRL: From → To
FORTH - Compact industrial research setup. A compact, fully equipped industrial research setup has been developed in Heraklion, Greece, directly linked with the local waste processing plant. The construction of this setup provides significant experience and know-how for the design and implementation of the prMRF.	TRL7 → TRL7 (WP2, WP6)
AIMPLAS – Waste collection. AIMPLAS has a broad experience in certification of recycled materials following European eco-labels and certifications of post-consumption waste. This method will be adapted for the collection and logistics of waste in scattered residential areas.	TRL7 →TRL8 (WP6)

PIL-4: Educational environmental gaming for social awareness and training

Gamification has gathered substantial academic and commercial interest in recent years for its ability to mobilize users in solving problems and in increasing their engagement. Playful interactions, themes inspired from games, and reward structures (gamification) have been extensively used in "serious games" for educational purposes[18] for citizen reflection[19] and for promoting a more ecologically-minded behavior[20]. Gamification principles have been applied extensively to motivate users/players to contribute their annotations and data in *human computation games*, in which player interaction is used to generate data or solve problems[21]. These games are not only built around increasing awareness, but directly capitalize on the desire of non-experts in contributing (small or larger parts) to address scientific challenges, a trend known as citizen science[22],[23]. Data games is another research strand focused on generating new games (or variations of games) from data, thus providing original gameplay experiences every time[24]. A core ambition of PIL-4 regards the development of **Recycling Data Games (RDG)** that pair human computation and data games in a way that real-world data is shown to users, who through their annotations contribute new knowledge models to the algorithms that select new content to show and that devise new challenges for users. Besides contributing ever-fresh knowledge and human computation to the challenges of recycling automation tasks, the above described closed-loop interaction can be particularly engaging for users of diverse age, gender, and background and may provide a framework to explain and teach users the basic principles of Data Science and AI.

Advancements to the state of the art

- Implement playful, intuitive interface paradigms for improving social awareness regarding recycling using good practices of gamification beyond extrinsic rewards.
- Implement playful, intuitive interface paradigms for human computation tasks which can lead to expanded and improved training data for use in PIL-1.
- Realize the first RDG instance that contributes to an iterative cycle of re-use, from real-world data to user annotations to improved baselines and better real-world data to show back to users.

The aforementioned advancements will be supported by well tested gamification methods which will be further advanced in terms of their Technology Readiness Level, as summarized in Table 1.1d.

Table 1.1d Asset description and advancement table with respect to PIL 4.

Partner - Asset description and advancement	TRL: From → To
UoM - DATA Agent[25]. Data game algorithms that consider the (re)use of real-world data to instantiate generated games. The algorithms will be used beyond entertainment (i.e. social awareness, human computation) and will be expanded to cover the recycling component using multi-player data to inform/ improve the data selection/reuse processes.	TRL4 → TRL6 (WP6)

[18] A. Domínguez, J. Saenz-de-Navarrete, L.Marcos, L. Fernández-Sanz, C. Pagés, J..Martínez-Herráiz. "Gamifying learning experiences: Practical implications and outcomes". Computers & Education, 2013.
[19] K. Kontiza, A. Liapis and C.E. Jones: "Reliving the Experience of Visiting a Gallery: Methods for Evaluating Informal Learning in Games for Cultural Heritage" in Proc. FDG 2020.
[20] R. Khoshkangini, G. Valetto, A. Marconi. "Generating personalized challenges to enhance the persuasive power of gamification" in Proc. of the Personalization in Persuasive Technology Workshop, 2017.
[21] K. Siu and M.O. Riedl. 2016. Reward Systems in Human Computation Games. in Proc. CHI PLAY 2016.
[22] Deterding, S., D. Dixon, R. Khaled, and L. Nacke (2011). From game design elements to gamefulness: Defining "gamification". In Proceedings of the 15th International Academic MindTrek Conference.
[23] L. Ceccaroni, J. Piera. "Analyzing the Role of Citizen Science in Modern Research". IGI Global, 2016.
[24] G. Barros, M. Green, A. Liapis, J. Togelius: "Data-driven Design: A Case for Maximalist Game Design," in Proceedings of the International Conference of Computational Creativity, 2018.
[25] Michael Cerny Green, Gabriella A. B. Barros, Antonios Liapis and Julian Togelius: **"DATA Agent,"** in Proceedings of the 13th Conference on the Foundations of Digital Games, 2018.

UoM - Iconoscope game[26]. This playful, online interface allows users to create novel drawings based on textual prompts, and also allows users to annotate others' drawings. The game will be transferred to the needs of material recognition allowing users to add data/annotations to existing datasets.	TRL4 → TRL6 (WP6)
RBNS - Quick-annotation tool. It allows users to select the entire pixel area of objects appearing in complex industrial images, with a single click. The tool will be gamified to develop a recycling data game that will engage the wider public in citizen science activities.	TRL5 → TRL7 (WP2, WP6)

1.2 Relation to Work Programme

The table below provides an overview of the HORIZON-CL4-2021-DIGITAL-EMERGING-01-09: AI, data and Robotics for the Green Deal (IA) scope and how it is addressed in the proposed work.

Table 1.2 Relation of **RECLAIM** to HORIZON-CL4-2021-DIGITAL-EMERGING-01-09: AI, data and Robotics for the Green Deal (IA).

Call text	*"... integrate and optimise AI, data and robotics solutions ... to demonstrate ... in actual or highly realistic operating environments, how they directly contribute to the Green Deal.[...]"*
RECLAIM generates new AI-driven computer vision models trained with real data, and a distributed multi-robot multi-gripper system for material recovery. The developed technology will be demonstrated in two real waste treatment use-cases to undertake material recovery and thus explicitly contribute to the Green Deal. **Objective 3, 4 – WP 3, 4, 5**	
Call text	*"... supported by industry or service relevant KPIs,.. and demonstrating scalability, and deployment potential.*
RECLAIM is supported by measurable service relevant KPIs (see Objectives). The development of multiple use cases with distinct characteristics demonstrates the scalability and proves the deployment potential of the developed technology. **Objective 3, 4 - WP 5**	
Call text	*"The environmental impacts of the proposed solutions should also be taken into account…"*
RECLAIM develops an eco-friendly prMRF that can be dynamically (re)configured based on environmental and economic criteria. The environmental impact of prMRF will be explicitly assessed and maximized. **Objective 1 - WP 2, 5**	
Call text	**"***... focus... on real-world scenarios, which can benefit in short to mid-term from the technology and solutions and demonstrate substantial impact on the Green Deal..***"**
RECLAIM introduces a new effective solution for material recovery in sectors with currently inefficient waste treatment thus accomplishing substantial positive impact on the Green Deal. It demonstrates the application of the developed technology in real world, difficult and highly demanding material recovery scenarios. **Objective 1, 5 - WP 2, 5**	
Call text	**"***while taking into account the maturity of the technologies to solve the problems at hand"*
The project exploits available, well tested technologies that are further improved and integrated into a new high-tech solution for decentralized, close to source material recovery. **Objective 3 - WP 3, 6**	
Call text	**"***Deep involvement of all relevant stakeholders.."*
The **RECLAIM** consortium involves a wide range of different stakeholders (Universities, RTOs, SMEs, NGOs, Local Authorities) who are particularly interested in the results of the project and are fully committed to its success. **Objective 6 - WP 1, 7**	
Call text	**"***Special attention will be given to including users of diverse age, gender and background."*
The implementation of reality-inspired environmental games and particularly the RDG provides an effective mechanism for involving users of diverse age, gender and background in the research activities of the project. **Objective 5 - WP 1, 6**	
Call text	**"***incorporate training programs for non-expert users of AI, data and robotics systems... including the basics concepts of Trustworthy and ethical AI."*
RECLAIM incorporates training sessions for the staff of the end-user participants to inform users who are not experts in AI, robotics and data management about the possibilities offered by new technologies, which are secured and framed by trustworthy and ethical AI. **Objective 1 - WP 1**	

[26] A. Liapis: **"Mixed-initiative Creative Drawing with webIconoscope,"** In Proceedings of the 6th International Conference on Computational Intelligence in Music, Sound, Art and Design (EvoMusArt), vol. 10198, LNCS. Springer, 2017.

1.3 METHODOLOGY

RECLAIM aspires to develop a compact, portable, automated and intelligent material recovery facility, the prMRF (Fig.1.2), which increases the scope of material recovery by covering areas with locally high waste treatment needs and enabling material recovery from waste streams that are not adequately treated by existing municipal systems. The project will deliver a technologically and scientifically sound solution and demonstrate its application in difficult real environments with high demands in terms of waste treatment and material recovery (see Box 2 summarizing project scenarios). Complementary to the above, **RECLAIM** develops a new gaming infrastructure that increases social sensitivity on recycling and promotes the active participation of citizens in the research activities of the project.

Overall, the proposed work is structured in three phases (Figure 1.4) that foresee implementation and release of three incrementally enhanced versions of the targeted technology. The first is the **Machinery Integration** phase [M1-M9] that will capitalize on available technology and the extensive expertise of the consortium in applied waste management solutions, to quickly implement the infrastructure and assemble the mechanical equipment that is necessary for the operation of the prMRF. This phase will additionally focus on the early integration of existing hardware and software robotic technology in order to promptly reveal/mitigate conflicts in the waste processing chain. Moreover, it will enable data collection to accelerate the training of the necessary AI modules and facilitate the design of environmental games targeting recycling and particular material recovery.

This rapidly created complete prMRF prototype will provide the basis for the second **Lean Research and Development**[27] phase[28] [M10-M30] that aims at implementing, testing in the field and continuously improving solutions for the AI-ILC and RoReWo-Team modules considering the local characteristics of each deployment site. A lean implementation that follows a test-fast/improve-fast approach for the immediate reveal and relief of problems in the field, is particularly appropriate for creating a prMRF that is functional and highly productive in real conditions. Moreover, the second phase will consider the use of real world data for RDG implementation to facilitate data collection by citizens.

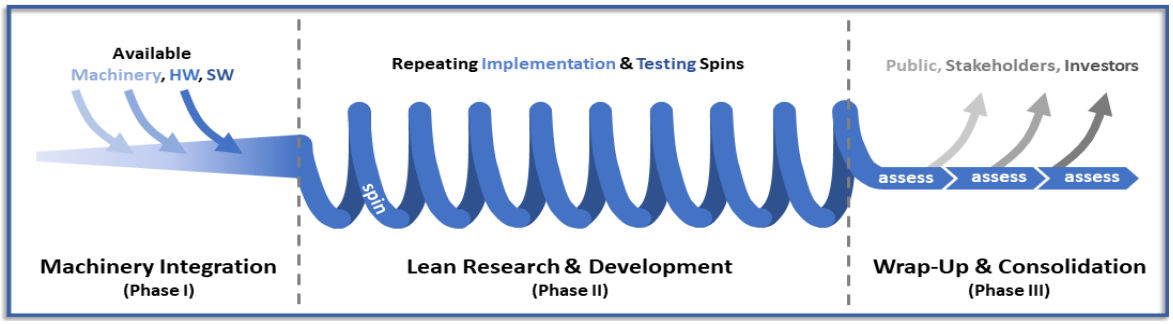

Figure 1.4. Schematic representation of the **RECLAIM** development plan.

The last, **Wrap-Up and Consolidation** phase [M31-M36] focuses on the continuous and uninterrupted operation of the deployed prMRF with small-scale optimizations on the combined operation of the mechanical and computing modules. This is the main result of the project that will be presented to the public complemented by the RDG environmental games to increase public sensitivity on recycling and support prMRF sustainability. The final prMRF is considered as the Minimum Viable Product (MVP) of the project that will be demonstrated to selected audiences such as the scientific community, stakeholders interested in waste management and potential investors for considering its commercial exploitation. Further to prMRF, the exploitation and potential commercialization of other project assets (RoReWos, AI-ILC, RDG) will be actively considered in the third phase of the project.

The following sections summarize the work to be done in the three development phases of **RECLAIM**, focusing separately on the four innovation pillars of the project.

1.3.1 Machinery Integration

The first phase focuses on the direct exploitation of the technology available to partners FORTH, KUL, IRIS, RBNS and the waste management/material recovery expertise available to HRRC, ION, AIMPLAS to develop an early version of the targeted prMRF, which will provide a solid basis for experimentation, thereby facilitating successful execution of the overall project work plan.

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[27] Kalinowski, M. et al. (2020) Lean R&D: An Agile Research and Development Approach for Digital Transformation. Int. Conf. Product-Focused Software Process Improvement

[28] Due to the significant expertise of the consortium on all background technologies, a “lean” development approach is followed in **RECLAIM** to emphasize focused, field-specific customizations.

PIL-1: Advanced AI for material identification, localization and categorization (AI-ILC)

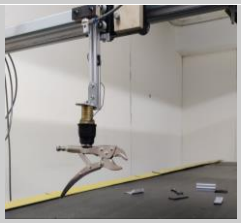
The first phase of the project will consider the purchase/installation of visual monitoring equipment and the use of available solutions contributed by project partners to enable the collection of recyclable material imaging data, using RGB and hyperspectral cameras. The data collected will be used to immediately start the training of Neural Networks for the AI-ILC modules, even in the first phase of the project. Data collection will be facilitated by the use of a quick annotation tool developed by RBNS, which will allow users with a single click on an image[29], to select the entire pixel area of the appearing object. The tool will be significantly improved in **RECLAIM** in terms of usability, accuracy and the ability to process hyperspectral data.

PIL-2: New, modular, low-cost, high performance, multi-robot/multi-gripper recyclable recovery

During the first months of the project, a targeted design and implementation of the RoReWo unit will be carried out. The RoReWo assumes a simplified architecture to achieve a significant reduction in hardware costs by trading off small reductions in robot efficiency and productivity. The RoReWo consists of a single axis closed loop motion control Cartesian robot that can be installed above any size conveyor belt transporting the waste, with its motion direction perpendicular to the movement of the belt. This allows the robot carriage to run along the entire width of the belt. A low-cost piston is mounted on the moving carriage of the robot, which moves vertically in relation to the conveyor belt (i.e. up/down), with the use of compressed air. The piston moves towards the belt in order to reach and grab a recyclable object and then away from the belt to pick and transfer the object to the material collection bins installed in the left and right side of the RoReWo. Attached to the piston is one of the different available end-effectors that grabs and holds the objects when transported to the collection bin. The synchronization of the moving robot carriage, the piston and the grabber is achieved using Newtonian mechanisms that combine the prediction of the location of objects on the belt to position the robot at the right time above the object to be sorted, and then timely activate the motion of the piston and the grabber. A particularly low cost RoReWo prototype has been already evaluated in relevant waste treatment conditions[30].

The RoReWo described above is of particularly low cost (approx. 4K €) and achieves ~45 picks per minute, when the commonly used Delta robots (approx. cost 25K €) achieve ~60 picks per minute. As a result, by using a 4-RoReWo Team working cooperatively for recyclable material recovery, we can triple the performance of the composite system and still have a lower implementation cost in relation to the Delta robot. Moreover, as shown in Figure 1.5, various grippers are already available by RBNS and KUL, which can be used as end-effectors of the different RoReWos in the team. The synthesis of a multi-robot/multi-gripper team is a significant advantage of the proposed modular **RECLAIM** approach. In contrast to the use of a single high-cost robot that carries only one end-effector, this approach expands the scope of the composite system and increases the utility and performance of the RoReWo Team for material recovery applications.

Figure 1.5. A variety of different grippers that can be used in each RoReWo

				
Long diameter suction cup for deformed packages	Short diameter suction cup for small plastic films	Fingered gripper for mid-volume and heavy objects	Pinned gripper for woods and soft heavy objects	Magnetic gripper for ferrous metals

PIL-3: Portable robotic Material Recovery Facility

For the rapid progress of the first phase there is already a cost plan for the Recycling Equipment (R-Eq) that will be installed in the **RECLAIM** deployments. During the first months of the project, the R-Eq features will be specified in detail, so that the machinery can be selected and purchased. At the same time, prMRF deployment sites will be examined in order to carry out preparatory actions necessary for prMRF operation.

Regarding prMRF deployment for material recovery from post-consumer waste, HRRC has already analyzed the population fluctuations of the Greek islands during a full year period and how these changes affect the amount of recyclables that should be recovered by the prMRF. Based on this analysis, it has already determined that the most suitable island for the installation of prMRF is Kefalonia with a second (alternative) option, the island of Lefkada.

[29] Video link: https://www.youtube.com/watch?v=HALgk1kb6Xk
[30] Video link: https://www.youtube.com/watch?v=ztffZwWAc44

Both of these islands face a problem in their waste management all year round, with the problem becoming very intense during the summer months. Thus, the double-shift operation of the prMRF during the tourist season and single-shift in the off-season months is expected to cover a significant part of the island's waste management needs. ION is the local authority responsible for the management of waste in both Kefalonia and Lefkada, being particularly interested to not only host the prMRF throughout the project offering the necessary infrastructure for its operation, but also consider the use of additional prMRFs to address the material recovery needs in other Ionian islands.

PIL-4: Educational environmental gaming for social awareness and data collection

The design of environmental games will proceed in parallel with the above described activities. The main focus will be on RDG and the collection of recycling data, specifying in detail operational scenarios (see Box 2). Additionally, we will consider the potential implementation of fun games to complement pure RDG, (e.g. a virtual recycling worker game, quiz questions on the reduction of CO2 emissions in relation to material recycling, familiarization with the recycling processes followed in European countries, etc.). UoM in collaboration with the consortium partners ISWA and AXIA that lead the communication exploitation and innovation activities, will examine new ideas and proposals to select games that better promote the results of the project to the public.

Box 3. Indicative RDG scenario. The users are shown real waste images captured by cameras installed in the prMRF and asked to complete one of many reviewing tasks to receive virtual prizes for their contribution in recycling research. Reviewing tasks may regard the annotation of materials appearing on images, the assessment of the outcomes of the currently best trained AI-ILC module (false positives/false negatives), the assessment of AI-ILC inferred masks, the correction of AI-ILC inferred masks to be re-used for training, etc. All inputs provided by individual anonymous users will be assessed by several other users to verify quality before they are included in the training dataset (experienced recycling staff will have the overall inspection of the gamified data-collection process). The improved dataset will be used for retraining the AI-ILC module which will afterwards be considered the new current AI-ILC to drive the game (i.e. produce new images for the users).

1.3.2 Lean Research & Development

The second phase includes the main part of developments and technological advancements considered in each innovation pillar, through repeated test-improvement cycles. During this phase, the second use-case scenario of the project will be considered (see Box 2). In particular, ION will initially bring citizen separated materials to the prMRF for testing and guided development/refinement of the software and hardware components. Subsequently, we will consider transferring the prMRF to Corfu to directly process the citizen separated waste. It is expected that making known the presence and the operation of the prMRF in Corfu will strengthen the interest of citizens to the circular economy and will strengthen their recycling activities.

PIL-1: Advanced AI for material identification, localization and categorization (AI-ILC)

To increase the efficiency of the AI-ILC module, two complementary means of waste imaging will be used. These regards (i) an ordinary RGB global-shutter camera to collect material information in the visual domain and (ii) a hyperspectral camera in the infrared domain (e.g. 1000-2000 nm) to collect spectral information of materials in wavelengths not visible to humans. The information captured from these two modalities will effectively complement each other to increase effectiveness and accuracy in material recovery applications. In particular, the visible light camera is important for the segmentation of materials in the case that two or more objects of the same material type are adjacent and thus not separable with hyperspectral imaging. On the other hand, the use of a hyperspectral camera offer significant possibilities in relation to material recognition and determination of objects consistency. In other words, using only information from the visual domain it is hard to identify the difference between objects with similar appearance but different material type (e.g. plastic films), while at the same time RGB computer vision may often face difficulties due to dirt or shape deformations. Accordingly, during the actual prMRF operation, both cameras will be simultaneously active, providing complementary information for the detection and categorization of materials.

In parallel to the above described joined visible and infrared imaging, **RECLAIM** will explore simpler cost-effective solutions solutions relying only on RGB imaging. This will exploit prior experience of FORTH and RBNS in urban environments (particularly in Crete and Cyprus) which has shown that satisfactory material recognition and categorization can be achieved by using RGB only imaging. This is expected to be appropriate for the second scenario considered in **RECLAIM** (see Box 2), implying that the manufacturing costs for the entire prMRF could be significantly lower in the case of negative separation applications.

In this phase of the project there will be extensive collection of both hyperspectral and RGB data, which will be used to train newly developed CNN solutions that combine RGB and/or hyperspectral information for the identification, categorization and localization of recyclables. Early annotation of collected data will be accomplished with the help of a new, quick annotation tool for recyclables, used by experienced recycling workers to collect, in a short time, a

large volume of data. Further annotation of collected data will be achieved with the participation of citizens using the RDG, as summarized below.

The collected data will be used to implement and evaluate different approaches to the optimal implementation of the AI-ILC. Research endeavors will consider (among others) the use of separate CNNs dedicated to the different material types vs multi-class CNNs categorizing multiple types of material, and will consider how AI-ILC performance is affected by other features of prMRF function (spatial distribution of materials, belt speed, frame processing rate, etc.). The repeating industrial experimentation and testing is described by the multiple "implementation & testing" spins shown in Figure 1.4.

To incrementally improve the operation of the AI-ILC, RGB and hyperspectral data recordings will be developed for items whose categorization has low confidence. The data of the poorly categorized objects, will be fed to the RDG for citizen-powered annotation and then included in the set of training data to repeatedly increase the performance of AI-ILC inferencing.

PIL-2: New, modular, low-cost, high performance, multi-robot/multi-gripper recyclable recovery

A key issue for prMRF operation is the efficiency of the robotic system used for the recovery and physical sorting of recyclables. Since the treatment of recyclables does not assume fine manipulations, the grippers to be considered in **RECLAIM** will all rely on pneumatic technology (small / big suction cups, pinned gripper, parallel gripper) and power supply (magnetic gripper, fingered gripper) so that they ease RoReWo re-configuration on the fly, according to the dynamic requirements of each deployment (e.g. seasonality). The industrial research that will be carried out in the second phase concerns the selection of the appropriate vacuum grip system (Venturi vs blower), the determination of suction cup according to the size and the deformations of materials, the comparison of the different grip options for each type of material, and the operational requirements of the system (number of materials/bins, relative proportions of materials in the waste stream, etc.). The motion control of RoReWos will consider the characteristics of the R-Eq installed in the prMRF (e.g. width/speed of the conveyor belt), as well as the characteristics of the different grippers (e.g. acceleration in relation to safe object transport).

An important part of second phase work will focus on RoReWos teamwork and the coordination of member actions to ensure that all RoReWos can move simultaneously without conflicts/restrictions, can efficiently share common resources (compressed air supply, vacuum supply). The selection of materials to be recovered will base on economic (secondary material price[31]) and environmental (carbon footprint[32]) criteria, which will be further combined with waste analytics to set-up the best team-configuration for the given mix of recyclables.

PIL-3: Portable robotic Material Recovery Facility

This part of the prMRF study will be carried out entirely in the field with the aim of customizing and optimizing its operation according to the individual needs of the use case scenarios considered in the project. The optimization will consider in combination the preparation of waste from the mechanical R-Eq accomplishing the placement of incomings in a single layer on the conveyor belt, the identification, categorization and localization of recyclables by the AI-ILC unit, their physical separation into different bins from the RoReWos team and finally the balling of the recovered materials before they are promoted in the secondary market. Besides mechanical adaptations, a major part of the PIL-3 work will focus on waste analytics to reveal the synthesis of incoming objects which will be further associated with the RoReWo Team configuration, the recovery actions performed for each material type and the number of items that have been actually recovered. The multifaceted consideration of the operation of prMRF units will provide significant know-how on the configuration of the composite system according to the specific requirements of each deployment site.

PIL-4: Realistic environmental gaming for social awareness and data collection

RDG development efforts will focus on producing games that are intuitive and simple to use, and that can be deployed on many platforms and devices, focusing on web-based solutions. This will leverage state-of-the-art work on data games combining them to create a recycling data-game pipeline for re-using and improving real-world data through human annotations. To implement RDG, established game engines will be exploited (e.g. Unity 3D), as they are powerful and deployable across a broad range of platforms. This part of work will additionally consider the formatting of the collected data to be directly exploitable by deep learning algorithms, and means to provide feedback to users acknowledging their contribution in **RECLAIM** scientific work. During this phase, the RDG will reach the community to support data collection and gain feedback with respect to its effectiveness, usability and user-friendliness.

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[31] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Recycling_-_secondary_material_price_indicator
[32] https://www.ipcc.ch/site/assets/uploads/2018/03/5_Waste-1.pdf

1.3.3 Wrap Up & Consolidation

This phase includes consolidating all the technical developments and data collected in the first 30 months of the project. The goal is to prepare exploitation plans for the market penetration of the individual project assets after considering stakeholders' feedback and the technical results and achievements of project developments. The assessment of project developments will proceed based on the measurable KPIs already specified in **RECLAIM** objectives. It is noted that during this phase, the third use case scenario (see Box 2) will be examined, which emphasizes the portability of the prMRF and its ability to reach areas with increased waste production in order to directly process and recover fresh, uncompressed, high quality materials.

PIL-1: Advanced AI for material identification, localization and categorization (AI-ILC)

In the last phase of the project, the final evaluation of the AI-ILC module will take place. Performance evaluation will consider the typical Mean Average Precision (MAP), Mean Average Recall (MAR) and F1-score that consider the relationship between false positive and false negative categorizations. Moreover, a comparison will be made between the two different approaches followed to implement the AI-ILC (rgb+hyperspectral vs rgb-only) in order to draw conclusions about the suitability of each approach in the different project scenarios and also reveal good practices for their use in material recovery and other applications beyond **RECLAIM**. In order to maximize the technological footprint of the project, the utilization of AI-ILC will be examined in applications that concern not only waste management but also the visual recognition and categorization of other products.

PIL-2: New, modular, low-cost, high performance, multi-robot/multi-gripper recyclable recovery

RoReWo evaluation will focus on the efficiency of the proposed solution in managing waste streams with multiple and varying characteristics. In addition to assessing the usability and robustness of RoReWo highly simplified architecture, it is important to evaluate the suitability and efficiency of the various grippers against the physical characteristics of each waste type (size, shape, deformation, etc.). RoReWo Teams that combine multiple robots and multiple grippers in a unified complex system, will be evaluated with emphasis on the balance between scope expansion (i.e. manage more waste types) and total number of picks offered by RoReWos, to maximize the value and ecological impact of recovered materials. The evaluation will focus not only on RoReWo productivity but also on the investment required to install a team in waste treatment environments compared to other available solutions, in order to draw up a business plan for its commercialization. RoReWos are expected to be by far the solution with the best pick per invested euro rate. New application areas will also be sought where RoReWos can act as new, highly cost-effective solutions managing items on conveyor belts (e.g. quality control in pharmaceutical/grocery industry).

PIL-3: Portable robotic Material Recovery Facility

The prMRF evaluation will focus on the chain operation of all installed subsystems to ensure their efficient and synchronized operation. Through small-scale refinements, the operation of each unit will be optimized so as to facilitate the operation of the next in line, and jointly compose a highly productive system that can recover multiple and different materials from the waste streams. Among others, the third phase of the project will reveal the portability of the prMRF to explore and demonstrate local, close to source robotic material recovery and sorting. In this context, prMRF will be deployed in the port of Kefalonia to directly process the waste of arriving ferries. ION will additionally consider other options for prMRF deployments in high visibility areas across the Ionian Islands. Overall, at least 15 days of live demos open to the public will be implemented in the last months of the project. Following this approach, **RECLAIM** will exploit the popularity of Ionian Islands as attractive touristic destinations, to demonstrate to the citizens and the visitors of the islands how EU research funding contributes to the solutions of daily problems and material recovery in particular, for the benefit of the environment, the society and the economy.

RECLAIM will undertake concrete activities that will pave the way for the commercial exploitation of the prMRF, which provides a timely solution to the strong global need for distributed material recovery. To this end, the needs and times required for the transfer and the installation of the prMRF in the application field will be studied and reported. Particular emphasis will be placed on the ease of use of the system by users who are not experts in artificial intelligence and robotics. Good practices will be reported regarding the configuration of the overall system and the combined operation of different grippers in order to make the prMRF directly useful in a broad range of material recovery applications. It is noted that the **RECLAIM** end-user participants (ION, HRRC) have already expressed their strong interest in distributed material recovery systems (in Greece there are dozens of islands that are in need of a prMRF) and would be among the first potential buyers of new prMRFs when they are placed on the market.

PIL-4: Educational environmental gaming for social awareness and data collection

The evaluation of RDG will consider the user friendliness/usability of the gaming environment, the engagement of users (repeating RDG use) and the accomplishment of its main purpose that is the collection of extensive recycling data via human contributions. To this end, questionnaires, interview and play-aloud sessions will be used to assess user satisfaction. Moreover, an action recording infrastructure will enable analytics on user behavior, which will

subsequently guide RDG refinements with respect to performance and usability so that users can easily provide recycling annotation data. At this final stage, it is important to ensure that the newly provided data are smoothly verified (e.g. through user voting) and pipelined into the confirmed datasets used for deep learning. This will facilitate the long-term sustainability of the deployed prMRF, but will also provide significant background contributions for future works adopting the citizen-science approach for the collection of scientific data.

1.4 OTHER KEY ASPECTS OF THE PROPOSED WORK

In addition to the prMRF development discussed above, a number of complementary activities are envisaged which address the requirements of the HORIZON-CL4-2021-DIGITAL-EMERGING-01-09 call, as summarized below.

Training Programs. **RECLAIM** will organize training programs focused on state-of-the-art waste management technologies (WP1, Task1.5). Besides being open to the public, these programs will be mainly designed for partners ION, HRRC, AIMPLAS, AXIA, ISWA that are staffed with waste management experts who need to know more about the differences of RGB and hyperspectral imaging, the power and the limitations of AI and Deep Learning, the available mechanical/ robotic systems involved in the automated treatment of waste, etc.

Open Science Practices. The **RECLAIM** partners fully support the Open Science practices in the sharing of knowledge, data and tools as early as possible in the project's lifetime. To this end, **RECLAIM** includes the implementation of the following actions:

- Use of open research infrastructures for knowledge and data sharing, such as the European Open Science Cloud (EOSC) that provides seamless access, FAIR (Findability, Accessibility, Interoperability and Reusability) management, and reliable reuse of research data and all other digital objects produced along the research life cycle (e.g. methods, software and publications).
- Early and open sharing of research data and results, through preprints, or informal publications in the project's website and public repositories.
- Direct and open sharing of research data collected by citizens with the help of recycling data-games through public repositories.
- Open access to research results and reports, either by publishing them with full open access in established, highly popular peer-reviewed scientific publications, or by rapidly sharing them at no cost, through focused publications in the Open Research Europe Publishing Platform (OREPP).
- Data Management Planning following the "as open as possible as closed as necessary" principle, which will ensure data and results are deposit in trusted repositories, they are physically or digitally accessible, they are licensed for public free use (e.g. under "CC BY"), with potential exceptions clearly specified.

Further to these general directions, **RECLAIM** already includes concrete actions that promote Open Science. In particular, to increase the quality, efficiency and impact of research and innovation activities, and accelerate responsiveness to societal challenges related to recycling, the RGB and hyperspectral waste imaging data will be made publicly available to create multiplicative effects in EU research focused on digital waste processing and particularly the automated identification and categorization of recyclables. The data will be supplemented with benchmarks so that developers can compare the performance of their newly developed algorithms. The relevant repositories will be elaborated in the Data Management Plan deliverable, D1.1. Moreover, to boost societal engagement in research and innovation and increase the trust of society in science, the citizen-annotated data via recycling data games will be made directly and openly available to their creators and the wider public for exploitation and re-use. The above actions are in full agreement with the Horizon Europe Key Impact Pathways that address open science practices, namely the 'Fostering diffusion of knowledge and open science' pathway focused on scientific impact, and the 'Strengthening the uptake of R&I in society' pathway focused on societal impact.

Management of data and other research outputs. The preparation of a Data Management Plan (DMP) and will ensure that research datasets will be managed in line with the FAIR principles (Findable, Accessible, Interoperable, Reusable). The DMP will follow the "as open as possible as closed as necessary" principle, which will ensure data and results are deposited in trusted repositories, are digitally (if necessary physically) accessible, and are licensed for public free use (e.g. under "CC BY") with potential exceptions clearly specified.

Types of data/research output: the project will collect, manage, and generate the following types of datasets: (i) waste imaging data with the use of RGB / hyperspectral cameras and annotations on these data gathered via recycling data games, (ii) the recycling data games and statistics on the use of these games, (iii) AI/Deep Learning models accomplishing the identification, localization and categorization of targeted recyclable materials appearing on these images, (iv) outputs regarding the performance of these models to be used as benchmarks for future research, (v) operational scenarios and performance statistics of the RoReWo Team and the complete equipment of the material recovery chain installed in the prMRF, (vi) videos of training programs on the use of digital and smart technologies for material recovery and citizen engagement.

Findability: the above datasets will be published as open data in relevant open repositories where a dedicated project space will be established, e.g. European Open Science Cloud (EOSC) and OpenAIRE's Zenodo. The datasets will be published following the FAIR Digital Object Framework. The project will use globally resolvable persistent identifiers that enable metadata indexing, such as Handles/DOIs for the FAIR indexing of data objects and the specification of data access and management policies associated with them.

Accessibility: The datasets will be made available on well-recognized repositories (e.g. EOSC, OpenAIRE's Zenodo) that ensure reliable access to research data relevant digital objects. Further to the above, recycling data-games will be freely accessible and usable through focused gaming networks and repositories (e.g. OS Game Clones, GitHub). Other physical project developments will be easily and freely accessible by interested stakeholders and the wider public.

Interoperability: To ensure the interoperability of published datasets, the project will design structure and publish the data using digital data models based on standards and ontologies (e.g. Industrial Ontologies Foundry). The project will also consider the introduction of new data types, in communication with Data Type Registries (e.g. Cordra, ePIC).

Reusability: The datasets will be made available for public access and reuse under Creative Commons Attribution 4.0 International Public License (CC BY 4.0) or a license with similar access permissions.

Curation and storage/preservation costs: The project aims to utilize zero-cost data repositories (EOSC, OpenAIRE's Zenodo. Moreover, the project coordinator has access to infrastructures that allow the hosting of data and the immediate, fast access to it. In any case, the project coordinator will bear all ongoing costs to dataset curation operations.

Dissemination/Communication. All partners will actively disseminate the expected/accomplished project results to the relevant stakeholders and target audience (see section 2.2.1). Additionally, with the support of all partners, ISWA will exploit the already established global communication channels to highlight, from project day-one, the new opportunities provided by prMRF for highly effective, distributed material recovery.

Relation to the `do no significant harm' principle (Article 17 of Regulation (EU) No 2020/852). The proposed work provides a clear and direct support to the fourth and fifth objectives of the EU Taxonomy Regulation, in particularly the "Transition to a circular economy" and the "Pollution prevention and control". Moreover, it provides an indirect support to the other four objectives of the same regulation without being against or harming any of the objectives by any means. Accordingly, **RECLAIM** fosters environmentally sustainable activities that contribute to the EU's environmental objectives, promotes sustainable investments and ultimately supports the implementation of the European Green Deal.

Gender Issues. Project planning and activities are compliant with Council Resolution 1999/C 201/01 which states that "the gender mainstreaming of research policy is not limited to promotion of women as research workers but should also ensure that research meets the needs of all citizens and when necessary contributes to the understanding of gender-relevant issues". The main research stream of **RECLAIM** focuses on material recovery with the minimum possible human intervention and therefore, it does not address research or technology topics related to human gender. In particular, upon successful completion of the project, all people regardless of gender, will be equally treated by the developed prMRF. On the other side, the recycling data-game work plan assumes active participation of users for testing and refinement. Besides the fact that the envisioned games do not assume any kind of functional differentiation between the sexes, **RECLAIM** will carefully and equally promote the game into men and women, considering the recruitment of a gender balanced mix of users, to provide unbiased guidance for planned developments.

Moreover, the consortium will balance the gender distribution of the research and technical staff involved in the project by following non-discriminating hiring practices and trying to address the EC goal of 40% participation of women at all levels, in implementing and managing EU-funded programmes.

Multidisciplinarity. **RECLAIM** ties together complementary technologies to promote and support the goals of the EU Green Deal. To accomplish its objectives, the proposal adopts a multidisciplinary approach combining expertise from different fields such as Data science, AI, robotics, mechanical engineering, optics, scientific gaming, environmental science, industrial management, innovation management, governmental authorities, citizen science to develop novel still feasible and directly applicable solutions on material recovery and to increase social awareness on recycling and the circular economy. The **RECLAIM** interdisciplinary approach is summarized in Table 1.3.

Table 1.3 Overview of the multidisciplinary approach adopted in **RECLAIM**

Discipline	Elements Involved in RECLAIM	Objective	Relevant Partner
Data Science	Data Collection & Annotation, Data Analytics	Obj.3, 5	IRIS, FORTH, RBNS, KUL
Artificial Intelligence	Computer Vision, Convolutional Neural Networks, ML/DL technology	Obj.3,4	FORTH, IRIS, KUL
Robotics	Real Time Control, Task level Programming, RoReWo Units, RoReWo Teams	Obj.2, 4	RBNS, FORTH, KUL
Mechanical Engineering	Pneumatic & Electric Grippers, Vacuum Technologies, Waste Pre/Post Processing	Obj.2, 4	KUL, AIMPLAS, HER
Optics	Hyperspectral Imaging, Computer Vision	Obj.3, 4	IRIS, FORTH
Scientific Gaming	Human Computation Games, Data Games, Citizen Science	Obj.5	UoM
Environmental science	Waste management, Waste economics, Material Recovery Automation, Logistics	Obj.1, 4	AIMPLAS, HER, RBNS, ISWA
Innovation Management	Research Result Commercialization, Exploitation Plan, Circular Economy Business Model	Obj.6	AXIA
Industrial Management	Waste Processing Chain, Operational Predictive Analytics	Obj.1, 4	ION, HER, AIMPLAS,
Governmental Authorities	Solid Municipal Waste Treatment, System Sustainability, Citizen Engagement	Obj.1, 5, 6	ION, ISWA

Readily available contributions from previous projects. The participants in the **RECLAIM** consortium bring into the project significant expertise from their past and ongoing projects. Several already available and successful outcomes from relevant national/international research and innovation activities will be exploited and further advanced to realize the objectives of **RECLAIM**. The following Table contains a list of indicative relevant projects, while further information is provided in the individual partner profile.

Table 1.4 European projects relevant to **RECLAIM**

InterRecycle (Interreg)	http://interrecycle.eu	FORTH
InterRecycle develops RoReWo Teams for material recovery deployed in two industrial MRFs, in Cyprus and Greece. **RECLAIM** will exploit the know-how on the RoReWo technology, the RGB image based computer vision for material identification, localization and categorization, the use of multiple grippers.		
FELICE (H2020)	https://www.felice-project.eu	FORTH
FELICE aims at increasing the agility of manufacturing production lines with the use of robots. **RECLAIM** will capitalize on operational analytics methods to link the characteristics of incoming waste streams with the optimization of the RoReWo Team configuration.		
ANASA (GSRT)	https://anasasorter.com/	FORTH, RBNS
ANASA has implemented delta-robot technology for material recovery driven by computer vision and hyperspectral imaging, which is applied with paired robots into Chania and Heraklion MRFs. **RECLAIM** will exploit the expertise on robot control, computer vision and hyperspectral imaging, vacuum gripping.		
AUSOM (H2020)	https://www.ausomproject.eu/	KU Leuven
AUSOM develops a robust cost effective robotic sorter based on deep learning, computer vision and laser induced breakdown spectroscopy (LIBS) for sorting shredded scrap metal. **RECLAIM** will exploit the AI categorization and robotic gripper technologies to accomplish advanced material recovery.		

ACROGRIP	https://acrogrip.be/	KU Leuven

The ACROGRIP has examined developed and tested robotic grippers for different application domains. The latter resulted in a wide variety (50+) of both commercial and in house developed robotic grippers which are available to be tested/used in the **RECLAIM** project.

ENVISAGE (H2020)	http://www.envisage-h2020.eu/	UoM

ENVISAGE implemented and tested a data-driven approach to design, adapt, revise and evolve virtual labs. Technologies and best practices for gamifying and monitoring virtual labs will be exploited to effectively design gamified applications for social awareness and human computation in **RECLAIM**.

AI4MEDIA (H2020)	https://www.ai4media.eu/	UoM

AI4Media creates a centre of excellence and a wide network of researchers across Europe, with a focus on delivering core AI advances to serve the key sector of Media. **RECLAIM** will exploit the open-sourced AI algorithms developed under AI4Media for computer vision, data monitoring and co-creation.

ROSSINI (H2020)	https://www.rossini-project.com/	IRIS

IRIS has developed and implemented cognition layers for the identification and tracking of different objects in manufacturing environments, while human and robots are collaborating in the same area. In **RECLAIM**, the object recognition experience gained from ROSSINI will be used for material identification in the wild.

MULTICYCLE (H2020)	http://multicycle-project.eu/	IRIS, AIMPLAS, AXIA, ISWA

MULTICYCLE develops and advanced and sustainable recycling processes and value chains for plastic-based multimaterials. **RECLAIM** will capitalize the lessons learned on hyperspectral monitoring of multimaterial packaging and complex composites from the automotive sector (IRIS), the pre- and post-processing technologies in recycled materials treatment (AIMPLAS), business cases development (AXIA) and proper dissemination (ISWA) to be applied in the context of this proposal.

DECOAT (H2020)	http://decoat.eu/project/	IRIS, AIMPLAS

A continuous recycling pilot plant demonstrates the novel DECOAT principle that allows upgrading existing mechanical recycling by adding tools for sorting by and activation of the trigger. **RECLAIM** will learn of advanced monitoring system deployment for textiles and painted plastic parts recycling (IRIS and AIMPLAS).

SOLREC2 (H2020)	https://cordis.europa.eu/project/id/101003532	AIMPLAS

Development and implementation of ground-breaking strategies for improving the sorting, separation and recycling of pharma blister packs and laminate consumer packaging waste consisting of multiple layers of polymers and aluminum. **RECLAIM** will learn of advanced sorting system deployment for identification and separation of plastic packages containing aluminum.

2 IMPACT

It is never too late to make a difference

2.1 RECLAIM's Pathway Towards Impact

ECLAIM is fully aligned with addressing the challenges associated with embracing transformational ideas to develop world-class smart technologies and increase the use of Robotics in Europe. The proposed work addresses complementary aspects of the HORIZON-CL4-2021-DIGITAL-EMERGING-01-09 call, and implements specific steps that greatly facilitate the wider adoption of robots on issues related to the EU Green Deal and the circular economy. In particular, it implements high-tech, eco-friendly, portable robotic Material Recovery Facilities that will be utilized for the decentralized recovery of recyclable materials. Accordingly, **RECLAIM** will strongly contribute to all three expected Innovation Actions impacts mentioned in the HORIZON-CL4-2021-DIGITAL-EMERGING-01-09 call topic, as described below.

Innovative AI, data and robotics solutions for resource optimisation and minimisation of waste in any type of sector (…), reduction of energy consumption and greenhouse gas emissions including exploitation of all data and information sources contributing to optimising applications for a greener planet. This includes among others contribution to enterprises' sustainability programs in the context of their CSR (Corporate Social Responsibility) strategies to reduce their ecological footprint, cutting costs and contributing to social welfare at the same time.

RECLAIM integrates state of the art AI, data and robotic solutions for the development of the first prMRF which will accomplish material recovery in distant, geographically isolated areas **not sufficiently covered by current waste management models.** In other words, it focuses on materials that, without the prMRF, are significantly less likely to be recovered. In **full support to the Green Deal**, enhanced material recovery will lead to a **significant reduction in GHG emissions**, since recycling of the sorted waste is enabled and thus emissions associated with the use of virgin materials are avoided. While the existing MRFs are usually energy-consuming, the proposed prMRF will be equipped with solar panels that will cover a substantial proportion of its power needs with renewable energy and hence **reduce the environmental footprint** of the prMRF. At the same time, the on-site processing of waste will lead to a **reduction in transportation emissions and costs**. Moreover, **RECLAIM** provides the opportunity to private organizations and SMEs to get involved in corporate **social responsibility activities** related to waste management research. **RECLAIM** participants have recognized the opportunity offered by the project to engage in research activities that support the circular economy model. Through their participation in **RECLAIM**, they **undertake significant resource costs** to implement the project objectives **for the benefit of the environment and society**. Future exploitation of the proposed technological solution is expected to contribute to **corporate sustainability beyond RECLAIM**. Companies could buy or rent and deploy the portable facility on-site, in their own scale and in full connection with their circular transition objectives and their contribution to the social welfare.

Optimised AI, data and robotics (including modular and adaptive solutions) to maximise contribution to the Green Deal in various applications such as environmental and waste management, including for instance waste clean-up (e.g. plastic collection, sorting), or in the circular economy value chain.

The development of the prMRF assumes the optimization and integration of state-of-the-art **AI-powered computer** vision with low-cost, maximum productivity robot-teams, for the recovery and sorting of recyclable materials. The project adopts a **highly modular** approach based on the use of multiple Robotic Recycling Worker (RoReWo) units to compose a team that undertakes material recovery. The number of RoReWos in the team may vary depending on the volume of waste and the number of material fractions that must be recovered. RoReWos will be equipped with different grippers for different types of materials to improve recovery speed and reliability. **RECLAIM** will develop technology that allows the change of the gripers attached on each RoReWo unit, thus providing a **flexible and adaptive solution** for material recovery. Adaptivity is planned both at a unit-level and at a team-level and will be driven by real-time **waste data analytics**. Overall, the proposed approach aspires to be the solution with the **highest "pick per invested euro" rate**, to date, thus providing a new boost to the Green Deal and its objectives. prMRFs can be directly used in the real world for the **recovery and sorting of multiple materials (not only plastic)**, in order to effectively implement the first nodes of the circular economy value chain. RoReWo-Team performance will be tested in two different demonstrators, handling different types of waste.

Advanced physical intelligence and physical performance of robotics solutions in diverse harsh environments serving the Green Deal.

The prMRFs developed in **RECLAIM** will take advantage of newly developed grippers that incorporate **physically intelligent gripping** to pick materials from the waste streams. Both demonstrations of the prMRF will show the **physical performance** of the RoReWo team, which will work long hours every day **in harsh conditions**, to recover and sort multiple recyclables. Accordingly, **RECLAIM** supports the key elements of the EU's environmental policy

which aim at managing waste in an environmentally sound manner and making use of the secondary materials they contain. The proposed work brings forward a new decentralized model of material recovery which practically contributes to the EU Green Deal goal[33] "no person and no place left behind from environment friendly activities" and at the same time, promotes a new economic model around recycling and creates new opportunities for innovation, investment and jobs, in full agreement with the EU Green Deal[34]. Additionally, the Recycling Data-Game that will be developed in **RECLAIM** will promote public awareness on recycling and will actively engage citizens of different ages, sexes and educational backgrounds in Green Deal activities.

2.1.1 Contribution to the wider Horizon Europe impacts

RECLAIM contributes to the four Key Strategic Orientations[35] for research and innovation in Horizon Europe, as described below.

Promoting an open strategic autonomy by leading the development of key digital, enabling and emerging technologies, sectors and value chains to accelerate and steer the digital and green transitions through human-centred technologies and innovations

Recent environmental challenges have highlighted the importance of digitalisation across several sectors in life. The current proposal introduces innovative technologies and solutions on waste management and material recovery, that make Europe more resilient and independent is strategic sectors of modern life. **RECLAIM** contributes to the open strategy of Europe and its global leadership in digital and emerging enabling technologies by promoting the widespread adoption of robotics in facing crucial environmental and social problems. It contributes new ideas and technological solutions that not only work tirelessly to address people needs but additionally accelerate digital and green transitions accomplishing the decoupling of resource use and emissions such as greenhouse gases (GHG) and pollutants from socio-economic development and growth. This goes hand in hand with the development and uptake of new technologies promoting and reinforcing the participation of citizens in the green and digital transformation of industrial ecosystems, providing services for the benefit of the environment, the economy and the society.

Restoring Europe's ecosystems and biodiversity, and managing sustainably natural resources to ensure food security and a clean and healthy environment

As human activities create pressure on natural resources that go far beyond sustainable levels, they affect ecosystems which in turn affect in a negative way all forms of life on earth. The European Union has the ambition to protect and preserve ecosystems, and manage natural resources, thereby ensuring the security of life. **RECLAIM** will advance knowledge, build capacities and provide innovative technologies and solutions for waste management, which support the state and functioning of ecosystems, to ensure a clean and healthy environment for all, and a sustainable management of natural resources that contributes to climate neutrality and adaptation. The transition to distributed, sustainable, low ecological footprint, solutions for material recovery will accelerate by enhancing governance, improving the volume of materials that is captured versus lost and developing new circular economy chains starting from distant or isolated areas in need of local, small scale material recovery plans. This will bolster the role of the European Union in the ecological transition as a solution provider for the benefit of all.

Making Europe the first digitally enabled circular, climate-neutral and sustainable economy through the transformation of its mobility, energy, construction and production systems

The European Union has the ambition to substantially reduce greenhouse gas emissions by 2030 and to become climate neutral by 2050, turning into a more sustainable climate-neutral, circular and competitive economy. This requires unprecedented changes in our lifestyle and the substitution of the traditional linear economy model in which products are consumed or used once and then discarded, with the circular economy model of production and consumption in which materials are reused and return back into the production cycle as much as possible. **RECLAIM** contributes in making EU economy climate-neutral, zero polluting, energy-efficient and globally competitive, through implementing breakthrough solutions and new business models on secondary material management. The distributed waste management concept promoted in **RECLAIM** will significantly contribute to improving the energy efficiency and decreasing the carbon footprint of waste management activities. Fostering a digitalized circular economy, contributes to sustainably supplying climate neutral materials, thus positioning Europe as an international technological and industrial leader in green transition.

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[33] https://ec.europa.eu/info/strategy/priorities-2019-2024/european-green-deal_en
[34] https://ec.europa.eu/info/strategy/priorities-2019-2024/european-green-deal/delivering-european-green-deal_en
[35] https://eeas.europa.eu/sites/default/files/horizon_europe_strategic_plan_2021-2024.pdf

Creating a more resilient, inclusive and democratic European society, prepared and responsive to threats and disasters, addressing inequalities and providing high-quality health care, and empowering all citizens to act in the green and digital transitions.

Social cohesion and inclusiveness together with the health, well-being, rights and security of its citizens are central aims of the EU Green Deal policies and programmes. As the EU is facing challenges related to, among others, the protection of nature, while at the same time the COVID-19 pandemic has demonstrated that existing systems are very fragile and can be easily affected by unforeseen factors putting under strain the well-being of citizens and communities, it is necessary to develop solutions that make the modern way of life more robust to external disturbances. The current proposal introduces new technologies on waste management and material recovery that contribute in keeping waste treatment and material recovery services running, being significantly less affected by unexpected challenges such as the ongoing pandemic or personnel shortages. Additionally, **RECLAIM,** through the RDG games, provides innovative approaches to connect citizens with emerging social and environmental needs, thus improving societal adaptation and integration in the green and digital transition.

2.1.2 Exploitable Assets

All innovation pillars considered in the project will deliver significant advancements in technologies for material recovery. These are identified as exploitable project assets that **RECLAIM** will produce as results upon completion.

The following table (Table 2. 1), summarizes these assets highlighting the key involved partners, their expected maturity level by the end of the project and the foreseen time needed for full maturity.

Table 2.1 The RECLAIM exploitable results

Asset	Type	Partners	Maturity Reached	Time to maturity
#1: prMRF The integrated portable robotic Material Recovery Facility that effectively recovers valuable materials from waste streams.	HW & SW	FORTH, IRIS, KUL, RBNS	80%	2 years
#2: Material recognition module for RGB-monitoring A component for the identification, localization and categorization of materials based on RGB images.	SW	FORTH, RBNS, IRIS	90%	1 year
#3: Material recognition module for Hyperspectral-monitoring A component for the identification, localization and categorization of materials based on hyperspectral imaging.	SW	IRIS, FORTH	90%	1 year
#4: Multi-material Gripper Newly developed grippers that can robustly grip/release materials of multiple different types	HW & SW	KUL	90%	1 year
#5: RoReWo Units & Team Low cost robotic units specifically designed and optimized for material recovery tasks	HW & SW	RBNS, KUL, FORTH	90%	1 year
#6: Recycling Data-Game A mixed, data collection & gaming module, to actively involve citizens in helping develop state of the art recycling technology	SW	UoM	90%	1 year
#7: Quick Annotation Tool A tool that simplifies and speeds up the annotation of RGB and hyperspectral images.	SW	RBNS	90%	1 year
#8: RGB waste imaging dataset A dataset of RGB waste images, annotated with respect to the presence of materials of multiple and different types.	DATA	RBNS, FORTH	100%	0 year
#9: Hyperspectral waste imaging dataset A dataset of hyperspectral waste images, annotated with respect to the presence of materials of multiple and different types.	DATA	IRIS, FORTH	100%	0 year

2.1.3 Business Case – Target Markets

A recent study[36] on the use of AI in the private sector of Western Europe has revealed that AI has great potential to drive transformational changes to businesses and generate significant benefits by optimizing companies' internal operations. The same is also true for the Robotics sector as summarized to the World Robotics Report[37] 2020, estimating 11% annual increase in industrial robot sales prior to the Covid-19 pandemic, with increased long-term perspectives. In view of these figures, **RECLAIM** addresses well organized target markets with high expectations for the coming years.

Robotic waste sorting. RoReWos have the potential to be the new trend in the area of robotic material recovery. They are low-volume, low-cost, modular, high performance robots with interchangeable grippers able to recover a wide variety of materials from waste streams and deposit them to material-specific bins. The use of RoReWos is expected to significantly increase cost-efficiency and productivity in material recovery facilities.

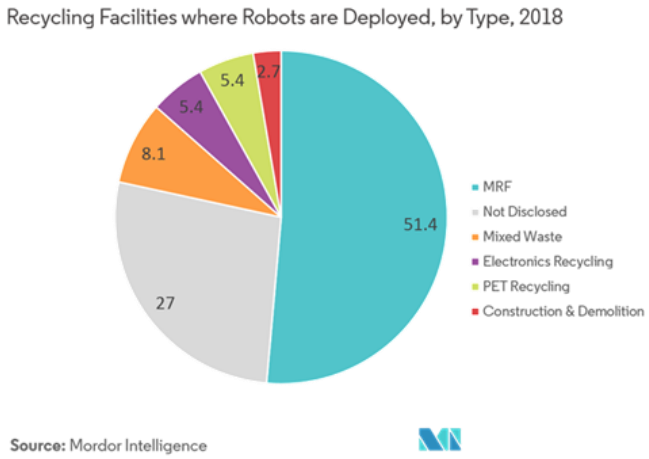

Figure 2.1. Recycling falilities where robots are deployed.

Scale: In the period 2021 to 2026, the global robotic waste sorting system market is expected to grow at a Compound Annual Growth Rate (CAGR)[38] of 19.5%. Currently, North America is the key player in this market, with more than 80 robots to be employed in sorting residential and commercial recyclable waste in the U.S. and Canada. Sorting robots can be useful in multiple industries; however, the recycling industry constitutes the largest segment[39] (52.1% in 2016).

Significance: Optimization of state-of-the-art robotic solutions will lead to an increased pick per invested euro index, and, particularly, more than 30% compared to current RoReWos. Additionally, the new robots will occupy 20% lower volume in comparison to the currently available solution and their picking speed can be twice as fast as humanly possible. The modular, flexible and efficient solution proposed is expected to have high potential market penetration and revenue generation.

Modular Robotic Systems. RoReWo Teams adopt an inherently modular and scalable architecture that is particularly appropriate for robotic tasks assuming the selection of objects transferred with belts. Due to the ability of RoReWo Teams to easily change their configuration according to the manufacturer's need or separation line requirements, their popularity is expected to increase in coming years.

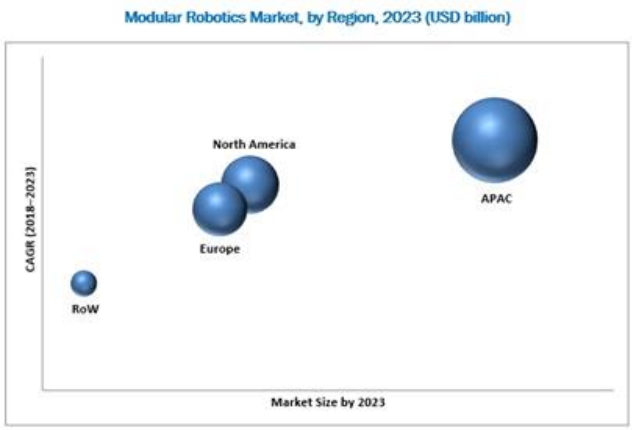

Figure 2.2. Modular robotics market by region (2023).

Scale: According to a market research report[40], the global modular robotics market is expected to reach USD 10.76 billion by 2023 from USD 4.7 billion in 2018. The regional assessment of the global Modular Robotics market includes a broad evaluation of top markets such as North America, Europe, Rest of the World (RoW) and Asia-Pacific (APAC) and it is shown that Modular Robotics market is expected to expand at a CAGR of 14%.

Significance: **RECLAIM** will implement technologies that facilitate the automated configuration of the team members and the composite RoReWo team after analyzing the characteristics of the input stream. Additionally, the combinatorial planning and control of the team will be explored to achieve effective use of shared resources and optimal team performance.

[36] https://pulse.microsoft.com/en/transform-en/na/fa1-articial-intelligence-report-at-a-glance/
[37] https://ifr.org/img/worldrobotics/Executive_Summary_WR_2020_Industrial_Robots_1.pdf
[38] https://www.mordorintelligence.com/industry-reports/robotic-waste-sorting-system-market
[39] https://www.goldsteinresearch.com/report/global-waste-sorting-robots-market
[40] https://www.fortunebusinessinsights.com/industry-reports/modular-robotics-market-100760

Improved AI system for material identification, localization and categorization. A combination of hyperspectral imaging, computer vision and CNN technologies will deliver the advanced AI-ILC system that will guide the RoReWos in the sorting process.

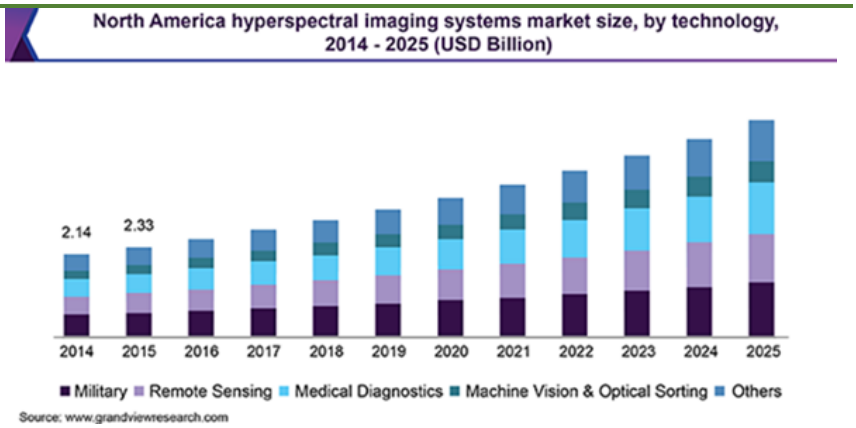

Figure 2.3. N. America hyperspectral impaging systems market size.

Scale: The global hyperspectral imaging systems market is growing from 10.66B€ in 2020 at a CAGR[41] of 19.3%. One of the many applications of hyperspectral imaging is machine vision and optical sorting. In addition, the computer vision market holds around 9.37B€ in 2019, growing with a CAGR of 7.8%

Significance: **RECLAIM** will improve existing technologies, aiming to achieve a waste recognition system that can separately recognize more than 90% of the treatable recyclables from the mixed-waste stream and store hyper-spectral/RGB images in a rwDB. More than 90% of the low confidence and misclassified waste images will also be stored in the database in order to enable the periodic retraining of the AI-ILC and lead to successful treatment of ~50% of the previously miscategorized recyclables.

Development and demonstration of portable robotic MRF. Currently, digital and robotic solutions are being examined by large MRFs, where a limited number of high-cost robots will be incorporated. Most MRFs, however, still rely on manual sorting. **RECLAIM** will deliver the first high-tech portable robotic MRF that can be used by waste management companies, demolition companies, local authorities and contractors.

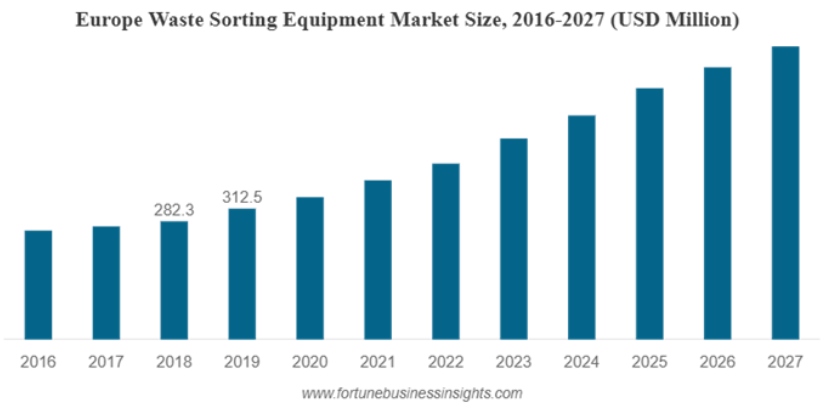

Figure 2.4. Eyrope waste sorting equipment market size.

Scale: The global waste sorting equipment waste market is estimated to amount to around 631.42M€ in 2019, with Europe's share being around 269.09M€. For the period 2019 to 2027, the global market will be growing with a CAGR of 12.4% due to the enforcement of environmental policies and automation advancements[42].

Significance: **RECLAIM**'s prMRF can be installed anywhere and will be fully operational in less than two days. Modifications in the digital systems and adjustments in the robotic teams in order to handle different waste materials can be performed within 30 minutes. The prMRF can be working continuously for over than 8hours daily with high performance rates, initially exceeding 80% of waste recovery. Recovery rates will be improved through the retraining of the AI system, resulting in high accuracy sorting equipment, necessary for the prevention of waste contamination. As a result, **RECLAIM**'s prMRFs are expected to recover a healthy return on investment due to the economical price point set for entry-level users and become the first 'static type plant' that can be rented and moved. Therefore, high revenues are expected from the commercialization of the developed technology.

[41] https://www.marketsandmarkets.com/Market-Reports/hyperspectral-imaging-market-246979343.html

[42] https://www.fortunebusinessinsights.com/waste-sorting-equipment-market-102861

Strategic autonomy and competitiveness of European industries. RECLAIM's prMRF allow the decentralized processing and recovery of waste, with high recovery rates and accuracy that will result in waste contamination prevention and allow the recycling of European waste in multiple locations within EU borders as well as the utilization of modular teams of robots that can process a variety of different waste types.

Scale: The European Union and the U.S. used to sell 95% and 70% of the plastics collected for recycling to China, respectively[43]. Aiming to reduce its GHG emissions, which remain higher than any other country, China introduced in 2018, stricter regulation measures regarding the allowable contamination levels for a number of wastes, resulting in a reduction of 99% of recovered plastic shipments heading to Chinese facilities[44]. As a result, the EU, North America, and other regions have to explore alternative ways of locally recovering their waste.

Significance: Europe has to take into consideration the variety and multidisciplinary of its members, adopting a more flexible, smaller scale, portable solution, such as the prMRF, that will address waste processing and recovery in an eco-friendly and smart way. Decentralized processing means high recovery rates of the fresh waste that has not been compressed, significantly reducing the contamination levels. Within this frame, new, compact, low-cost, low-volume robotic teams will provide an alternative to large scale, high-cost robots used in the U.S. market. Therefore, Europe's competitiveness within the global market will be increased and autonomous waste management will be achieved.

2.1.4 Requirements, potential barriers and mitigation measures

The successful exploitation of **RECLAIM** results will be determined by the requirements and potential barriers discussed in the following. We also identify the implemented strategy for the mitigation of barriers:

1. Lack of alternatives in strategic policies to support decentralized waste management.

In Europe, political measures and policies at local, national and European level determine the framework and regulations of waste management processes. Following China's recent restrictions, there is an increasing need for local material recovery, far beyond the metropolitan centers. European and national leaders have to explore alternatives.

RECLAIM will reach out to waste management companies, public bodies, local authority contractors and policy-makers in order to disseminate the project's results and highlight the new, decentralized material recovery opportunities that will be implemented. Accordingly, the proposed work will facilitate European waste management autonomy and will significantly contribute in protecting the European ecosystems ensuring a clean and healthy environment for all.

2. Achieving industry acceptance.

A typical risk for Innovation Action projects regards the unsuccessful connection of scientific results to industrial and societal needs (i.e. technological advancements do not provide long-term solutions to real-world problems).

RECLAIM aims to address this barrier by working from day one on real world problems and implementing key technological advancements on site, therefore yielding results that are directly evaluated in real and demanding conditions. Through the organization of training seminars, the advantages of the developed solutions will be explained to interested stakeholders. Additionally, the proposed work considers activities to enforce the sustainability of the prMRF technology and ensure its long term use after the end of the project (Task 5.4). In agreement to the HORIZON-CL4-2021-DIGITAL-EMERGING-01-09 call, **RECLAIM** adopts a concrete problem-solving approach, which sets the market upscale of the developed solution as a key objective. The proposed work is designed in a completely applied and industry driven framework, and the consortium aims at innovative products with high industrial value and great potential for post-project commercialization.

3. Cost competitiveness of the proposed solution.

Currently, MRFs are either large-scale facilities near large cities, equipped with costly robots, or small scale mobile units relying extensively on manual sorting. Compared to the high-cost large scale MRFs, prMRF is an extremely cost competitive solution. Small scale, manual MRFs on the other hand might be cheaper in terms of facility, equipment and maintenance cost, but they have much lower efficiency and higher operating costs.

RECLAIM follows a particularly low cost approach by adopting the use of RoReWos that achieve a particularly high "picks per invested euro" ratio. Concerning the supplementary R-Eq to be installed in prMRF, it will be selected after thorough market search for identifying low-cost solutions that reduce the initial burden. Overall, prMRF is expected to be a promising investment in the long run. In addition, in the way to its market introduction, alternative business models may be chosen, such as renting the facility for a specific period.

[43] https://eandt.theiet.org/content/articles/2021/09/ai-driven-robotics-key-to-recycling-s-challenges/
[44] https://resource-recycling.com/recycling/2019/01/29/china-plastic-imports-down-99-percent-paper-down-a-third/

4. Public acceptance and engagement.

Waste facilities are usually not welcomed by the local communities, due to odors, noise and traffic issues that might be associated with them. What is more, AI and robotics might also be examined with skepticism regarding ethical issues that might arise and the potential threat to employment.

RECLAIM will organize activities to engage the local communities and properly explain the technologies and the operational conditions, addressing questions and uncertainties. In addition, it will be made clear that the digitalization and automation of processes and facilities do not exclude the experienced workers, who will provide their knowledge and know-how to the entire process, but it will shift their duties towards the handling and control of the machines and other management tasks. Most importantly, **RECLAIM**'s proposed waste management decentralization will offer additional job creation at local levels, bringing back jobs that have currently been outside Europe. Finally, the public's engagement through the RDG is a key objective to this project and it will be addressed responsibly, ensuring that all groups (based on age, gender, etc.) will be encouraged to participate.

2.2 Measures to Maximize Impact

The **RECLAIM** partners have already planned a credible path to maximize the impact of project actions. This plan is comprised of five primary and interdependent elements, namely: (i) specification of target groups, (ii) dissemination activities, (ii) communication channels and activities, (iii) exploitation strategy for market penetration, (iv) intellectual property and regulatory assessment.

Before describing in detail the individual elements, it is noted that both dissemination and communication activities will be **led by ISWA, the largest waste management network worldwide**, with the mission to promote and develop sustainable and professional waste management and catalyse the transition to a circular economy. ISWA has already established an extensive network of different stakeholders in the waste management sector that will be actively informed about **RECLAIM**, its activities and results. Along this line, **the project will be regularly presented in the annual ISWA World Congress** which is the industry's leading solid waste management event. Exploiting direct access to the focused, global waste management and circular economy community, **RECLAIM** will assure maximum impact for the proposed work and high potential to promptly include prMRF in the material recovery technologies used globally.

*It is noted that, during the last phase of the project (Wrap up & Consolidation phase), the consortium will consider the transfer and **live demonstration of the prMRF in the ISWA World Congress** and/or other high-visibility exhibitions in the circular economy domain. This unique demonstration opportunity will be possible due to the portability and quick installation of the prMRF that permits the implementation of demonstration tours to increase project visibility and spark the interest of stakeholders and investors to prMRF and the relevant project technologies.*

Exploitation activities will be led by AXIA, which has long experience in designing strategies and implementing activities that overcome the barriers for the market uptake of research results. Before the end of the project, AXIA will organize a major event to present **RECLAIM** results to stakeholders and demonstrate prMRF to investors and potential customers.

2.2.1 Target audience

The **RECLAIM** partners have identified different communities with an expected high interest in the targeted innovations. These are summarized in Table 2.2.

Table 2.2 RECLAIM Target Audience

Target Audience	Sector Description
Industry	End users in sectors of interest: ▪ Robotic waste sorting system manufacturers such as Machinex Industries, AMP Robotics Corporation, ZenRobotics, and waste sorting equipment manufacturers such as STEINERT GmbH, TOMRA, REDWAVE, Buhler Group ▪ Recycling Industry (both mechanical and chemical recycling) ▪ Waste management companies, demolition companies ▪ Local authorities and contractors ▪ Hyperspectral imaging systems developers such as Specim, Spectral Imaging Ltd. ▪ Computer vision companies such as Basler, Baumer Optronic ▪ Secondary end users (plastics, paper, metal and minerals industry, but also wood and food waste industry ▪ Gaming industry

European associations	EXPRA (Extended Producer Responsibility Alliance), PRO-EUROPE (Packaging Recovery Organization Europe), EPDA (European Plastics Distributors Association), Plastics Europe (Association of plastics manufacturers), EPRO (European Association of Plastics Recycling and Recovery Organisation), PRE (Plastics Recyclers Europe), Zero Waste Europe, EGBA (European Gaming and Betting Association)
Scientific Community	Academics & researchers, Universities, research institutions, students, living labs
Developers	professionals, open source, hobbyists in the area of AI, robotics, and gaming
EU projects and Digital Innovation Hubs	Funded projects from the CSA HORIZON-CL4-2021-HUMAN-01-02, partnerships from Horizon Europe Clusters 4 and 6, collaboration with the AI-on-Demand platform, the EU Digital Industrial Platform for Robotics, the BigData Europe and similar follow up projects
General Public	Citizens, communities facilitating the demonstrators, communities with the potential to facilitate prMRF operation, environmental NGOs, students
Policy-makers, Public Bodies	Standardization and certification bodies, national Governments, environmental ministries, municipalities and local regional authorities, authorization bodies, EC parliament
Innovation Communities	Investors, start-ups, entities in charge of funding research programs, funding schemes, financial stakeholders

2.2.2 Dissemination Activities

The **dissemination plan** aims to inform EU stakeholders and the largest possible audience **about the project's results** and enable the future market penetration of the solutions proposed. The dissemination activities, starting after the first project results are obtained, will promote the project's outcomes to identified target groups (see Table 2.2), through complementary means, enabling their use, uptake and further evolution/application in various fields. Under this scope, all **RECLAIM** partners will closely cooperate, using all the successfully tested practices in line with the EC Guidelines and Communication Strategies of the European regulation, and taking advantage of their established networks to raise global awareness on the obtained results. More specifically, **RECLAIM** will actively disseminate project results to the targeted communities by implementing activities aligned with the needs of the various constituencies that will benefit from project outcomes. Dissemination actions addressing important communities include the following:

Table 2.3 Dissemination actions.

Tool description and scope	Target Audience
Open access of data, computational models, scientific publications	Scientific community, Developers
On-site demonstrators	General Public, Industry, Policy-makers
Communication with networks and associations	European Associations, Academia & Research
Presentation of Results in Scientific Conferences	Academia & Research
University courses, trainings, thesis	Academia & Research
New and ongoing projects of partners	Academia & Research
Training Programs	General Public, Academia & Research
Results presentation in Exhibitions and fairs	General Public, Academia & Research, Policy-makers, Public Bodies, Innovation Communities
Networking with EU funded activities	EU projects and Digital Innovation Hubs, Academia & Research, Innovation Communities

Dissemination activities in **RECLAIM** will be constantly monitored to evaluate the effects of the implemented activities and plan future efforts. The following table provides a list of Dissemination Key Performance Indicators (D-KPIs), setting a baseline for verifying that key project dissemination objectives have been met.

Table 2.4 Dissemination KPIs

KPI Description	Threshold
Training of non-specialist partners staff on robotic, AI, hyperspectral imaging, data analytics and scientific gaming technologies	At least 2 trainings of non-specialist **RECLAIM** partners staff and professionals on waste management (including reference to trustworthy and ethical AI)
Open online educational events on state of the art robotic, AI, hyperspectral imaging, data analytics and scientific gaming technologies	At least 2 online webinars open to the public on recent technology developments, their benefits/restrictions and potential applications (including reference to trustworthy and ethical AI)
Recycling Data-Game users	At least 2000 unique registered users

Open dataset of annotated RGB waste images	At least 50.000 images of waste annotated with respect to the presence of recyclable materials
Open dataset of annotated Hyperspectral waste images	At least 30.000 hyperspectral images of waste annotated with respect to the presence of recyclable materials
Open access model for recyclable material recognition based on RGB imaging	At least one DL-trained CNN model on material recognition based on RGB imaging
Open access model for recyclable material recognition based on hyperspectral imaging	At least one DL-trained CNN model on material recognition based on hyperspectral imaging
Open access papers	At least 5 international high-quality, peer-reviewed journal/conference publications
Special Issues	At least 1 special issue in a high-impact international journal
Networking with EU-funded projects	Seek networking and joint activity organization with at least 4 EU funded projects
Staff exchanges	At least 3 weekly visits/staff exchange between **RECLAIM** partners (Covid-19 pandemic permitting)
Participation in events	Participation in at least 8 major physical or virtual events (conferences and fairs) with presentations, booths

RECLAIM partners will carefully select publication venues based on their scientific excellence and impact privileging, with a preference to open access publishing. An *indicative*, non-restrictive list of high impact venues to be targeted for publication by the project consortium are:

- *Robotics and AI:* IEEE Intern. Conf. on Robotics and Automation (ICRA), IEEE International Conference on Intelligent Robots and Systems (IROS), IEEE Conference on Computer Vision and Pattern Recognition, International Conference on Pattern Recognition, RSS, ICRK SPR, Parallel, RA-L, T-RO, RAM, Annual International Conference on AI and Data Science, ROBOMECH Journal, Engineering Applications of Artificial Intelligence, Frontiers in Robotics and AI, Autonomous Robots, AI & Big Data Expo Europe, EmTech Europe, Annual International Conference on AI and Data Science.
- *Waste Management:* ISWA World Congress, World Conf. on Waste Management (WCWM), Intern. Conf. on Science, Technology, Engineering & Management (ICSTEM), Intern. Conf on Environment and Life Science (EUCELS), Sustainable Waste Management Conference (AIChE), Waste Recycling and Management (ICWRM), Journal of Waste Resources and Recycling, Open Waste Management Journal.
- *Circular Economy:* Circular Economy and Sustainability, International Conference on Circular Economy and Sustainability, Journal of Shipping and Trade, International Conference on Circularity in the Built Environment, European Recycling Conference, Annual event WCEF2021.

2.2.3 Communication Channels and Project Activities

The **RECLAIM** communication plan will include strategic and targeted measures for promoting the project itself (including its results/accomplishments) to multiple audiences, the media and the wider public. Planned activities will raise awareness about the project but also the wider topics of recycling and the circular economy and how these are supported by **RECLAIM** technologies, namely robotics, AI, RGB/hyperspectral imaging, data analytics and scientific gaming.

Key goals of the communication activities are to demonstrate how EU funding contributes to tackling environmental, societal and economic challenges in relation to waste management, recycling and transition to a circular economy. Particular emphasis will be provided in revealing how **RECLAIM** and the development of prMRF in particular support **material recovery and the circular economy in the current Covid-19 pandemic**.

RECLAIM has already specified a communication toolbox to support and assure the effectiveness of communication activities, which are listed below:

- Visual identity and brand of the **RECLAIM** project (e.g. attractive, easily identifiable project logo)
- Project website, blog, newsletters
- Regularly prepared informative material, both multimedia (videos, presentations, digital flyers) and printable (banners, posters, flyers)
- Use of social media to communicate project achievements (create campaigns for targeted audiences)
- Participation in industry events (e.g. ISWA World Congress, IFAT)
- Presentations to local, national and international media (TV broadcasts, radio, newspapers)
- Participation to European and International events demonstrating scientific and technological developments to the public (e.g. European Researchers Night, European Research and Innovation Days, etc.)

Communication activities in **RECLAIM** will be constantly monitored and evaluated against the list of Communication Key Performance Indicators (C-KPIs) summarized in Table 2.5.

Table 2.5 Communication KPIs

Communication Channel	Threshold
Project website *(D&C)*: General information about the project, objectives, consortium, news, progress and publications	▪ Annually updated project flyer ▪ 20 visits per day after year 2 ▪ 3,500+ downloads of reports and dissemination material
Social Media *(Twitter, FB, LinkedIn)*: Raise awareness of general public, attract stakeholders and potential industrial partners	▪ 1000+ followers in each platform ▪ 5 posts/platform every month ▪ 20+ re-tweets per month ▪ 2+ targeted campaigns annually to promote prMRF and RDG
Videos	▪ project dedicated YouTube channel after month 3 ▪ 2 videos uploaded in YouTube and website annually ▪ 500+ views per video
Articles (in magazines/newspapers)	▪ 6 articles in 6 different countries on the use of AI and robotics for material recovery in the Covid-19 era, in the first year ▪ 6 articles in 6 different countries on the use of RDG for scientific data collection in the second year
e-Newsletters *(D)*: Project progress and updates, upcoming events and publications	▪ 3+ newsletters annually ▪ 4,000+ recipients after year 2
Press releases *(C)*: Awareness of impacts for techno-economic, societal, environmental benefits	▪ 2 press releases annually, communicating relevant achievements in TV, magazines
Printed material (leaflets, flyer, brochures, posters) *(D&C)*: Distribution in conferences, fairs, B2B, presenting aims, expected impact.	▪ 6 printed concepts ▪ Distribution of 1,500 leaflets in events
Partners' websites (*D&C)*: Dedicated section for the project in all partners' pages	▪ A total of 20 visits/month in the dedicated section
European and International events	▪ 2 participations of the project participants annually ▪ 2 presentations and/or video demonstrations annually
Open Live Demo	▪ 15+ days of live prMRF demo in public spaces

Within the frame summarized in the last two sections, ISWA, supported by all partners, will deliver a Plan for Dissemination and Communication (D7.1), developed in M6. This plan will be annually updated.

2.2.4 Exploitation Activities

Exploitation activities have a more practical scope, including actions for utilizing the project results in activities other than those envisaged by **RECLAIM**. These regard using, advancing and marketing the product and assets of the project, creating and providing a log-term services, or include project outcomes to steer EU and national policies and refine regulations on material recovery. The exploitation plan of **RECLAIM** is centered on the open-source distribution of project results to accelerate developments on smart, automated material recovery. Project partners will ensure that technologies will be readily accessible to the communities of interest, enabling them to use and potentially benefit from the project's results.

It is expected that the **RECLAIM** partners will actively exploit the achievements and results of the project by their own efforts. Relevant activities include:

- Extend the capabilities of the prMRF, the RoReWos, the AI-ILC, the RDG with additional features to support additional types of production processes and allow their use in new applications.
- Conduct further research to address new desirable features identified in the final stages of the project.
- Exploit **RECLAIM** technologies in ongoing or new national and EU projects with related research activities and technology developments.
- Utilise existing technology transfer programmes to make project technologies available to industrial actors in Europe.
- Register project results to European initiatives to maximize visibility and potential interest of interested stakeholders and investors (e.g. Horizon Results Booster, Horizon Results Platform, Horizon Impact Awards, EU Web awards, Enterprise Europe Network, European Cluster Collaboration Platform, Innovation Radar)
- Evaluation of the ability of **RECLAIM** technologies to support additional types of production scenarios and configurations beyond those addressed within the project.

- Create synergies and transfer knowledge from/to relevant projects and technology networks such as the AI4EU: European AI On-Demand Platform, EU Digital Industrial Platform for Robotics, the Munich Innovation Hub for Applied AI, Digital Innovation Hub in Artificial Intelligence & Robotics for Sustainable Development Goals

The exploitation strategy will pave the way for the market uptake of the project outcomes comprising hardware modules, namely RoReWos and the prMRF enhancing their potential for post-project commercialization. In order to transfer them into real innovation - a new economic activity that creates jobs and wealth- the project participants will present the implemented fully operational prototype to investors, waste management enterprises and other stakeholder with a focused interest and economic activities related to the circular economy. Moreover, to support the market up take of RoReWos and the prMRF, **RECLAIM** will contact innovation incubators (e.g. European BIC Network) providing established channels for technology commercialization and other clustering services. The software results of the project will be made available as open source, for re-use and potential advancement by researchers and developers.

A detailed exploitation plan will be created regarding the key innovation results of the project, as well as partners' individual pathways, leading to the maximization of the project's impact. Key Results (KR) of **RECLAIM** are presented in Table 2.6, along with a categorization of their type, IP ownership and exploitation route. Four main categories of KR type are identified: (i) new HW&SW products, (ii) SW processes, (iii) prototypes, (iv) data. These innovative and exploitable results will result from the work conducted by each partner and will be further updated during the project implementation.

Table 2.6 RECLAIM Key Results (KR)

Key Result	Type	IP Owner	Exploitation route
KR1: Development of high-performance AI-ILC	SW Process	IRIS, FORTH	Open Source, collaborative research
KR2: Optimization of low-volume, low-cost, high performance RoReWos	HW&SW Product	RBNS, KUL	Patenting, collaborative research or licensing
KR3: Development of eco-friendly, high-tech prMRFs	HW&SW Product	FORTH, IRIS RBNS, KUL	Patenting, collaborative research or licensing
KR4: Development of recycling data games	SW Process	UoM	Open Source
KR5: Demonstration of prMRF in three different use cases	Prototype	ION, HRRC	Sustainable operation beyond **RECLAIM**
KR6: Annotated dataset of RGB and hyperspectral waste images	Data	RBNS, IRIS, FORTH	Open Source, benchmarking

Furthermore, it is worth mentioning that the exploitation strategy will include a general plan for each partner's foreseen exploitation route. Within this frame, the benefits from the involvement of partners in the project and their contribution to **RECLAIM**'s impact are highlighted. The above-mentioned exploitation strategy will be included in the Plan for the Innovation and Exploitation Activities (D7.2), originally submitted at M6 and updated twice at M18 and M36.

2.2.5 Management of Intellectual Property

The Consortium Agreement (CA) will be signed by the EC and the consortium, setting the legal basis for the management of IPR, ownerships, access to key knowledge and research data. The background IPR of each partner will be identified and clearly defined in the CA, and remain the property of the originator partner. A well-defined IPR strategy will allow handling aspects related to ownership, development, exploitation and confidentiality of IP arising from the project's results. Particularly, the IPR generated within the project will be the property of the partner responsible for its development. In the case of joint invention, partners collaborating for the achievement of the project's objectives will have joint ownership of results, defined and protected in a bilateral agreement. All background knowledge necessary for the implementation of the project will be available to the relevant partners. Whenever background IP is needed for the post-project exploitation, the owner will choose whether royalty-free access will be granted, or a license fee will be charged, under fair and reasonable terms. The IPR strategy will ensure the protection of IP during the exploitation of the key results. Originally defined in the CA and Grant Agreement (GA), IPR strategy will be updated by AXIA in the Plan for the Innovation and Exploitation Activities (D7.2). Additionally, AXIA will conduct a patent mapping analysis in order to ensure infringement is avoided and to identify the patent potential of the project.

2.3 SUMMARY

A summary of the Impact section is presented in the canvas bellow, highlighting the key elements of the **RECLAIM** impact pathway and of the measures to maximise its impact.

SPECIFIC NEEDS

- Many **material recovery** activities are currently underway which are however mainly **focused on serving urban centers**.
- There are several cases with pressing waste treatment and **material recovery needs** that are still **not covered** efficiently by this model (rural areas, islands, transport hubs and more).
- **Portable material recovery units** are used to accomplish remote waste treatment, which are still **human powered**, lacking the incorporation of available intelligent, automated components that could significantly enhance their productivity.
- The **Green Deal** aims to transform the EU into a resource-efficient and competitive economy where ..."**no person and no place is left behind**".
- There are increasing **global needs** for material recovery in distant and isolated areas.
- Difficulties in material recovery due to the **Covid-19 pandemic**.

EXPECTED RESULTS

- An AI module based on Deep Learning, which combines imaging in the visual and infrared spectrums to **identify, localize and categorize recyclables**.
- A **modular multi-robot/multi-gripper** robotic system for material recovery that is based on low cost, high productivity Robotic Recycling Workers (RoReWos).
- A portable, robotic Material Recovery Facility (prMRF) equipped with an AI-ILC module and a RoReWo Team to provide **local-scale material recovery with industrial-level efficiency and low cost**.
- A **Recycling Data-Game** (RDG) that enables and encourages citizens to participate in RECLAIM's RTD activities by providing data annotations, to be used in Deep Learning training.
- **Three large scale demonstrations** of prMRF applied to different material recovery scenarios that are common for distant, less accessible areas.
- A **benchmark dataset** of RGB and hyperspectral images that provides a testbed for comparing and assessing future R&D on recyclable material recognition.

D & E & C MEASURES

Key Exploitation Measures

- Patenting the prMRF; sell or lease the use of the prMRF to waste management companies, local authorities and contractors
- Open source distribution of all implemented models for recyclable material recognition based on RGB and/or hyperspectral imaging
- Open dataset of annotated RGB and hyperspectral waste images
- Open source RDG to raise awareness and facilitate incorporating a data-driven citizen science approach into future projects

Key Dissemination Measures

- Open access scientific publications presenting all RECLAIM technical accomplishments
- Participation to conferences and fairs, including live demonstrations of the prMRF
- Synergetic actions and collaboration with DIHs and other relevant EU and national projects
- Training of non-experts in AI, robotics, hyperspectral imaging and citizen-science gaming

Key Communication Measures

- Project website, social media, videos, presentations, open live demos
- Promote the extensive use of the RDG by
  - Citizens informed about and actively involved into EU Green Deal activities
  - Citizens informed on how EU research funding contributes in dealing with environmental, societal and economic challenges

TARGET GROUPS

- **Waste management** & Material recovery companies
- Local government authorities and contractors
- Circular Economy & Recycling Industry

- **Robotic** waste sorting system manufacturers
- Application domains managing items on conveyor belts (e.g. agri-food, pharmaceutical, grocery industry, etc)

- **Hyperspectral and Computer vision** system producers, scientists, researchers
- Hyperspectral imaging systems producers, scientists, researchers

- **Gaming** Industry (data games, serious games)

- Communities promoting **citizen science**

- **Academics, researchers** in the fields of environment protection, robotics, computer vision, hyperspectral imaging, scientific gaming, data analytics, machine learning

OUTCOMES

Decentralized waste management practices

- The prMRF deployed in Ionian islands will remain operational and productive for at least 3 years after RECLAIM completes
- It is expected that prMRF will be widely adopted to cater for local material recovery needs in many Mediterranean, European islands, but also in isolated, less accessible areas all over the world

EU Green Deal

- Major boost in making Europe zero polluting, climate-neutral, sustainable and globally competitive
- Raise awareness of a broad range of citizens on the Green Deal and targeted activities

Economic/Technological achievements

- New, high-performance solution for modular, AI-driven, low-cost robot-team applications

Societal

- Non experts trained on AI, robotics, data exploitation, hyperspectral imaging and socially sensitive gaming

IMPACTS

Robotic Material Recovery

- Breakthrough in material recovery by enabling close to source, intelligent and automated waste processing
- New, modular and low cost multi-robot/multi-gripper approach for material recovery
- Significant advancement in the "picks per invested euro" ratio in the field of material recovery robotics
- Widespread adoption of robotics in tackling urgent environmental and social problems
- New market for decentralized material recovery based on portable and digitally advanced systems, with Europe obtaining the first mover advantage

EU Green Deal

- Significant contribution in achieving the goal of "no person and no place left behind"
- Significant contribution in circular economy activities making the EU more sustainable and resource independent

Scientific/Technological Impact

- Significant advancements in RGB and/or hyperspectral imaging for environmental applications
- Extensive waste image dataset supporting future developments in material identification and recovery
- Significant advancements in data gaming technology that is tightly coupled with state of the art AI solutions

3 IMPLEMENTATION

3.1 WORK PLAN - WORK PACKAGES, DELIVERABLES AND MILESTONES

3.1.1 Components and interdependencies

The work planned in **RECLAIM** will be articulated in 7 work packages. The work breakdown into work packages is based on gathering related work, to ensure synergistic cohesion between related tasks and maximise the efficiency of the R&D work. Figure 3.1 provides a graphical representation of the interactions among work packages which are further discussed below.

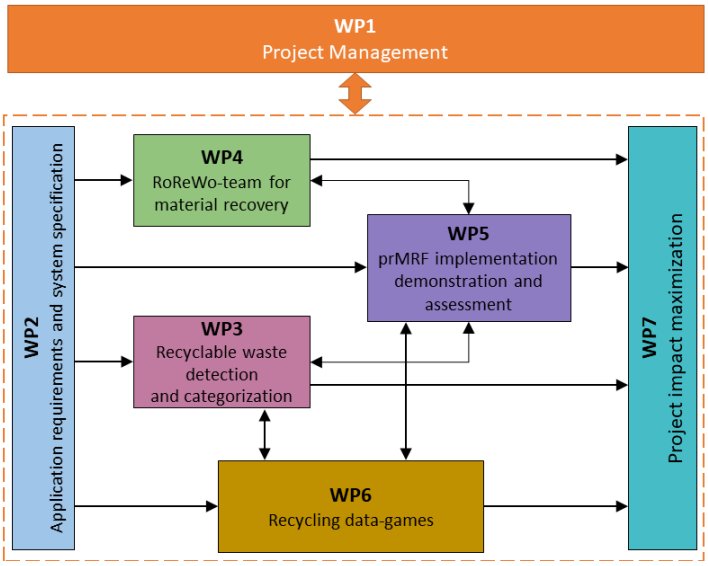

Figure 3.1. The interaction of work packages in **RECLAIM.**

Project Planning - Timeline and Effort Distribution. The project duration will be 36 months. As already discussed in Section 1.2, the project development will be based on early prototyping, followed by iterative "implementation and real-world testing" spins that will ensure we build the right tools the right way, and finally the system assessment, consolidation and asset exploitation. This is reflected to the work package structure summarized below. An overview of how the different parts of the planned work span over time is shown in the Gantt chart (Figure 3.2).

It is reminded here that the first use case scenario "Material recovery from mixed recyclables streams" will be addressed during the whole duration of the project, the second scenario "Cleaning of citizen separated, material-specific streams" will be considered in the second and third project years, while the third scenario "Immediate, close to source material recovery" will be addressed in the last semester of the project (see Box 2).

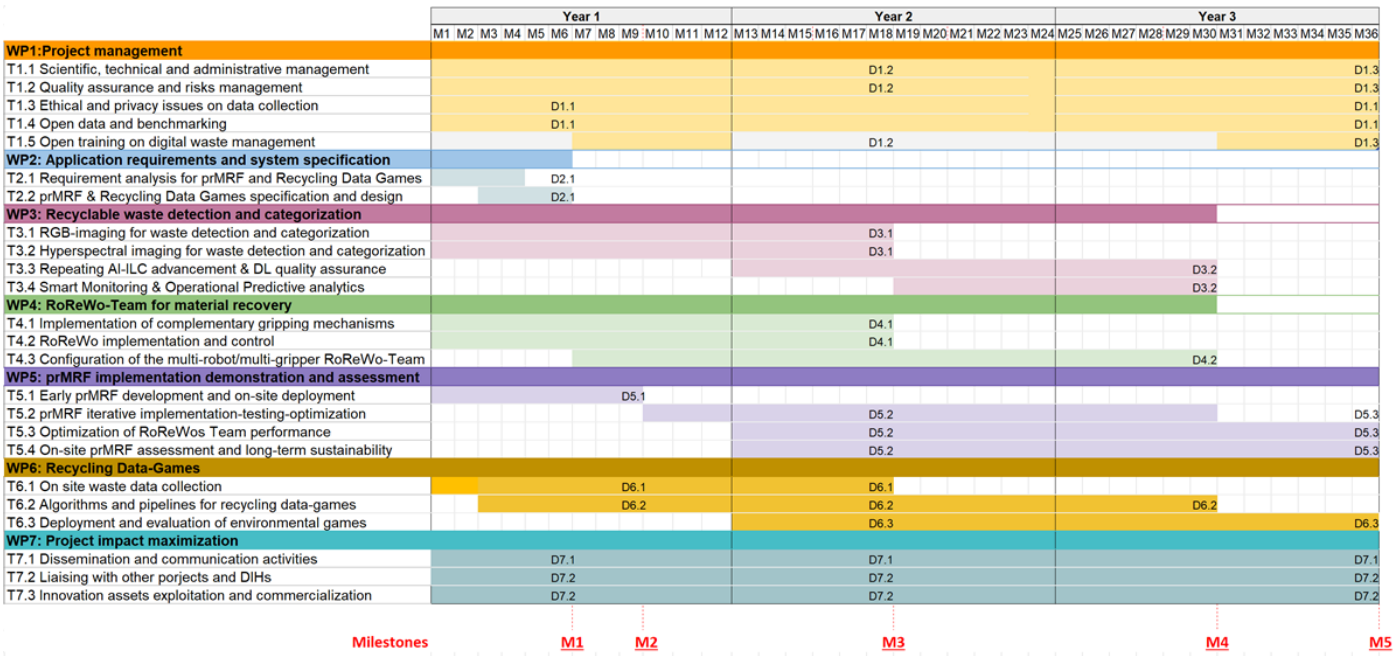

Figure 3.2. The **RECLAIM** Gantt Chart

Additionally, a summary of effort distribution in the work packages is provided bellow, in Table 3.1a.

Table 3.1a. List of **RECLAIM** Work Packages (WPs).

WP No	WP Title	Lead Part. No	Lead Particip.	Person Months	Start Month	End month
WP1	Project management	1	FORTH	30	M1	M36
WP2	Application requirements and system specification	7	AIMPLAS	40	M1	M6
WP3	Recyclable waste detection and categorization	5	IRIS	73	M1	M30
WP4	RoReWo-Team for material recovery	3	KUL	71	M1	M30
WP5	prMRF implementation, demonstration & assessment	1	FORTH	99	M1	M36
WP6	Recycling Data-Games	2	UoM	51	M1	M36
WP7	Project impact maximization	8	AXIA	80	M1	M36
Total person months				444		

The work included in each work package is summarized in the Tables bellow.

Table 3.1b. RECLAIM work package description.

WP Number	1	WP Leader		FORTH		Start Month	1	End Month		36
WP Title	Project Management									
Participant No	1	2	3	4	5	6	7	8	9	10
Participant Name	FORTH	UoM	KUL	HRRC	IRIS	RBNS	AIMPLAS	AXIA	ISWA	ION
PMs/Participant	9	2	2	4	3	2	2	2	2	4

Objectives

The first WP will monitor and coordinate project activities with the following objectives:

- Implement and maintain an effective management infrastructure for monitoring and controlling progress.
- Manage resources to achieve the overall project objectives and implement quality management procedures.
- Coordinate and communicate with the European Commission on the status of the project, including financial reports and all documentation necessary to meet the contractual obligation of the project.
- Oversee and coordinate implementation risks and ethical and privacy issues related to **RECLAIM**.
- Organization of training activities for the consortium partners and the general public.

Description of Work

T1.1 Scientific, technical and administrative management [M1-M36] FORTH, All

This task is dedicated to ensuring collaboration between partners and high quality of all project outputs. It includes: (a) coordinating the joint efforts of the consortium to ensure smooth progress of the work plan and the fulfillment of the consortium's contractual obligations, (b) providing the necessary liaisons between the consortium and the EU, and (c) monitoring the progress of the project according to the project schedule and all related resource usage, and budget allocations, (d) ensuring the on-time completion of the project objectives, milestones, and deliverables. Project coordination meetings and technical meetings will be organized regularly. At least two technical meetings are foreseen every year to review and agree on project developments. Extensive use of teleconferencing facilities will be adopted in all cases where physical meetings are hard to realize, such as the current COVID-19 pandemic crisis.

To ensure smooth and successful implementation of the project, distinct managerial roles have been identified and assigned to participants with significant experience in European projects. The Project Coordinator (Dr. Michail Maniadakis - FORTH) will be responsible for overseeing the entire project, (administrative, technical, and scientific coordination). The Exploitation Manager (Dr. Ioanna Deligkiozi -AXIA) will co-ordinate the exploitation activities of the project. The Ethics & Privacy Manager (Ioannis Kakogiannos - IRIS) will oversee data protection strategy on data collection, storage and processing procedures, to ensure compliance with GDPR requirements.

T1.2 Quality assurance and risks management [M1-M36] AIMPLAS, All

This task aims at periodically comparing the performance of the prMRF with the KPI metrics described in Section 1 Obj 1, 4, to assure that the project objectives are fully accomplished. Risks will be linked to specific parts of the project as, for instance, tasks, deliverables or milestones, in order to make it easier to monitor them. This will help potential problems to be identified early and addressed promptly and effectively without the risk increasing and shifting to the next stages of the project. Foreseen risks regarding the management and technical WPs are discussed in Section 3.1.3. Moreover, **RECLAIM** considers the use of an External Advisory Board (EAB) that will be actively involved in the strategic steering of the project's innovation endeavors. Three high profile experts in areas relevant to **RECLAIM** have committed their active support to the project with Letters of Intent, provided in the Appendix.

T1.3 Ethical and privacy issues on data collection [M1-M36] IRIS, All

This task is responsible for ensuring that any ethical or privacy concerns occurring during the implementation and evaluation of technical developments that involve user data (mainly from playing RDG) are addressed appropriately,

with full respect to EU regulations. This task that spans the whole duration of the project will oversee (1) the compliance of **RECLAIM** data collection and usage to user privacy regulations (2) the preservation of the confidentially of personal data (3) the achievement of a high-level security regarding potential risks (mainly from using the RDG).

T1.4 Open data and benchmarking [M1-M36] IRIS, RBNS, FORTH

This task aims to benchmark technology results achieved in the project in order to provide a testbed for comparing and assessing future works targeting the recognition of recyclable materials in difficult real-world conditions. The consortium aims to make the collected data and the developed AI models that accomplish the identification, localization and categorization of targeted recyclable materials, available for open use. It is expected that the data and benchmark created in **RECLAIM** will be established as the standard for future works to compete, in relation to well-specified performance indicators. To support benchmarking activities **RECLAIM** will seek collaboration with the EU DataBench project.

T1.5 Open training on digital waste management [M7-M12] [M31-M36] ION

RECLAIM will organize training programs that present the latest developments in AI, Robotics, Data exploitation, Hyperspectral imaging and socially sensitive gaming, how they can support circular economy, and how Trustworthy AI can support B2B interactions and joined innovation endeavors. The training programs will invite renowned experts from the CSA projects funded under the call Human-Centered and Ethical Development of Digital and Industrial Technologies (Horizon-CL4-2021-Human-01-02) that focus on the adoption of AI technologies in the public and private domain. The training programs will be open to the public but designed with a focus on the ION, HRRC, AIMPLAS, AXIA, ISWA staff who are non-experts in these topics but would like to know how they can exploit state-of-the-art technologies. **RECLAIM** will incorporate two different training programs, one in the first six months of the project to make people aware of what they can expect from **RECLAIM** and one at the end of the project, to discuss recent advancements of and future opportunities from smart digitization.

Deliverables

D1.1: Data management plan and ethics/privacy manual [IRIS; M6, M36; (T1.3, T1.4)] Report documenting the consortium's data management plan in compliance with the EC "Guidelines on Data Management in Horizon Europe" and ethics/privacy manual to be consulted over all project stages to ensure privacy and security of data and subjects.

D1.2: 1st Periodic Project Report [FORTH; M18; (T1.1, T1.2, T1.5)] Report on technical progress, dissemination and resource usage and cost claims for the first contractual reporting period.

D1.3: Final Project Report [FORTH; M36; (T1.1, T1.2, T1.5)] Report on technical progress, dissemination and resource usage and cost claims for the final contractual reporting period.

WP Number	2	WP Leader		AIMPLAS		Start Month	1	End Month		6
WP Title	Application Requirements and System Specification									
Participant No	1	2	3	4	5	6	7	8	9	10
Participant Name	FORTH	UoM	KUL	HRRC	IRIS	RBNS	**AIMPLAS**	AXIA	ISWA	ION
PMs/Participant	5	4	4	5	3	4	**6**	2	2	5

Objectives

The main purpose of this WP is to prepare the ground for the development and innovation WPs, through the:

- Definition of the prMRF requirements for processing waste on all use case scenarios.
- Specification of the characteristics of the prMRF units comprising the material recovery chain.
- Definition of the RDG operation, connectivity and usability requirements.
- Specification of the RDG data collection procedures and user interface.

Description of Work

T2.1 Requirement analysis for prMRF and Recycling Data Games [M1-M4] HRRC, All

Analysis of the waste streams in each deployment site to determine the seasonal characteristics and the daily volume of the waste flows, the variance of waste, the percentages of each material type, the deformation of waste during transport to the sorting site, and the frequency that new unknown materials may appear. Specify the operational requirements and priorities in each site, to design sustainable solutions with maximum social impact after taking into account the environmental burden of existing practices.

T2.2 prMRF & Recycling Data Games specification and design [M3-M6] AIMPLAS, All

Design of the prMRF to be developed in the Ionian islands, considering the individual characteristics and requirements of the three application scenarios assuming intensified waste processing (T2.1). Consider alternative methods for the pre- and post-post processing of the waste streams and definition of mechanical processes to ensure

that the requirements for robotic material sorting are met and efficient cooperation along the processing chain is accomplished. As part of this task the prMRF equipment will be specified and purchased.
Moreover, this task involves the specification and design of playful activities that increase social awareness regarding recycling and solicit data annotation from citizens. The design efforts will involve all partners through a number of project-wide workshops in order to collect requirements and discuss affordances of how games can assist other objectives and tasks in **RECLAIM**. In addition, we will solicit end-users' feedback in terms of interaction paradigms and preferences through dedicated focus groups with gaming enthusiasts and citizen scientists. Therefore, the design process will follow a user-centered and participatory design approach, including all relevant stakeholders.

Deliverables

D2.1: prMRF and RDG requirements and systems specification [AIMPLAS; M6; (T2.1, T2.2)] Report on the operational requirements of the prMRF and the RDG, and their specified (targeted) characteristics.

WP Number	3	WP Leader		IRIS		Start Month	1	End Month		30
WP Title	Recyclable Waste Detection and Categorization									
Participant No	1	2	3	4	5	6	7	8	9	10
Participant Name	FORTH	UoM	KUL	HRRC	**IRIS**	RBNS	AIMPLAS	AXIA	ISWA	ION
PMs/Participant	13	9	4	4	**27**	6	5			5

Objectives

This WP will develop perception mechanisms which will process flows of RGB and/or Hyperspectral images to recognize recyclable materials. Its objectives are:

- Exploit state of the art Deep Learning (DL) methods that address the instance segmentation problem
- Create DL modules for material identification, localization, categorization on RGB & hyperspectral images
- Design and implement methods for the systematic advancement of the DL modules.
- Implement sensors network to monitor full-prMRF performance

Description of Work

T3.1 RGB-imaging for waste detection and categorization [M1-M18] FORTH, IRIS,RBNS

Exploit the extensive image data collected in T6.1 for the development of a RGB-based visual recognition system for recyclable materials, using modern AI and machine learning techniques. Comparison and optimization of alternative CNN solutions in identifying, categorizing and localizing recyclable materials in waste stream, in real and difficult environments. The inference results of the model will guide the actions of the RoReWo Team in simplified operational scenarios (see for example scenario 2, Box 2) and will provide the basis for implementing the quick annotation tool to be used for the annotation of collected data. In particular, early CNN models will be employed as part of the annotation tool engine, which will allow the collection/annotation of new data. Then, the collected data will be used to improve CNN models, which will enable better annotation and so on.

T3.2 Hyperspectral imaging for waste detection and categorization [M1-M18] IRIS, FORTH,

The task considers the implementation of a prototype based on optical and spectroscopic techniques to correctly detect and categorize materials. In particular, NIR/SWIR-HSI and XRF systems will be designed and assembled on a conveyor test bench. Different configuration modes will be tested such as absorption, diffuse reflectance, interactance and transflectance. The selected core devices will be acquired using commercial components that have been optimized to identify the main target materials in terms of spectral and dimensional resolution, lighting system, scanned size, speed, etc. Samples of all recyclable materials materials of interest will be collected and supplied together with their offline composition characterization results. Using these reference samples, IRIS will develop hybrid models based on chemometric and machine learning principles to provide compositional information about the materials. Different frame rates of the cameras, conveyor speed, camera-conveyor distances and collection times will be tested to maximize the accuracy of the distinction between different materials.

T3.3 Repeating AI-ILC advancement & Deep Learning quality assurance [M13-M30] RBNS, FORTH, IRIS

To ensure the viability of prMRF, a process of gradual expansion of the material identification capabilities is necessary, so that the visual system can recognize materials that have not been processed so far (e.g. new commercial products). To this end, this task will develop procedures that facilitate the collection of data and the training of CNN models by non-AI and robotics users. The process will be based on the tool already developed by RBNS for the fast annotation of waste images. The tool will be extended in terms of handling hyperspectral data and improved in terms of usability and interfacing to the CNN training. In parallel to the above FORTH will design, implement and test quality assurance methods for DL outputs that examine the automatically derived behavior after training (i.e. without logical design by engineers). The method will provide a means to compare past/new candidate solutions to assure satisfactory performance.

T3.4 Smart Monitoring and Operational Predictive Analytics [M19-M30] IRIS, HRCC, AIPLAS, FORH

The data to be acquired by the systems developed during the previous tasks provide significant information about how much waste is created and processed in a determined period of time, what kind of plastic is used the most etc. The current task aims to ensure long-term, high quality performance for prMRF. It will leverage the extensive volume of data processed every day to perform joint operational analytics on waste streams and prMRF performance to actively and informatively reconfigure prMRF to maximize productivity and facilitate predictive maintenance (e.g. suction cups will soon need replacement). To this end, a data acquisition platform will be developed. All the data gathered will be cleared, visualised and then combined and analyzed to return valuable insights regarding the production while providing predictions for future incoming waste streams in the short and long term.

Deliverables

D3.1: Material recognition based on RGB and Hyperspectral imaging [IRIS; M18; (T3.1, T3.2)] Report on DL methods for recyclable material recognition applied on RGB and Hyperspectral images.

D3.2: prMRF operation monitoring and repeating advancement [FORTH; M30; (T3.3, T3.4)] Report on smart sensors and processes to assure long-term high quality prMRF performance for both RGB and Hyperspectral imaging.

WP Number	4	WP Leader		KUL		Start Month	1	End Month		30
WP Title	RoReWo-Team for Material Recovery									
Participant No	1	2	3	4	5	6	7	8	9	10
Participant Name	FORTH	UoM	**KUL**	HRRC	IRIS	RBNS	AIMPLAS	AXIA	ISWA	ION
PMs/Participant	14	2	**20**	5	2	18	5			5

Objectives

The WP considers the implementation of the modular robotic system to undertake the physical separation and sorting of recovered materials. The planned work is driven by the following objectives:

- Assess market-available and newly developed gripping mechanisms in picking recyclables of different type
- Implement and test in the filed fully functional RoReWo units amenable to mount several grippers
- Adopt a multi-robot/multi-gripper approach to compose RoReWo-Teams that recover multiple materials

Description of Work

T4.1 Implementation of complementary gripping mechanisms [M1-M18] KUL, FORTH

The nature of waste sorting tasks is significantly different compared to most grasping tasks in industry and agriculture, as the grasp can be destructive, placing accuracy is unimportant and variation is significant in both geometry and material properties. Novel gripping solutions will be investigated, prototyped and tested in the direction of combined gripping mechanisms, such as the use of both vacuum (or high flow lifting) and finger gripping for picking, and use of blowing for fast release.

T4.2 RoReWo implementation and control [M1-M18] RBNS, FORTH

Implementation of low-cost cartesian robot-based RoReWo units, which use closed loop motion control in a single axis to achieve precise movements across the width of the conveyor belt that carries the waste, and use of a vertical motion piston to approach and pick materials (see operational prototype). This task will also explore how RoReWo behavior should be adjusted according to the operation of the grippers (see T4.1), so that together they efficiently recover recyclable materials. The required adjustments will be encoded in parameterized ready to use behaviors, which are expected to significantly facilitate the synthesis of RoReWo-Teams and their integration to prMRF operation.

T4.3 Configuration of the multi-robot/multi-gripper RoReWo-Team [M7-M30] KUL, RBNS, FORTH

This task will improve flexibility and sorting efficiency of the prMRF by facilitating a quick and easy customization to respond to changes in the material supply and waste characteristics. For this purpose, different methods to characterize the geometry of the waste stream based on the images of the computer vision system will be explored and the suitability of different grippers considering these characteristics will be mapped. Considering that multiple robots will cooperate for sorting, we will investigate methods to define the optimal combination of grippers to be used. To obtain the required data and to validate these methods a looping system will be implemented to measure the rates of successful gripping for different RoReWo configurations, as well as the influence of other systems settings, such as the conveyor belt speed.

Deliverables

D4.1: Gripping mechanisms and RoReWo units for material recovery [KUL; M18; (T4.1, T4.2)] Report on the implementation of material grippers, RoReWo units and their combination.

D4.2: Multi-robot/multi-gripper RoReWo-Team configuration [RBNS; M30; (T4.3)] Report on the dynamic configuration of RoReWo-Teams according to the type of materials to be recovered.

WP Number	5	WP Leader		FORTH		Start Month	1	End Month		36
WP Title	prMRF implementation, demonstration and assessment									
Participant No	1	2	3	4	5	6	7	8	9	10
Participant Name	**FORTH**	UoM	KUL	HRRC	IRIS	RBNS	AIMPLAS	AXIA	ISWA	ION
PMs/Participant	**17**	2	15	9	10	15	15	2	2	12

Objectives

This WP aims at the implementation of the complete prMRF through the integration of all relevant enabling technologies. The objectives of the WP are:

- Implementation of the complete waste processing chain for smart and automated material recovery.
- Provision of a complete and fully functional prMRF for validation in challenging real world conditions
- Exploitation of feedback gained by verification tests to fine-tune the modules and the system.
- Definition of maintenance strategy for sustainable prMRF operation beyond **RECLAIM**.

Description of Work

T5.1 Early prMRF development and on-site deployment [M1-M9] FORTH, ION, ALL

Early installation of purchased mechanical equipment (T2.2) and integration of the software and robotic technology already available to **RECLAIM** partners to develop two semi-functional prMRF prototypes powered by hybrid eco-friendly technology (e.g. solar panels). The prMRF will enable extensive image and hyperspectral data collection in particularly realistic conditions in both deployment sites (T6.1). This early version of the prMRF will complete by the end of the first project phase (section 1.3) providing an end-to-end implementation of the waste treatment chain.

T5.2 prMRF iterative implementation-testing-optimization [M10-M30] FORTH, IRIS, KUL, RBNS

The prMRF implemented in T5.1 will operate as industrial experimental platforms that enable the direct integration and testing of the newly developed AI-ILC (WP3) and RoReWo-Team technologies (WP4) with real waste streams. The incremental integration of the gradually advanced AI-ILC and RoReWo-Team in the prMRF and their direct evaluation in material recovery in real conditions, will reveal potential risks and will guide informed iterative optimizations to improve the robustness of the integrated system. By the end of this task a fully functional prMRF will be developed, properly customized in terms of AI-ILC training and RoReWo-Team configuration, to meet the needs of each application scenario.

T5.3 Optimization of RoReWos Team performance [M13-M36] KUL, RBNS, HRRC, AIMPLAS

Implement efficient collaborative behaviors for multi-robot/multi-gripper RoReWos Teams. This includes the distribution of roles and tasks among the available RoReWos and the management of shared resources (e.g. use of timed valves in common vacuum supply) in a way that maximizes global efficiency and balances the work and wear of the different grippers and robots. Compare alternative approaches for creating highly efficient collaborative work plans (spatial waste characteristics, waste location, proximity to collection bin, RoReWos/grippers ordering, common RoReWos roles, commercial value of recovered materials).

T5.4 On-site prMRF assessment and long-term sustainability [M13-M36] FORTH, IRIS, KUL, RBNS, HRRC

The implemented prMRF will be evaluated on material recovery according to the KPIs described in section 1.1. This latest version of the prMRF, is considered the Minimum Viable Product (MVP) of the project, featured with highly desirable characteristics, such as easy portability, fast installation, readily operation and eco-effectiveness regarding waste management for material recovery. The current task will demonstrate both the expandable capacity of the AI-ILC to recognize new materials of interest and the modularity of the RoReWo-Team, as well as the informed, waste stream driven on-the-fly reconfigurability of the individual RoReWos. This task will also demonstrate the ease of use of prMRF by non-robotics experts, in order to promote it as a stand-alone solution capable of covering the need for remote material recovery.

Deliverables

D5.1: Early prMRF development based on available enabling technologies [FORTH; M9; (T5.1)] Report on the development of a first prMRF prototype to set the ground for subsequent implementation steps.

D5.2: Preliminary assessment of prMRF performance [FORTH; M18; (T5.2, T5.3, T5.4)] Report on the repeating implementation-testing spins targeting the optimization of the prMRF.

D5.3: Final assessment of prMRF and sustainability plan [KUL; M36; (T5.2, T5.3, T5.4)] Report on the final assessment of prMRF performance on each deployment site and plan for sustainable prMRF use beyond **RECLAIM**.

WP Number	6	Lead beneficiary		UoM		Start Month		1	End Month	36
WP Title	Recycling Data-Game									
Participant No	1	2	3	4	5	6	7	8	9	10
Participant Name	FORTH	**UoM**	KUL	HRRC	IRIS	RBNS	AIMPLAS	AXIA	ISWA	ION
PMs/Participant	4	**23**		3	4	6	2	2	2	5

Objectives

WP6 considers the implementation of a gamified approach for user participation in recycling data collection, in order to facilitate the training of DL models. The planned work is driven by the following objectives.

- Collect and exploit real waste RGB and Hyperspectral imaging for the development of the RDG.
- Pipeline deep learning and gaming algorithms to create repeatedly new gaming experiences for users.
- Implement an RDG that is easy to use and engaging for citizens of diverse age, gender, and background.

Description of Work

T6.1 On site, waste data collection [M1-M18] RBNS, UoM, IRIS, HRRC

The early development of the targeted prMRF (T5.1), will allow the direct visual and hyperspectral examination of waste streams under the same conditions as the final operation of the prMRF after considering constant and efficient lighting on the waste. Open source versioning tools (e.g. DVC) will be used to efficiently organize the data captured along iterative data collection procedures, to assure quality results. It is planned that data collection will be carried out periodically every 2 months for at least a whole year to meet the expected seasonal characteristics in the appearance of waste (e.g. raindrops in winter, dust in summer). The tool developed in T3.3 will be used by experienced workers to quickly annotate the collected data to be used for RDG development and train the early versions of the AI-ILC module.

T6.2 Algorithms and pipelines for recycling data-games [M3-M30] UoM, ION

This task will create a recycling data-game pipeline for re-using and improving real-world data through human annotations. In this novel paradigm which advances research in game generation (an indicative scenario is provided in Box 3). This will leverage already available machine learning techniques for user modelling, procedural content generation and recommender systems. The algorithmic advances will be integrated into intuitive and simple to use games that can be deployed on many platforms and devices, focusing on web-based solutions. Established game engines such as Unity 3D will be used for development, as they are powerful and can deploy to a broad range of platforms such as windows, android, web, etc. To ensure that RDG evaluation in T6.3 is carried out to best effect, a series of usage analytics will be implemented so that they can be collected online while the users play.

T6.3 Deployment and evaluation of environmental games [M13-M36] UoM, ION, ISWA

This task will validate the RDG, through a series of dedicated pilots with focus groups as well as in-the-wild experiments carried out online. This will allow (a) targeted and specific feedback regarding the incremental advancement of AI-ILC, and rising recycling awareness (through focus groups) and (b) test RDG usability and data collection potential at scale (through in-the-wild experiments) and the exploitation of gathered data by other project developments (T3.3). Focus groups studies will use a combination of quantified metrics collected during playthroughs and qualitative feedback through semi-structured interviews before and after the playthrough. In-the-wild experiments will deploy the games online to engage a large set of users through broad-scale dissemination campaigns. Evaluation in these large-scale experiments will focus on quantified game analytics collected within the game, and will also explore how the user annotations have impacted AI-ILC performance and the circulating user-RDG interaction.

Deliverables

D6.1: Waste Data for material recognition and Recycling Data Game [UoM; M9, M18; (T6.1)] Report on the on-site data collection procedures. Data available for internal and public use.

D6.2: Algorithms and pipelines for Recycling Data Games [UoM; M9, M18, M30; (T6.2)] Report on the implementation of the RDG and its interaction with other project activities.

D6.3: Assessment of the Recycling Data Game [UoM; M18, M36; (T6.3)] Report on the assessment of the RDG, plan for its supportive role on prMRF use beyond **RECLAIM**.

WP Number	7	WP Leader		AXIA		Start Month	1	End Month		36
WP Title	Project impact maximization									
Participant No	1	2	3	4	5	6	7	8	9	10
Participant Name	FORTH	UoM	KUL	HRRC	IRIS	RBNS	AIMPLAS	**AXIA**	ISWA	ION
PMs/Participant	6	6	5	4	4	6	5	**28**	12	4

Objectives

This WP will develop **RECLAIM** outreach activities with the following objectives:

- Develop a plan for and implement dissemination and communication activities, including target groups and channels, to facilitate the outreach of project results.
- Develop a plan for and implement innovation and exploitation activities, demonstrating the potential business opportunities of **RECLAIM** results.
- Develop an IPR strategy, including actions to preserve the IPR generated within the project.
- Carry out networking/clustering activities with other relevant projects and DIHs
- Provide performance metrics and results to benchmark **RECLAIM** solutions

Description of Work

T7.1 Dissemination and communication activities [M1-M36] ISWA, All

This task will monitor the progress of the dissemination and communication strategy summarized in Section 2. Communication activities will start from the very beginning of the project and will continue with concrete actions (images, posts, blogs, articles, presentation, videos etc.) until the end of **RECLAIM**. In addition to the broader picture of the project, the obtained results in all four innovation pillars, will have their own dissemination and communication identity. Noticeably, ISWA has well established contacts with multiple recycling organizations and enterprises all over the world, which will help promote the idea of prMRF-powered distributed material recovery and will help the project achieve substantial global impact.

T7.2 Liaising with other projects and DIHs [M13-M36] AXIA, FORTH, All

This task will focus on establishing and coordinating links with projects and initiatives in AI, Data and Robotics in Horizon Europe and other programmes (e.g. Networks of excellence centres, DIHs, pilots, data platforms), in view of sharing knowledge, developing synergies, and coordinating efforts (e.g. organization of joint events). It is expected that many initiatives will express their interest in both the AI-powered robotic material recovery and the RDG, as tangible examples to promote public awareness and addressing acceptability and trustworthiness of AI. **RECLAIM** will highlight the value generated by the combination of AI, data and robotics in the GreenDeal and the circular economy in particular, to showcase the contribution of state-of-the-art technologies to social welfare and sustainability. Within this frame, particular efforts will be made in order to establish and refine standardization protocols among the relevant projects. Besides the general public, **RECLAIM** will target the business community, with a particular focus on SMEs and governmental authorities to foster vast deployment of prMRF for distributed, decentralized material recovery and additionally the broader domain of smart high-tech solutions.

T7.3 Innovation assets exploitation and commercialization [M13-M36] AXIA, ISWA, ALL

The exploitation plan will be presented in M6 and updated throughout the project implementation, to guide the efforts of the consortium in demonstrating the potential of **RECLAIM** solutions to meet market needs. The business plan report that will be prepared to highlight the business opportunities for the Key Results Owners, will include: (i) the identification of the exploitation routes that partners want to pursuit, (ii) the identification of market segments, (iii) a competition analysis, (iv) a patent mapping analysis, using PatSnap platform, (v) a risk analysis and (vi) a financial analysis. The exploitation activities will take into consideration the need to raise awareness of potential investors and reach out to interested stakeholders, policy-makers and public bodies. In addition, a strategy for the protection of IPR and knowledge management will be established. All IPR principles will be managed from an early stage, through the Consortium Agreement (CA) and the Grant Agreement (GA), where ownership, access rights, publications, confidentiality and liability will be examined. However, a concrete IPR strategy will clarify the ownership of the results and identify the most promising exploitation routes that will secure the collaboration among partners.

Deliverables

D7.1: Plan for the dissemination and communication activities [ISWA; M6, M18, M36; (T7.1)] Report on the implemented and planned dissemination and communication activities to ensure a substantial impact both at the EU level and international level.

D7.2: Plan for the innovation and exploitation activities [AXIA; M6, M18, M36; (T7.2, T7.3)] Report on the implemented and planned innovation and exploitation activities including Open Data and Benchmarking, IPR management and patent mapping analysis.

Table 3.1c. List of **RECLAIM** Deliverables in order of delivery dates.

Del.	Del. Name	WP	Lead	Type	Level	Month
D1.1	Data management plan and ethics/privacy manual	WP1	IRIS	DMP + ETHICS	PU	M6, M36
D2.1	prMRF and RDG requirements and systems specification	WP2	AIMPLAS	R	EU-R	M6
D5.1	Early prMRF development based on available enabling technologies	WP5	FORTH	R+DEM	PU	M9
D6.1	Waste Data for material recognition and Recycling Data Game	WP6	RBNS	R+ DATA	PU	M9, M18
D1.2	1st Periodic Project Report	WP1	FORTH	R	PU	M18
D3.1	Material recognition based on RGB and Hyperspectral imaging	WP3	IRIS	R+O	PU	M18
D4.1	Gripping mechanisms and RoReWo units for material recovery	WP4	KUL	R+DEM	PU	M18
D5.2	Preliminary assessment of prMRF performance	WP5	FORTH	R+DEM	PU	M18
D6.2	Algorithms and pipelines for Recycling Data Games	WP6	UoM	R	PU	M9, M18 M30
D7.1	Plan for the innovation and exploitation activities	WP7	ISWA	R+ DEC	SEN	M6, M18 M36
D3.2	prMRF operation monitoring and repeating advancement	WP3	FORTH	R+O	PU	M30
D4.2	Multi-robot/multi-gripper RoReWo-Team configuration	WP4	RBNS	R+DEM	PU	M30
D5.3	Final assessment of prMRF and sustainability plan	WP5	KUL	R+DEM	PU	M36
D6.3	Assessment of the Recycling Data Game	WP6	UoM	R+DEM	PU	M18, M36
D7.2	Plan for the innovation and exploitation activities	WP7	AXIA	R+ DEC	EU-R	M6, M18 M36
D1.3	Final Project Report	WP1	FORTH	R	PU	M36

3.1.2 Milestones

There are five major milestones in **RECLAIM**, listed in Table 3.1d. These work as checkpoints in order to verify that research and development is progressing in the right direction.

Table 3.1d List of **RECLAIM** Milestones.

No	Milestone	Related WP(s)	Due Date	Means of verification
MS1	**Consortium and project start up**: Partner responsibilities specified, repository procedures established, project e-presence & website, dissemination commenced, prMRF & RDG requirement analysis and specifications complete.	WP1, WP2 ALL WPs	M6	D1.1, D2.1
MS2	**Early integration of available technologies - RECLAIM Phase I complete:** Semi-functional prMRF implementations, data management plan and ethics/privacy manual available	WP5, WP6	M9	D5.1, D6.1
MS3	**Key computer-vision and robotic components available and tested in the wild.** AI-ILC complete, RoReWos and grippers complete, prMRF & RDG implementation in good progress, data publicly available, preliminary innovation exploitation.	WP1-WP7	M18	D1.2, D3.1, D4.1, D5.2, D6.2, D7.1
MS4	**Early integration of available technologies - RECLAIM Phase II complete:** Smart production monitoring complete, RoReWo Team complete, prMRF & RDG complete and ready for final assessment.	WP3, WP4	M30	D3.2, D4.2
MS5	**RECLAIM Sustainability and asset Exploitation.** prMRF and RDG refined and assessed, open access pilot demonstrations finalized, sustainability plan, project assets ready for exploitation and commercialization. Project End.	WP1, WP5 WP6, WP7	M36	D5.3, D6.3, D7.2, D1.3

3.1.3 Project risks

An initial list of identified risks is presented in Table 3.1e; more detailed assessment of risks will regularly be carried out during the execution of the project. As a general prevention measure, the consortium will closely monitor the progress of the project in order to detect and react early to any problems that may occur.

Table 3.1e. Identified **RECLAIM** project risks and mitigation measures.

Risk Description (Likelihood/Severity)	WP(s) involved	Risk-mitigation measures
Technical Risks		
The development of prMRF faces limitations in the formation of the production chain (L:low/S:med)	WP5	FORTH is experienced in implementing compact recycling units (see Figure 1.3). The parallel development of two slightly different versions of prMRF will facilitate the exchange of know-how and good practices to ensure the efficient performance of the mechanical R-Eq across the production chain.
The operation of prMRF requires equipment that could not have been foreseen at proposal submission	WP5, WP3, WP4	This is rather unlikely due to the experience of FORTH in implementing similar compact robotic material recovery units. The end-user ION is fully committed to the project and will provide any minor equipment that might be needed, undertaking the expenses.
The AI-ICL module underperforms in terms of accuracy (Lamed/S:med)	WP3, WP5	Mitigation will try to find a trade-off between performance and complexity. Instead of targeting the categorization of many material types the problem will be simplified, targeting the recovery of less types to increase accuracy rates.
Newly developed grippers do not improve picking skills (L:med/S:low)	WP4, WP5	**RECLAIM** has already available many material specific grippers that perform successfully in material picking. New grippers will focus on universality (i.e. be applicable in many materials). If this is not possible, the available classic grippers will be used to showcase successful prMRF performance.
RoReWos underperform in material picking due to single axis motion. (L:high/S:low)	WP4, WP5	This is a known issue that appears due to the simplified very-low-cost RoReWo architecture. This issue is mitigated by slightly slowing-down the conveyor belt to significantly increase RoReWo picking success.
Material flow is too high and the RoReWos-Team can recover only a part of it (L:High/S:med)	WP5	Due to the modularity of the RoReWo Team this issue can be easily mitigated by adding more RoReWos in the operating Team. To assess that, we will gather all (eight) RoReWos in a single prMRF to assess the performance of the extended RoReWo Team.
Delays in the development of individual tools. The complete solution is not sufficiently tested and complete (L:Low/S:med)	WP3, WP4, WP5, WP6	**RECLAIM** aims at implementing a functional prMRF very early in the project. Interim versions of the tools will be integrated continuously so that it does not hinder the development of other parts and even if the final version of a tool is not yet ready, a tested version can still be included in the solution.
Not many citizens engage with the use of the RDG (L:Low/S:low)	WP6, WP3	RDG will be assessed and refined to ensure user friendliness and usability. The RDG will be actively promoted in social media to gain popularity. All participant organizations will encourage their employees to use the game. UoM, KUL, FORTH has direct access to a large pool of students who are potential users of the RDG
The RDG does not contribute as expected in the collection of annotated data (L:med/S:low)	WP6, WP3	A major part of data collection is planned very early in **RECLAIM**. AIMPLAS, HRRC and ION staff will annotate this data to enable AI-ILC training and satisfactory performance. If necessary, additional resources will be allocated to data collection & annotation. Nevertheless, citizen data will help in the optimization of the AI-ILC, but prMRF can be effective even without them.
Management Risks		
Communication problems may cause delays in the delivery of results (L:low/S:med)	WP1-7	Kick-off meeting will be held to establish personal contacts. Regular monthly communication between WPs will be established. To do Handbook for the day-to-day management of the project will be set up.

Lack of resources for a partner (L:low/S:high)	WP1-7	The issue will be mitigated either by rearranging resources among the partners, or re-planning work distribution to support the partner in need of resources.
Personnel changes may impact the quality of the outcomes (L:low/S:high)	WP1-7	Multiple staff members from each partner are involved in R&D disciplines, ensuring smooth transition for the newly hired staff. Additional support will be provided by partners working in closely related areas.
COVID-19 Pandemic (L:med/S:high)	WP1-7	Consortium partners will utilize online services to launch the project and collaborate in project tasks. prMRF implementation will proceed in slower steps with all partners contributing their parts one after the other for integration in the main platform. We don't foresee any negative impact of Covid-19 to RDG activities.

3.1.4 Summary of effort and costs

A summary of the overall effort is provided in Table 3.1f. The table presents the distribution of person months over the duration of the project for each WP and for each Participant.

Table 3.1f. Summary of **RECLAIM** staff effort.

	WP1	WP2	WP3	WP4	WP5	WP6	WP7	Total PMs
1 FORTH	9	5	13	14	17	4	6	**68**
2 UoM	2	4	9	2	2	23	6	**48**
3 KUL	2	4	4	20	15		5	**50**
4 HRRC	2	5	4	5	9	3	4	**32**
5 IRIS	3	3	27	2	10	4	4	**53**
6 RBNS	2	4	6	18	15	6	6	**57**
7 AIMPLAS	2	6	5	5	15	2	5	**40**
8 AXIA	2	2			2	2	28	**36**
9 ISWA	2	2				2	16	**20**
10 ION	4	5	5	5	12	5	4	**40**
Total PMs	**30**	**40**	**73**	**71**	**99**	**51**	**80**	**444**

3.1.5 Subcontracting costs

None of the partners in the consortium intend to use subcontracting to carry out project tasks.

3.1.6 Purchase costs

RECLAIM regards the construction of complete prMRF unit, which assumes increased costs in terms of purchasing and integrating the recycling equipment used for waste pre- and post- processing, the waste monitoring units and the RoReWo team, into a container box. Additionally, travel costs are occasionally high due to frequent travels of the technical partners to the on-site implementation, testing and deployment sites. Accordingly, for some partners, the budget of "Other Costs" is exceeding the 15% of the personnel costs. For these partners, Table 3.1g provides details on how the relevant "Other Costs" have been determined.

Table 3.1g. Other direct cost' items (travel, equipment, other services, infrastructure).

1 FORTH	**Cost (€)**	**Justification**
Travel and subsistence	45,000	Multiple long travels for on-site prMRF implementation, deployment, assessment + participation to project meeting, technical meetings and conferences + travel of the External Advisory Board
Equipment	40,000	PC + Hyperspectral camera + lighting
Other goods/works/services	15,000	Open Access Fees + Consumables + Audit
Total	100,000	

4 HRRC	**Cost (€)**	**Justification**
Travel and subsistence	12,000	Travel costs for on-site prMRF implementation and testing + participation to project meetings, technical meetings, dissemination events
Equipment	130,000	Container box + Mechanical recycling equipment to be integrated with the prMRF (e.g. trommel, flip-flop sorter, conveyor belts, waste press)
Other goods/works/services		
Total	142,000	

5 IRIS	**Cost (€)**	**Justification**
Travel and subsistence	18,000	Travel costs for project meetings, technical meetings, dissemination events
Equipment		
Other goods/works/services	88,000	Materials and consumables for building the prototype real-time monitoring solution: camera, electromechanical parts, electronics, PC, software licensing etc. + consumables + Open access fees + dissemination events services
Total	106,000	

6 RBNS	**Cost (€)**	**Justification**
Travel and subsistence	30,000	Multiple long travels for on-site prMRF implementation, deployment, assessment + participation to project meeting, technical meetings and conferences
Equipment	40,000	Robotic equipment used for implementing the RoReWo Team. For each RoReWo includes linear positioning unit + motor drive system + gripper. For the whole team includes PC + air blower + valves
Other goods/works/services	20,000	Robotic and other consumables, which includes mainly: piping for pneumatic gripping, screws, suction cups, I/O communication hubs, wiring, sensors (vacuum, temperature, power consumption) and more.
Total	90,000	

9 ISWA	**Cost (€)**	**Justification**
Travel and subsistence	20,000	Travel costs to disseminate RECLAIM achievements + participation to project meetings and conferences
Equipment	0	
Other goods/works/services	12,000	Services for organizing demos in major international events + dissemination material
Total	32,000	

10 ION	**Cost (€)**	**Justification**
Travel and subsistence	20,000	Travel costs for project meetings + technical meetings + participation to live prMRF demos
Equipment	15,000	Supporting equipment to facilitate prMRF deployment and assessment in three different use case scenarios
Other goods/works/services	35,000	Consumables + prMRF setup services + services for waste collection & transport to testing sites + services for prMRF transport to demo sites and conferences
Total	70,000	

3.2 CONSORTIUM AS A WHOLE

The consortium is comprised of a balanced blend of academic & research organizations (FORTH, KUL, UoM, AIMPLAS), a new start-up in the area of AI-driven robotic material recovery (RBNS), an SMEs with significant experience in optics (IRIS), an innovation consultancy service provider (AXIA) a local governmental authority on waste management (ION), a major national material recovery provider (HRRC) and the leading global association on recycling and the circular economy (ISWA). The partners provide complementary expertise to facilitate the project meet its objectives as depicted in Table 1.3.

3.2.1 Partner roles in the project & contributing expertise

FORTH will undertake the coordination of the project (WP1), and will exploit its long experience in computer vision, ML/DL and state-of-the-art robotic material recovery to lead R&D activities regarding prMRF development. It will be involved in data collection, material recognition and material recovery. FORTH will lead WP5, ensuring that enabling technologies are effectively integrated and prMRF is successfully demonstrated.
UoM is very experienced in scientific AI-driven tools for game design and development. UoM will lead WP6 and will undertake the implementation of user-friendly Recycling Data-Games, which will engage non-expert citizens in project activities and particularly the collection of annotated material-imaging data.
KUL is specialized in industrial robotic systems with a focused expertise on gripping mechanisms. KUL will lead WP4 and will investigate multi-material grippers and how they can be optimally attached to effective RoReWo Teams undertaking material recovery.
HRRC has extensive expertise in industrial material recovery. It will undertake a major part of prMRF implementation costs and will contribute in the synthesis, implementation and on-site assessment of the prMRF. HRRC is keen in using multiple prMRFs in material recovery activities in many Greek islands beyond the Ionians.
IRIS is the WP3 leader and contributes in the project expertise in computer vision, hyperspectral imaging, ML/DL for material recognition and the use of non-visual sensors for the control of manufacturing processes.
RBNS is a new start up specialized in material recovery robotics. It brings to the project the already developed and assessed RoReWo units and expertise in robotic deployments for material data collection in the wild.
AIMPLAS provides significant expertise in plastics recovery and the circular economy. It will contribute in prMRF requirements and equipment specification, prioritizing the recovery of different plastics, and optimizing system performance for the different use case scenarios.
AXIA brings expertise in innovation management and will lead WP7. AXIA will undertake activities to exploit the **RECLAIM** products by revealing their value in the context of the Circular Economy and paving the way for their potential commercialization.
ISWA provides an extensive waste management network of experts and associations that spans the globe. ISWA will disseminate **RECLAIM** technological advances worldwide, revealing their positive impact on waste management and the transition to a circular economy.
ION is the local authority responsible for waste management in Ionian Islands, Greece. ION will host the prMRF undertaking material recovery from post-consumer waste providing all facilities necessary for experimentation in the context of the three use case scenarios considered in the project.

10/11/2021

To: Foundation for Research and Technology Hellas FORTH, *RECLAIM proposal Coordinator*

Re: Letter of Support to the RECLAIM project proposal

As a senior scientist and project manager at ABB Corporate Research in Västerås, Sweden, I would like to express my support for the RECLAIM project proposal, coordinated by FORTH, in response to the EU call HORIZON-CL4-2021-DIGITAL-EMERGING-01-09: AI, data and Robotics for the Green Deal (IA).

To accelerate the transition to the circular economy, robots must adopt a key role in material recovery. This is because robots can work non-stop around the clock to sort and clean recycled materials. The recovered high-purity fractions can then be converted into high-quality secondary raw materials in line with the circular economy concepts.

Interestingly, the RECLAIM proposal extends the above even further, aiming at the development of intelligent and robotic mobile units to undertake material recovery in distant areas where waste is mistreated.

I found this idea attractive, workable and useful for the environment, the society and the economy. Therefore, I am pleased to offer my support to the RECLAIM proposal as a member of the Advisory board.

Sincerely,

Pietro Falco, Ph.D.
Senior Scientist
ABB AB, Corporate Research
Forskargränd 7
SE 72178, Västerås
Email: pietro.falco@se.abb.com

NYU | TANDON SCHOOL OF ENGINEERING

Letter of Support for the RECLAIM Project Proposal

I would like to express my support for the RECLAIM project proposal, coordinated by FORTH, in response to the EU call HORIZON-CL4-2021-DIGITAL-EMERGING-01-09: AI, data and Robotics for the Green Deal (IA).

The work proposed in RECLAIM aims at knowledge and technology advancements through data collection and monitoring done by the public. The development of the Recycling Data-Game in particular, provides a nice example of how scientific gaming can be used to facilitate community-powered research, support the solution of modern environmental problems and stimulate innovation.

The AI and Robotics work proposed in RECLAIM is nicely coupled with state-of-the-art research and applications in Game Artificial Intelligence, and manages to directly and effectively exploit the results of collective work to empower the portable robotic Material Recovery Facilities, making them more efficient and flexible to adapt to the needs of the individual deployment sites.

Following the above, I believe that the work proposed in RECLAIM is important, well focused and particularly promising, focusing on the implementation of complementary systems that will operate in the wild to facilitate the solution of modern societal problems.

For these reasons and more, I am pleased to offer my support to the RECLAIM proposal as a member of the Advisory board.

Sincerely,

Julian Togelius
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Department of Computer Science and Engineering
370 Jay Street, Brooklyn, NY 11201
United States of America
Editor-in-Chief, IEEE Transactions on Games
e-mail: julian@togelius.com
Personal web page: http://julian.togelius.com

18/10/2021, Buenos Aires, Argentina

To: RECLAIM proposal Coordinator
Re: Letter of Support to the RECLAIM project proposal

I would like to express my support for the RECLAIM project proposal, coordinated by FORTH, in response to the EU call HORIZON-CL4-2021-DIGITAL-EMERGING-01-09: AI, data and Robotics for the Green Deal (IA).
Clearly, the recent advances in robotics and AI provide new opportunities for the digital transformation of the waste management sector.
RECLAIM proposes important and innovative work in this direction. In particular, the investigation and development of portable robotic Material Recovery Facilities, has the potential to significantly support material recovery in distant areas with poor recycling activities.
Additionally, the implementation of Recycling Data-Games that promotes the engagement of users in recycling research, will help accelerate innovation through public participation.
Following the above, I believe that the research proposed in RECLAIM is both timely and particularly promising, focusing on the implementation of complete and real-world applicable solution that will effectively support recycling.
For this reason, I am pleased to offer my support to the RECLAIM proposal as a member of the Advisory board.

Promoting Sustainable Waste Management Worldwide and a Transition to a Circular Economy

Sincerely,

Marcelo Rosso, Member of the ISWA Board

Ing. Marcelo E Rosso

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